

Automated experimental design with highly parameterized ansätze

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Abstract

Experimental design has been historically driven by scientists' knowledge and creativity. As there are endless ways to arrange components—far too many for a human to explore—scientists rely on their experienced intuition to navigate the vast space of designs. This thesis follows an alternative, top-down approach: instead of generating experiments from scratch, we rely on experimental templates, or *ansätze*. They are configurations of multiple components, typically deployed in regular patterns, with plenty of parameters to tune. These systems are optimized to achieve quantifiable goals, like generating entanglement or detecting gravitational waves. Maximizing a success metric, our algorithms search for novel experiments and provide useful scientific tools. Moreover, as the new proposals often display unexpected tricks, we can gain a deeper understanding of the systems we are tinkering with.

The three articles included in this thesis showcase different applications of the method and some of the challenges we must address. The first article, in [Chapter 3](#), is perhaps the most enlightening. We present an abstract representation of quantum optics experiments which expands the features of its predecessor. Using a graph representation, we find more than 100 new experiments, some almost accidentally. The patterns among solutions are often so clear that any attentive eye can extrapolate them, even without knowledge of quantum physics. This is the power of an intuitive representation.

A further version of the graph framework, introduced in [Chapter 2](#), is used in the second paper, in [Chapter 4](#), to create more efficient experiments. This project presents a common challenge of building experimental *ansätze*: the richer the physics we model, the more expensive the simulations are and the harder it is to explore new designs. Moreover, rather than searching for high-quality designs, which yield exactly the outputs we want, we now trade quality for efficiency, as the output of these and the graph-encoded experiments is probabilistic. Hence, our exploration methods must adapt to the new multiobjective task.

In the final chapters—the introductory [Chapter 5](#) and the following paper, in [Chapter 6](#)—we tackle a new challenge: designing interferometric gravitational wave detectors. Building on the quasi-universal interferometer, a highly expressive grid-like template, our goal is to accelerate the design by surrogating the simulator. We do so by training neural networks, which are also highly parameterized *ansätze* for functions.

Together, the three papers illustrate the power and challenges of expressive *ansätze* to design new experiments. This blooming field benefits enormously from new differentiable simulators, GPU acceleration, and advances in artificial intelligence. We discuss some of these in [Chapter 7](#), as we look forward to new research horizons.

Zusammenfassung

Das experimentelle Design wurde historisch von Wissen und Kreativität der Forschenden getrieben. Da es endlose Möglichkeiten gibt, Komponenten anzuordnen—viel zu viele, als dass ein Mensch sie erkunden könnte—verlassen sich Forschende auf ihre erfahrene Intuition, um sich im riesigen Designraum zurechtzufinden. Diese Dissertation verfolgt eine alternative Top-down-Methode: Anstatt Experimente von Grund auf zu generieren, stützen wir uns auf experimentelle Vorlagen, sogenannte Ansätze. Diese sind Konfigurationen mehrerer Komponenten, die typischerweise in regelmäßigen Mustern angeordnet sind und über zahlreiche einstellbare Parameter verfügen. Diese Systeme werden optimiert, um quantifizierbare Ziele zu erreichen, etwa die Erzeugung von Verschränkung oder die Detektion von Gravitationswellen. Indem sie eine Erfolgsmetrik maximieren, suchen unsere Algorithmen nach neuartigen Experimenten und liefern nützliche wissenschaftliche Werkzeuge. Da die neuen Vorschläge zudem oft unerwartete Kniffe aufweisen, gewinnen wir ein tieferes Verständnis der Systeme, mit denen wir hantieren.

Die drei in dieser Dissertation enthaltenen Artikel zeigen verschiedene Anwendungen der Methode sowie einige der Herausforderungen, denen wir uns stellen müssen. Der erste Artikel, in Kapitel 3, ist vielleicht der aufschlussreichste. Wir stellen eine abstrakte Darstellung von Quantenoptik-Experimenten vor, die die Funktionen ihres Vorgängers erweitert. Mithilfe einer Graphdarstellung finden wir mehr als 100 neue Experimente, einige davon fast zufällig. Die Muster zwischen den Lösungen sind oft so klar, dass ein aufmerksames Auge sie extrapolieren kann, selbst ohne Kenntnisse der Quantenphysik. Das ist die Stärke einer intuitiven Darstellung.

Eine weiterentwickelte Version des Graph-Frameworks, eingeführt in Kapitel 2, wird im zweiten Artikel, in Kapitel 4, verwendet, um effizientere Experimente zu erzeugen. Dieses Projekt zeigt eine häufige Herausforderung beim Aufbau experimenteller Ansätze: Je reichhaltiger die Physik ist, die wir modellieren, desto teurer werden die Simulationen und desto schwieriger wird es, neue Designs zu erkunden. Statt nach hochwertigen Designs zu suchen, die genau die gewünschten Ergebnisse liefern, tauschen wir zudem Qualität gegen Effizienz, da die Ausgabe dieser sowie der graph-kodierten Experimente probabilistisch ist. Daher müssen sich unsere Explorationsmethoden an die neue multikriterielle Aufgabe anpassen.

In den abschließenden Kapiteln—dem einführenden Kapitel 5 und dem darauffolgenden Artikel, in Kapitel 6—widmen wir uns einer neuen Herausforderung: dem Design interferometrischer Gravitationswellendetektoren. Aufbauend auf dem quasi-universellen Interferometer, einer hochgradig ausdrucksstarken, gitterartigen Vorlage, wollen wir das Design zu beschleunigen, indem wir den Simulator surrogieren. Dies erreichen wir, indem wir neuronale Netze trainieren, die ebenfalls hochgradig parametrisierte Ansätze für Funktionen darstellen.

Zusammen verdeutlichen die drei Artikel die Stärke und die Herausforderungen ausdrucksstarker Ansätze zum Design neuer Experimente. Dieses aufblühende Feld profitiert enorm von neuen differenzierbaren Simulatoren, GPU-Beschleunigung und Fortschritten in der künstlichen Intelligenz. Einige davon diskutieren wir in Kapitel 7, während wir neuen Forschungshorizonten entgegenblicken.

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Part I.

Introduction

Theory and experiment have been the pillars of science for centuries. In this classic view, scientists are keen observers who stare at nature trying to find out how it works. Eager for understanding, they elaborate theories, models, equations..., explanations of what they have seen that allow them to predict what may happen. This picture has radically changed in less than a century, as computer simulations have established a new paradigm in scientific research, becoming the third pillar of science.

Nowadays, we set the rules of virtual physical systems, fix the conditions of our "experiments", and observe their evolution and properties. The implications are far beyond having fast calculators [1]. The precise control of the studied systems and the ability to tinker with them in endless ways provide priceless insights and intuition. Undoubtedly, these predictions must be experimentally tested, but also in designing such experiments, computers can guide us. With a focus on quantum optics, this thesis shows how to automate the search for new experiments, obtaining useful configurations and a deeper understanding.

1.1 Automating experimental design

In science and engineering, design is driven by measurable goals. We may want materials with certain mechanical properties [2], circuits that implement specific transformations [3], or drugs that interact with cells in a more or less deadly manner [4]. Often, we aim for multiple, and even competing, objectives. Domain expertise and intuition are common paths towards such objectives, but alternatives exist. Instead of trial and error from scratch, by proposing and testing designs, we can define solution candidates and tweak them until they meet our requirements. Our initial guesses, or *ansätze*¹, are designs with tunable parameters, continuous or not, from which we compute a set of properties. The space of possible configurations is then explored by a search algorithm until successful proposals are found or some stopping criteria are met. This approach is known as inverse design.

Applying these principles to the design of experiments presents multiple challenges. First, we need the *ansätze*, the models to be tuned. We must be able to simulate a variety of setups, to then propose parameterized designs expressive enough to include useful proposals. For that, "usefulness" needs to be defined. Given a set of parameters (inputs), the model has multiple associated properties (outputs). We must choose which properties to foster, which to avoid, and to what extent. These choices subsume into a loss function that our favorite optimization algorithm will minimize by adjusting the available parameters. Finally, when new solutions are found, we want to understand how they work and ensure they are feasible. If the simulator is flawed or we did not properly constrain the search space, the optimizer may exploit the loopholes and yield unachievable designs with a very low loss. Moreover, even when correct, inverse design solutions are often counterintuitive. Learning how they work and extrapolating the findings deepens our understanding of the physics we model, and it can accelerate discovery [5].

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1: as Spaniard, I first ran into an *ansatz* when learning about variational methods, where a parameterized function is optimized to minimize certain quantities. Machine learning operates on similar principles. From polynomial fitting with a few variables to transformer architectures with millions, they are all parameterized functions optimized to reach some goal.

The described blueprint can be loosely split in three stages: simulation, exploration, and understanding. Within this framework, the next sections offer an overview of the works presented in [Chapter 3](#), [4](#), and [6](#), the three main papers of this thesis. Here, we focus on the types of systems we model, the degrees of freedom we can tune, and the goals we want to reach. We also offer some preliminary insights from the designs' structure, while deeper analysis are left for the following chapters.

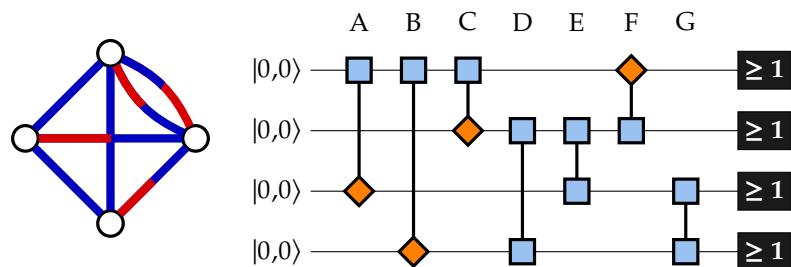
1.1.1 Simulation

Physics simulators, as theoretical models, offer a simplified description of nature. We can describe physical systems in arbitrary detail, but this comes with a higher computational cost, and the same applies to our *ansätze*. There is plenty to model, not only the mechanisms we can leverage, also the environment we cannot fully control. However, as the computational demands increase, the search becomes slower, and finding good designs gets harder. There is a tension between capturing physical details and offering a simple parameterized design that we can successfully optimize. The development of PyTheus [\[6\]](#), presented in [Chapter 3](#), exemplifies how simple representations can boost discovery.

Comparing the graph-based abstraction of PyTheus with Melvin [\[5\]](#), its predecessor, it is hard to imagine that both model photonic experiments. Melvin proposed a toolbox with possible initial states and experimental transformations one could apply to them. Using these resources, Melvin assembled experiments and then computed the final state and transformation. While some elements had continuous tunable parameters, the choices were mostly discrete. The path identity framework simplified the procedure [\[7\]](#), as one just had to choose in which optical paths photons were emitted from different sources. Shortly after, this was shown to be equivalent to weighted undirected graphs [\[8–10\]](#), in which edge colors represented photon modes and nodes represented optical paths. Certainly, the graphs omit implementation details, but because different experimental platforms can reproduce our proposals, the possibilities extend beyond Melvin's toolbox.

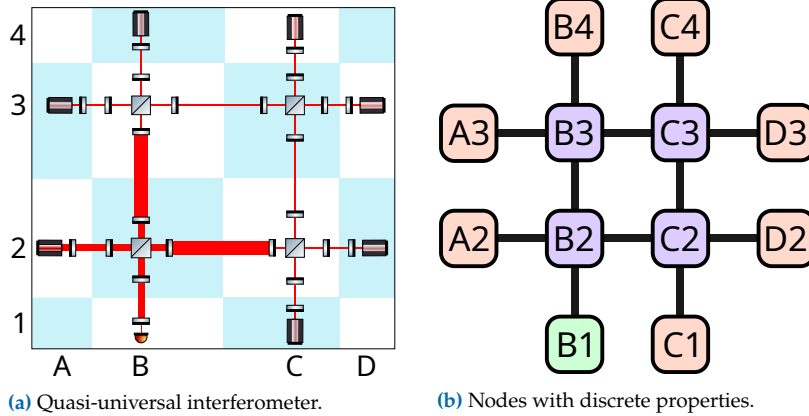
A major advantage of the graph representation, shared by the following *ansätze*, is the possibility of a top-down design strategy. Melvin's setups grow as the algorithm progresses, so one needs to limit the available resources or set some other stopping criteria. But what if we were a few elements away from a groundbreaking design? Or maybe our early choices already ruled out useful experiments? Instead, a top-down approach starts with a highly expressive and redundant design, which is then optimized and simplified as much as possible. For the graph, we make an initial choice of the number of photons (nodes) and their modes (colors), and build a fully connected graph with these properties. From there, computations get simpler as the graph is repeatedly pruned.

Figure 1.1: from a single graph to multiple possible circuits. If the sources' ordering is considered, a graph representation of 7 edges can have up to 5040 permutations.



The Pytheus algorithm largely avoids discrete search, as a sufficiently expressive graph would include the target designs for the right choice of parameters. Our main limit is the size of the graphs we model, as we need to compute the hafnian for each color combination, which is a #P problem [11]. The challenge increases for the second ansatz, as it is a generalization of the graph representation where the order of the sources (the edges) must be taken into account. Similar to the original path identity, these designs can be represented as circuits or, equivalently, directed acyclic graphs. Accordingly, for each initial graph described by Pytheus, we must now consider multiple permutations (see Fig. 1.1). Moreover, as we see in Chapter 4, the computational costs of a single circuit are significantly higher than for the equivalent (unordered) graph.

From the generation and manipulation of quantum states, we move to gravitational wave detection with the *quasi-universal interferometer*, or UIFO [12]. Unlike the previous cases, this ansatz was developed on top of an existing interferometer simulator, Finesse, and the main goal in Chapter 6 is to reduce its dependence by using a much faster neural surrogate. However, even with surrogates or faster simulators, the UIFO designs still require a discrete search. With the previous graphs and circuits, we could avoid discrete choices by adding more elements, even if computational costs may be prohibitive. In contrast, the rectangular grid structure of the UIFO requires us to select the optical elements at specific nodes and the discrete features they may exhibit (see Fig. 1.2). We cannot have them all, even if we could compute them. Moreover, when placing extra mirrors, the most common element, these introduce losses into the system and additional constraints² the optimizer must consider. In short, in the UIFO grid, extra elements may hinder³ performance, and adding them is not always possible.



2: they must not be reached by more light power than they can resist. The same applies to beamsplitters.

3: in previous designs, this was not a problem. Their elements effectively "disappeared" if their properties were zero.

Figure 1.2: the grid structure of a quasi-universal interferometer. (a) Our ansatz for interferometric gravitational wave detectors is a rectangular grid of beamsplitters, surrounded by a detector (cell B1) and multiple light sources. Multiple mirrors are placed between these elements. (b) At each node, a discrete property must be set. In the bulk, beamsplitters must be oriented. The detector must be placed on the border, and the type of light, squeezed vacuum or a laser pulse, must be chosen. In this simple configuration, all beamsplitters are aligned, and only laser light is used.

1.1.2 Exploration

Enabling a top-down optimization approach is one of the primary advantages of the presented ansätze. We replaced most of the costly discrete search with continuous parameter optimization, which allows us to leverage gradient-based methods and (sometimes) automatic differentiation. Details on the optimization methods, such as how stochasticity helps us avoid local minima, are covered in the following chapters. Here, we briefly review the different objectives we pursue and how constraints and/or penalties are enforced.

4: the graph may describe a transformation, but the representation is the same.

5: this is described in [Chapter 2](#) and is closely related to cosine similarity.

The outcome of the Pytheus' graphs are quantum states⁴, i.e., an array of complex values. In most cases, we search for specific states and use a similarity metric as a loss function, typically fidelity⁵. While we also aim for other properties, such as entanglement, we always look for one at a time. Thus, there are no conflicting properties we must trade off. As a rule of thumb, we search for small graphs, as they usually lead to simpler implementations and less noisy outputs. In Pytheus, this is achieved by iterating optimization and pruning, usually removing the lower weighted edges. Additionally, certain nodes should not be directly connected for some applications. This constraint, the only one we have, is enforced in the initial topology. Hence, no penalty is required, and vanilla gradient descent suffices.

The circuits from [Chapter 4](#) also output quantum states, but these are described in more detail than those from Pytheus. Again, we aim for particular states, and use the fidelity as a distance metric towards our goals. However, from the new state description, we also extract the probability that our experiments yield the desired outputs. This probability, which we aim to maximize, conflicts with fidelity. The higher the outcome quality, the less likely it becomes. Accordingly, we look for the best compromise between both, the Pareto front. We face the task with an important advantage: Pytheus' results show that an almost perfect similarity can be achieved. Thus, we can turn this multiobjective optimization into a search for designs that maximize the probability for a given fidelity, which is enforced with a strong penalty in the loss function. Additional properties can be favored with softer penalties, such as sparsity, and topology constraints can be placed by construction, just as with the previous graphs. Due to the large fluctuations in the probability values, the penalty weights must be tuned for different states and fidelities.

6: as values fluctuate by several orders of magnitude, we average their logarithms. This is the so-called geometric mean.

Finally, the interferometers are the systems with the richer physics, further described in [Part III](#). From the modeled phenomena, we focus on the sensitivity curve and the light power at each optical element. The former are the smallest perturbations we can detect for different wave frequencies, an array of values whose (geometric) average⁶ we minimize. While the sensitivity of the designs usually peaks within a certain frequency range, high performance can be achieved over a broad frequency range [12]. There is no clear tradeoff as before. We also monitor the light power at each optical element to ensure they are not damaged. Here, we are conflicted: high powers can boost sensitivity, but when a single element is damaged, the design becomes unusable. This is addressed by adding a penalty for each optical element, using a differentiable approximation of the step function. This function must be steep enough for powers to approach the threshold, and high enough that a single penalty can rule out the design. Unlike the previous cases, no further constraints are added to simplify the design.

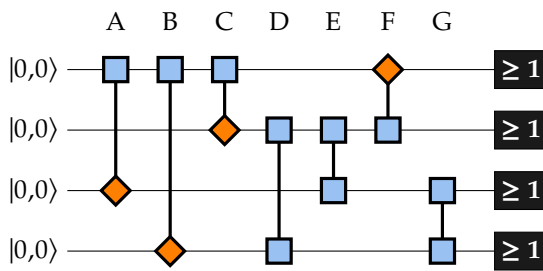
1.1.3 Understanding

The presented ansätze are conceptually similar to modern machine learning architectures. Both are parameterized models optimized to perform a task, and we must select certain discrete features (or hyperparameters) to then optimize their continuous parameters. They share a common risk too: expressive models provide useful tools that we may not understand [13, 14]. While understanding is an elusive concept [15], here we refer to the ability to predict, with some precision, the models'

behavior without computing them. This may not seem like a high bar for scientists. However, one could see our designs as means to higher ends—studying new forms of entanglement or extracting information from gravitational waves—and disregard deeper insights. That would be scientifically unsatisfactory, but also a mistake with very practical implications.

Our design proposals are meant to be deployed⁷, so they must be reliable. This is clear for interferometric detectors, where failing to assess the optical powers flowing across the mirrors can lead to very expensive damage. Of course, additional analysis can and should be performed. For example, the beam profile, which is not accounted for in our ansatz, plays a key role in the light power damage. Meticulous studies of the proposals must take place, often with more expensive simulations and careful human examination. Yet, we need to understand why designs fail and incorporate that knowledge into our search⁸ pipelines. Even if design miscalculations do not lead to costly damages, pursuing wrong proposals would cost time.

Another strong, pragmatic motivation to seek a deeper understanding is to improve the search process and even the ansätze. Already in the Melvin algorithm [5], the toolbox was expanded as useful configurations were identified. Similar structures emerge in graph-based designs [19, 20], where combinations of sources act as independent tools. Intuitive representations⁹ and visualizations can help us identify such structures [21], which can then be reused in new designs or incorporated into the ansatz. The mathematical structure of the ansätze can also guide the search. Many of our designs are invariant under group transformations, and the solutions in Chapter 4 also present partial commutation relations, which can be leveraged to reduce the discrete search space (see Fig. 1.3). Similar principles could be applied to continuum parameter spaces, where identifying fundamental domains¹⁰ can mitigate the dimensionality curse and boost the search for larger and more complex experiments.



1.1.4 A never-ending pursuit

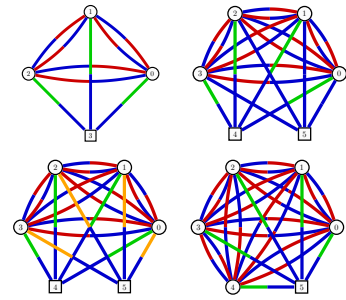
This has been a preview of the thesis. We have seen the common aspects of our ansätze, especially the sharp preference for top-down optimization, largely avoiding discrete search. Further details of the experiments' physics and the different objectives are left to the following chapters. As a final remark, it is worth stressing the nonlinear nature of our approach and how strongly its 3 main "steps" influence one another.

From the start, the ansätze are designed to favor continuous optimization and gradient-based methods to facilitate the search. Extracting design principles from the solutions is another valuable tool for exploration, providing new solutions without resorting to brute-force search. Moreover, even when our algorithms yield infeasible solutions, we may gain

7: some already were [16–18].

8: like with the beam profile, this can still be done in a separate stage [12]. Yet, even if finetuning is required, ansätze must be reliable for the approach to work.

9: one can find and extrapolate patterns in graph solutions without knowledge of quantum optics, as in Fig. A.8:



10: that is, for a function f , the smallest input region that hits every possible output, without redundant copies from the group's symmetries.

Figure 1.3: among the 5040 possible permutations of the circuit's sources, only 54 are canonical sortings. The rest are equivalent to one of these 54, under partial permutations and symmetry transformations. More details in Appendix B.

valuable insights to reshape our proposals and exploration strategies. Sometimes we want to model phenomena that were previously ignored, and we must step back from our simplifications. Inversely, we may find that some physics details are not as relevant as we initially thought, and can be dropped out. Hence, our methods are in constant evolution, as are the experiments we model and design.

1.2 Thesis' outline

This document is organized in two main parts, each with an introductory chapter followed by the papers that present the thesis' contributions.

- **Designing quantum experiments.** Expanding the graph based representation of quantum optics experiments, the Pytheus paper finds 100 diverse quantum experiments [6]. Then, reconsidering some of the simplifications of the graph representation, we propose circuit-like experiments able to generate different types of entanglement more efficiently [22].
- **Faster search of interferometers.** Quasi-universal interferometers are a promising tool to design high-quality gravitational wave detectors, but they are expensive to compute and optimize. To speed up the search, we train surrogates that, after learning from the simulator, guide us across the space of interferometer designs [23].

To conclude, we synthesize the contributions of this thesis' and look forward to new research directions.

Part II.

Quantum Optics Experiments

Quantum optics | 2.

This chapter introduces the path identity experiments and the different ways in which we can represent them. Aiming to make it self-contained, the chapter also includes a brief introduction to quantum physics and the description of the experimental building blocks we use. The next two chapters are built upon these foundations.

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2.1 Quantum crash course

This is a very succinct introduction to quantum theory. It shows the basic elements of quantum theory from a matrix-centered perspective, much inline with quantum computing textbooks [24].

2.1.1 Quantum states

The polarization of a single photon can be described as a coherent *superposition* of 2 physically distinguishable states, horizontal and vertical, respectively described as

$$|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle, \quad \text{and} \quad |V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle. \quad (2.1)$$

Such system, often called *qubit*, can be fully described with a unitary *vector state* in a two-dimensional complex *Hilbert space*, $\mathcal{H}_2$, as¹

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \quad (2.2)$$

where $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$. Taking the parameters θ and ϕ , we can represent a qubit as a point in the surface of the *Bloch sphere*.

1: multiplying the vector state with a *global phase*, $e^{i\alpha}$, has no measurable effects. Thus, equation 2.2 describes any qubit state unambiguously.

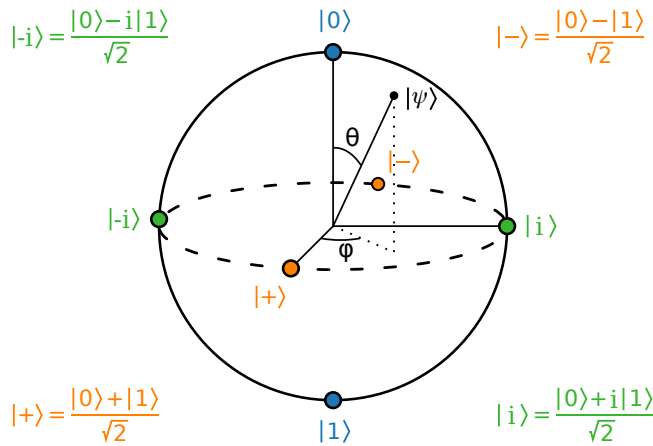


Figure 2.1: the Bloch sphere, orthogonal states are located at antipodal points. The basis $\{|+\rangle, |-\rangle\}$ and $\{|i\rangle, |-i\rangle\}$, are often used to describe qubits.

We can compute the *fidelity* between 2 states, $|\alpha\rangle = a_0|0\rangle + a_1|1\rangle$ and $|\beta\rangle = b_0|0\rangle + b_1|1\rangle$, using the *inner product* of their Hilbert space

$$F(\alpha, \beta) = |\langle\alpha|\beta\rangle|^2 = |a_0^*b_0 + a_1^*b_1|^2 = F(\beta, \alpha), \quad (2.3)$$

where $\langle \alpha |$ is the *complex conjugate* of $|\alpha\rangle$. This is a measure of similarity between states.

Beyond qubits, quantum systems can have an arbitrary number of orthonormal states. Even single photons can be in a coherent superposition of infinitely many *orbital angular momenta* or *optical paths*, among other properties. If a system can be in a coherent superposition of d different states, we call it *qudit*. A *multipartite system* (or *composite*) of several qudits is described by a *joint Hilbert space*, defined as the tensor product of the Hilbert spaces of each subsystem. For example, if we have 3 qubits, A , B and C , they can be described within the Hilbert space

$$\mathcal{H}_{ABC} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C, \quad (2.4)$$

as

$$|\psi_{ABC}\rangle = \sum_{i,j,k} c_{ijk} |\alpha_i\rangle \otimes |\beta_j\rangle \otimes |\gamma_k\rangle, \quad (2.5)$$

2: $|\alpha_i\rangle \otimes |\beta_j\rangle \otimes |\gamma_k\rangle$ is typically written as $|\alpha_i\rangle |\beta_j\rangle |\gamma_k\rangle$ or $|\alpha_i, \beta_j, \gamma_k\rangle$ with or without commas. We stick to the latter option.

where $\sum_{ijk} |c_{ijk}|^2 = 1$, and $\{|\alpha_i\rangle \otimes |\beta_j\rangle \otimes |\gamma_k\rangle\}_{ijk}$ are arbitrary states² from the 3 subspaces that form the 8-dimensional space $\mathcal{H}_{ABC}$.

The vector states we have seen so far, also called *pure states*, offer a complete description of an isolated physical system. However, our knowledge about the system might be partial. We might know that a qubit was prepared in the state $|0\rangle$ or $|-\rangle$, with probabilities 0.6 and 0.4, respectively. Such a state is described as a *statistical ensemble* of pure states, or *mixed state*

$$\rho = \frac{3}{5} |0\rangle\langle 0| + \frac{2}{5} |-\rangle\langle -| = \frac{1}{5} (4|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|), \quad (2.6)$$

typically described with a *density matrix*. For an arbitrary Hilbert space, we can define a mixed state as

$$\rho = \sum_i p_i |\phi_i\rangle\langle \phi_i|, \quad (2.7)$$

3: any mixed state can be originated from infinitely many ensembles. Using a *unitary matrix* U , i.e. $U^\dagger U = U U^\dagger = I$, we can define $\{|\phi_i\rangle\}$ such that $\sqrt{p_i} |\phi_i\rangle = \sum_j u_{ij} \sqrt{q_j} |\varphi_j\rangle$, and therefore $\rho = \sum_i p_i |\phi_i\rangle\langle \phi_i| = \sum_j q_j |\varphi_j\rangle\langle \varphi_j|$.

where $p_i \geq 0$ such that $\sum_i p_i = 1$, and $\{|\phi_i\rangle\}$ are normalized states, not necessarily orthogonal³ between them. When $p_i = 1$, i.e. $\rho = |\phi\rangle\langle \phi|$, we have a pure state. In the case of qubits, we can describe mixed states as points within the Bloch sphere shown in Fig. 2.1. Using the angles $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ from equation 2.2, together with a modulus $r \in [0, 1]$, we can describe any qubit as

$$\rho(\vec{r}) = \frac{I + \vec{r} \cdot \vec{\sigma}}{2} \quad \text{where} \quad \begin{aligned} \vec{r} &= r(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \\ \vec{\sigma} &= (\sigma_x, \sigma_y, \sigma_z) \end{aligned} \quad (2.8)$$

The *Bloch vector*, $\vec{r}$, represents a point within the Bloch sphere. It is often written as $\vec{n}$ for pure states, when $r = 1$. I is the identity matrix, and $\{\sigma_x, \sigma_y, \sigma_z\}$ are the *Pauli matrices*,

$$\sigma_x = |+\rangle\langle +| - |-\rangle\langle -|, \quad \sigma_y = |i\rangle\langle i| - |-i\rangle\langle -i|, \quad \sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (2.9)$$

Mixed states are useful to describe part of a composite system. The state described in equation 2.6, let us rename it ρ^B , can be part of a larger system described by

$$|\psi_{AB}\rangle = \sqrt{\frac{3}{5}} |0\rangle_A |0\rangle_B - \sqrt{\frac{2}{5}} |1\rangle_A |-\rangle_B. \quad (2.10)$$

ρ^B is the *reduced density matrix* from $\rho^{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|$, and is obtained with the *partial trace*

$$\text{tr}_A(\rho^{AB}) = \frac{3}{5} \text{tr}(|0\rangle\langle 0|) |0\rangle\langle 0| + \frac{2}{5} \text{tr}(|1\rangle\langle 1|) |- \rangle\langle -| = \frac{3}{5} |0\rangle\langle 0| + \frac{2}{5} |- \rangle\langle -|. \quad (2.11)$$

Conversely, we can represent any mixed state as the a pure state within a larger Hilbert space, with a process is called *purification*. Taking the state ρ from equation 2.7, of arbitrary dimensions, we can describe a bipartite pure state

$$|\Phi_{SA}\rangle = \sum_i \sqrt{p_i} |\phi_i\rangle_S |a_i\rangle_A, \quad (2.12)$$

where $\{|a_i\rangle\}$ is an arbitrary⁴ orthonormal set of states from an ancillary space $\mathcal{H}_A$. Then, by construction

$$\text{tr}_A(|\Phi_{SA}\rangle\langle\Phi_{SA}|) = \sum_i p_i |\phi_i\rangle\langle\phi_i| = \rho. \quad (2.13)$$

Finally, the fidelity between 2 arbitrary states, ρ and σ , is defined as⁵

$$F(\rho, \sigma) = \left(\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2. \quad (2.14)$$

While this expression might look very different from the fidelity described in equation 2.3, if the states are pure, we recover the old description

$$F(\rho, \sigma) = \left(\text{tr} \sqrt{|\rho\rangle\langle\rho| \sigma |\rho\rangle\langle\rho|} \right)^2 = |\langle\rho|\sigma\rangle|^2 \left(\text{tr} \sqrt{|\rho\rangle\langle\rho|} \right)^2 = |\langle\rho|\sigma\rangle|^2. \quad (2.15)$$

On the other hand, the fidelity between a pure state $|\psi\rangle$ and a mixed state ρ can be interpreted as the weighted average of fidelities between the pure state $|\psi\rangle$ and the states from the statistical ensemble $\{p_i, \phi_i\}$ that constitute ρ .

$$\begin{aligned} F(\psi, \rho) &= \left(\text{tr} \sqrt{|\psi\rangle\langle\psi| \rho |\psi\rangle\langle\psi|} \right)^2 = \langle\psi|\rho|\psi\rangle \\ &= \langle\psi|\left(\sum_i p_i |\phi_i\rangle\langle\phi_i| \right)|\psi\rangle = \sum_i p_i |\langle\psi|\phi_i\rangle|^2. \end{aligned} \quad (2.16)$$

The last two equations suffice for this project, since we always compare pairs of states from which at least one is pure.

Finally, states can mutate, and the *transformation* of a state ρ into a new one ρ' can be described by a *unitary operator* U , such that

$$\rho' = U\rho U^\dagger, \quad \text{where} \quad U^\dagger U = U U^\dagger = I, \quad (2.17)$$

or $|\psi'\rangle = U|\psi\rangle$ for pure states. These operators can describe continuous transformations, usually in time, of a certain initial state⁶

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle. \quad (2.18)$$

In physical systems, these operators arise from *Hamiltonians*, H , that dictate the evolution of a system according to the *Schrödinger equation*⁷

$$H |\psi(t)\rangle = -i\hbar \frac{d}{dt} |\psi(t)\rangle \quad \rightarrow \quad |\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle. \quad (2.19)$$

4: since the orthonormal states from $\mathcal{H}_A$ are arbitrarily chosen, there are infinitely possible purifications related by unitary transformations.

5: by definition, a density matrix A is a positive operator of trace 1. Thus, $\sqrt{A} = \sum_i \sqrt{\lambda_i} |\lambda_i\rangle\langle\lambda_i|$, where $\{|\lambda_i\rangle\}$ is the orthonormal eigenvectors, and $\lambda_i \geq 0$ the eigenvalues, with $\sum_i \lambda_i = 1$.

6: in the context of quantum computing, transformations are implemented by gates, which are continuous transformations for fixed times.

7: we assume the Hamiltonians are time-independent.

2.1.2 Measurements

Now that we know what a state is, we want to know how it is doing, we want to *measure* it. A measurable attribute $\mathcal{A}$ of an arbitrary state, $\rho \in \mathcal{H}$, can be described with an *observable* operator A . That is, a hermitian matrix acting on the same Hilbert space $\mathcal{H}$ to which the state belongs. Since A is hermitian, we can *spectrally decompose* it as⁸

$$A = \sum_n a_n P_n, \text{ where } P_n = \sum_{m=1}^{g_n} |a_{nm}\rangle\langle a_{nm}|, \text{ and } \sum_n P_n = I. \quad (2.20)$$

The orthonormal states $\{|a_{nm}\rangle\}$ are the basis of a subspace of dimension g_n , which contains all eigenstates with a well defined attribute value a_n , which is a real eigenvalue⁹. The probability of each outcome of an observable A is given by the *projectors* $\{P_n\}$. For example, the probability of observing a_n when measuring the state ρ is

$$\text{prob}(\mathcal{A} = a_n) = p_n = \text{tr}(P_n \rho) = \sum_{m=1}^{g_n} \langle a_{nm} | \rho | a_{nm} \rangle. \quad (2.21)$$

Accordingly, the expected value of the observable A is

$$\langle A \rangle = \sum_n a_n p_n = \sum_n a_n \text{tr}(P_n \rho) = \text{tr}\left(\rho \sum_n a_n P_n\right) = \text{tr}(\rho A). \quad (2.22)$$

8: we only describe and use *projective* measurements. These can be generalized by *positive operator-valued* measurements.

9: we assume that the measurement outcomes form a *discrete spectrum*, an assumption that holds for the whole thesis.

10: in the experiments we will see, the photons are no longer described by a coherent state after they are detected.

After the measurement, if the state is not destroyed¹⁰, it will become

$$\rho_n = \frac{P_n \rho P_n}{p_n} = \frac{P_n \rho P_n}{\text{tr}(P_n \rho)}. \quad (2.23)$$

Similarly, a vector state $|\psi\rangle$ becomes

$$|\psi_n\rangle = \frac{P_n |\psi\rangle}{\sqrt{p_n}} = \frac{P_n |\psi\rangle}{\sqrt{\langle \psi | P_n | \psi \rangle}} = \frac{P_n |\psi\rangle}{\|P_n |\psi\rangle\|}, \quad (2.24)$$

where $\|P_n |\psi\rangle\|$ is the norm of the projected vector.

The expressions for the probabilities are very similar to those for the fidelities. If we measure the polarization of a photon ρ the probability of obtaining $|+\rangle$, given by the observable σ_x , is the same as the fidelity between ρ and $|+\rangle$. Moreover, two arbitrary states can be reliably distinguished only when their fidelity is 0, no matter the measurements we use.

2.1.3 Entanglement

In Section 2.1.1, we saw states formed by different parts, each from a different Hilbert space, collectively described within a composite Hilbert space. Let us revisit the concept with a 2 qubit system,

$$\begin{aligned} |\psi_{AB}\rangle &= \frac{1}{2} |0\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |1\rangle_B \\ &= \frac{1}{2} \sum_{i,j=0}^1 |ij\rangle_{AB} \in \mathcal{H}_{AB} \end{aligned} \quad (2.25)$$

where $\mathcal{H}_{AB} = \mathcal{H}_2 \otimes \mathcal{H}_2 = \mathcal{H}_2^{\otimes 2}$. This lengthy state can be rewritten as

$$|\psi_{AB}\rangle = \left(\frac{1}{\sqrt{2}} \sum_{i=0}^1 |i\rangle_A \right) \left(\frac{1}{\sqrt{2}} \sum_{j=0}^1 |j\rangle_B \right) = |+\rangle_A \otimes |+\rangle_B = |++\rangle. \quad (2.26)$$

Therefore, $|\psi_{AB}\rangle$ is a *separable state*, as it can be rewritten as a tensor product of its different parts. This does not apply to the *Bell state* [25],¹¹

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad (2.27)$$

as it is a non-separable, and therefore *entangled state* [26]. This mathematical distinction has huge implications. The measurements on different subsystems of a separable state yield independent outcomes. The output of a measurement σ_z in the first qubit of $|++\rangle$ tells us nothing about the state of the second¹². Contrarily, if we repeat an arbitrary measurement, $N = |\vec{n}\rangle\langle\vec{n}| - |-\vec{n}\rangle\langle-\vec{n}|$, on the qubits of a Bell state $|\Phi^+\rangle$, their outputs will be the same, 1 or -1. To see this, we can rewrite $|\Phi^+\rangle$ as¹³

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|\vec{n}, \vec{n}\rangle + |-\vec{n}, -\vec{n}\rangle}{\sqrt{2}}.$$

Another remarkable property of the state $|\Phi^+\rangle$, common to any bipartite *maximally entangled state*

$$|\Phi_{AB}\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^d |i\rangle_A |i\rangle_B, \quad (2.28)$$

is that applying a unitary operator U on one qudit, is equivalent to applying the transpose of such operator on the other

$$(U \otimes I)_{AB} |\Phi_{AB}\rangle = (I \otimes U^T)_{AB} |\Phi_{AB}\rangle. \quad (2.29)$$

These *non-local* properties, that hold even if we separate the particles, are widely used in quantum communication tasks, such as key distribution or teleportation. Yet, entanglement alone does not allow one party to tell the state of the other. Each subsystem of a maximally entangled pair looks completely random when considered in isolation, and its reduced state is *maximally mixed*

$$\rho^A = \text{tr}_B(|\Phi^+\rangle\langle\Phi^+|) = \rho^B = \text{tr}_A(|\Phi^+\rangle\langle\Phi^+|) = \frac{I}{2}. \quad (2.30)$$

To extract useful correlations between the measurement's outputs, one must know which measurement (or other operations) have been performed on each particle. Thus, quantum communication protocols must share classical information about the local transformations on each qubit¹⁴ to obtain a non-local description of the pair.

We can quantify the entanglement of composite systems knowing how mixed the reduced states are. The *von Neumann entropy* tells us this. For an arbitrary state ρ , we can define its entropy as¹⁵

$$S = -\text{tr}(\rho \ln \rho) = -\sum_i \lambda_i \ln \lambda_i, \quad (2.31)$$

where $\{\lambda_i\}$ are the eigenvalues of ρ . If the state ρ^A , part of a state ρ^{AB} , is pure, then its entropy is zero and ρ^{AB} is a separable state. The opposite

11: there are 4 of them: $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$, and $|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$.

12: in both cases the output is 1 or -1, with 1/2 chance each.

13: $\vec{n}$ and $-\vec{n}$ are two orthogonal states in the Bloch sphere (see equation 2.8).

14: this keeps your grandfather safe.

15: the right expression is often referred as *Shannon Entropy*, where $0 \ln 0 \equiv 0$.

16: AME states do not exist for all systems of N qudits. Some low-dimension AME are listed here:
<https://huberfe.github.io/ame>.

happens with maximally entangled states of dimension d , where the reduced density matrix is I/d and the entropy, $S = \ln d$, is maximal. This also apply to systems with more than 2 parties. We call a state *absolutely maximally entangled* (AME) if for any bipartition of the system, the reduced density matrix is maximally mixed. For example, if a 3-qudits state $\rho^{ABC} \in \mathcal{H}_d^{\otimes 3}$ is absolutely maximally entangled, then¹⁶

$$\rho^A = \text{tr}_{BC}(\rho^{ABC}) = \rho^B = \text{tr}_{AC}(\rho^{ABC}) = \rho^C = \text{tr}_{AB}(\rho^{ABC}) = \frac{I}{d}. \quad (2.32)$$

Many other entanglement metrics exist, yielding both continuous and discrete values. For example, the *Renyi entropies*,

$$S_\alpha = \frac{1}{1-\alpha} \ln \sum_i \lambda_i^\alpha, \quad (2.33)$$

generalize the von Neumann entropy, which reappears when $\alpha \rightarrow 1$.

Finally, a popular discrete metric is the *Schmidt Rank Vector* (SRV). That is, the rank of the reduced density matrix after tracing out different particles. In a 4-particle system $\rho \in \mathcal{H}_{ABCD}$, the SRV (r_A, r_B, r_C, r_D) is given by

$$r_X = \text{rank}(\text{tr}_X \rho). \quad (2.34)$$

The values r_X indicate the minimum local dimensionality of the state. For example, a possible state with SRV(4,4,2) is, up to normalization,

$$|000\rangle + |111\rangle + |220\rangle + |331\rangle. \quad (2.35)$$

These and other metrics will be further explored in chapter 3, where experiments are designed to maximize such quantities.

2.1.4 Fock space

In previous sections, multipart states were combinations of *distinguishable* entities. Even when two photons were exchangeable, like two horizontally polarized photons $|H\rangle_A |H\rangle_B$, there was an implicit assumption of distinguishability. One photon is in the *mode* A, and the other in the mode B. They must have some observable property from which they can be told apart, e.g., they are in different optical paths—one heading to Alice, another to Bob. This is not always the case, multiple photons (or any other *boson*) can be in the same state, being *indistinguishable* from one another. One can have, for example, two horizontally polarized photons in the same mode A, described as number state or *Fock state*,

$$|H\rangle_A |H\rangle_A = |2;0\rangle_{AB}, \quad (2.36)$$

where the mode A is occupied by two photons and the mode B remains at vacuum. Similarly, the Bell state can be rewritten as

$$|\Phi^+\rangle = \frac{|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B}{\sqrt{2}} = \frac{|1,0;1,0\rangle + |0,1;0,1\rangle}{\sqrt{2}}, \quad (2.37)$$

where each mode, A and B is split in 2 possible polarizations. With this notation we consider an arbitrary number of photons on each of the 4 modes, a infinite dimensional Hilbert space defined by the basis $|n_{A,H}, n_{A,V}, n_{B,H}, n_{B,V}\rangle$, the so-called *Fock space*. The new basis opens the

door to new entanglement forms such as the $N00N$ states, superpositions of N particles occupying d different modes:

$$|N00N\rangle_d^N = \frac{1}{\sqrt{d}} (|N, 0, \dots, 0, 0\rangle + e^{i\theta_1} |0, N, \dots, 0, 0\rangle + \dots + e^{i\theta_{d-2}} |0, 0, \dots, N, 0\rangle + e^{i\theta_{d-1}} |0, 0, \dots, 0, N\rangle)_{1, \dots, d}. \quad (2.38)$$

In the infinite Fock basis, operators are described in terms of *annihilation* and *creation* operators (or ladder), respectively defined as

$$a|n\rangle = \sqrt{n}|n-1\rangle \text{ \& } a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \text{ where } [a, a^\dagger] = I. \quad (2.39)$$

Similarly, states can also be described by ladder operators acting on vacuum $|0\rangle$, often referred as $|\text{vac}\rangle$. With this notation, the (by now familiar) Bell state becomes

$$|\Phi^+\rangle = \frac{|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B}{\sqrt{2}} = \frac{a_H^\dagger b_H^\dagger + a_V^\dagger b_V^\dagger}{\sqrt{2}} |\text{vac}\rangle, \quad (2.40)$$

where the operator $a_H^\dagger$ ($b_V^\dagger$) generates a horizontal (vertical) photon in the mode A (B). While this state was originally described in finite Hilbert space, the Fock basis is infinite, and as long as they are normalized, states can also have infinite amplitudes: $|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$. The *coherent states*, the eigenstates of the annihilation operator, $a|\alpha\rangle = \alpha|\alpha\rangle$, are remarkable examples of such states. They are described as¹⁷

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad (2.41)$$

and model the photon-number distribution of laser sources. That is, a Poisson distribution with an expected photon number $\langle n|n\rangle = |\alpha|^2$. The vacuum is a limit case of coherent states and different coherent states are related by the *displacement operator*, $|\alpha\rangle = D(\alpha)|0\rangle$, a unitary operator described as¹⁸

$$D(\alpha) = e^{\alpha a^\dagger - \alpha^* a}. \quad (2.42)$$

Similar operators are introduced in section 2.2, and coherent states will return in section 2.3.3.

17: one can derive the expression computing $\langle n|\alpha\rangle$, where $\sqrt{n!}|n\rangle = (a^\dagger)^n|0\rangle$ and $\langle\alpha|\alpha\rangle = 1$.

18: notice that $|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} |0\rangle$. Yet, for $D(\alpha)$ to be unitary we can (and must) add $|0\rangle = e^{-\alpha^* a} |0\rangle$, which using the *Baker-Campbell-Hausdorff formula* leads to $e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a} = e^{\alpha a^\dagger - \alpha^* a}$.

2.2 Photonic building blocks

Using the Fock basis we introduced states with infinite amplitudes as well as the operators that transform them. With the same tools, this section describes the most important operators for the path identity formalism, those that model the creation and measurement of photon states. We also show beamsplitters, as they are widely used in quantum optic setups and their description applies to many other linear optical devices.

2.2.1 Spontaneous parametric down-conversion

All the quantum experiments presented in this chapter use probabilistic photon-pair sources. The photons are created by non-linear crystals via *spontaneous parametric down-conversion* (SPDC), where a coherent *pump* photon from a laser source is transformed into pairs of correlated

photons, the *signal* and *idler* photons [27]. This process is described by the Hamiltonian

$$H_{\text{int}} = i\hbar \left(\zeta^* a_s a_i - \zeta a_s^\dagger a_i^\dagger \right), \quad (2.43)$$

being the coupling term ζ proportional to the second-order nonlinear susceptibility of the crystal. Moreover, the coherent pump field, $|\alpha\rangle_p$, is assumed undepleted. That is, the photon number is so high, $|\alpha|^2 \gg 1$, that it remains (approximately) unchanged. This leads to the operator

$$S_2(\zeta) = \exp\left(\zeta^* ab - \zeta a^\dagger b^\dagger\right), \quad (2.44)$$

where the signal and idler operators have been relabeled into the modes $a^{(\dagger)}$ and $b^{(\dagger)}$. This unitary is typically referred as 2-mode *squeeze operator*, and tuned with the laser's pump, proportional to the *squeezing parameter* $\zeta = r e^{i\theta}$. When such crystals act on the vacuum, they produce

$$S_2(\zeta) |0, 0\rangle = \frac{1}{\cosh r} \sum_{n=0}^{\infty} (-e^{i\theta} \tanh r)^n |n, n\rangle, \quad (2.45)$$

and for an arbitrary initial state, they lead to [28, 29]

$$\begin{aligned} S_2(\zeta) |p, q\rangle &= \sum_{k=0}^{\infty} \sum_{n=0}^{\min(p,q)} \frac{(-e^{i\theta} \tanh r)^k (e^{-i\theta} \tanh r)^n}{(\cosh r)^{p+q-2n+1}} \\ &\quad \times \sqrt{\binom{p}{n} \binom{q}{n} \binom{p-n+k}{k} \binom{q-n+k}{k}} \\ &\quad \times |p-n+k, q-n+k\rangle. \end{aligned} \quad (2.46)$$

19: the $1/2$ is introduced to simplify the equation 2.48.

Finally, if $a = b$, the degenerate case, the single-mode operator¹⁹

$$S_1(\zeta) = \exp\left(\frac{1}{2} \left(\zeta^* a^2 - \zeta (a^\dagger)^2 \right)\right), \quad (2.47)$$

acts on an arbitrary initial state as

$$\begin{aligned} S_1(r, \theta) |p\rangle &= \sum_{k=0}^{\infty} \sum_{n=0}^{\lfloor p/2 \rfloor} \frac{(-e^{i\theta} \tanh r)^k (e^{-i\theta} \tanh r)^n}{2^{k+n} (\cosh r)^{(p-2n+1/2)}} \\ &\quad \times \sqrt{\binom{2n}{n} \binom{2k}{k} \binom{p}{2n} \binom{p+2(k-n)}{2k}} \\ &\quad \times |p+2(k-n)\rangle. \end{aligned} \quad (2.48)$$

2.2.2 Different detection schemes

To measure a photonic state, detectors absorb the photons and thus destroy them. While this holds for all our experiments, there are significant differences across designs. Some of our experiments employ *photo-number resolving* (PNR) detectors, described by the projectors $P_n = |n\rangle\langle n|$, which reveal how many photons reach the detector. Yet, most of our proposals rely on *threshold* detectors that click when any photon reaches them, but do not show how many did it. Such measurement is described by

$$P_0 = |0\rangle\langle 0| \quad \text{and} \quad P_{\geq 1} = \sum_{n=1}^{\infty} |n\rangle\langle n| = I - P_0. \quad (2.49)$$

The photons that reach these devices disappear as soon as we measure them. Yet, they may have been entangled with other photons. This is exploited in *heralded* detection, where information from one or more detectors allows us to characterize the unmeasured photons. This is clear for the photon pairs generated by SPDC crystals (see equation 2.45). If we measure the number of photons in one mode, we can tell how many are in the other. More generally, given a bipartite system, ρ_{AB} , the remaining state ρ_A after measuring the subsystem B is given by

$$\rho_A = \frac{1}{p_k} \text{tr}_B ((I_A \otimes P_B^{(k)}) \rho_{AB}), \quad (2.50)$$

where $P_B^{(k)}$ represents a projective measurement on the system B , leading to an output k with probability p_k . This holds for arbitrary (projective) measurements on one or more photons, independently of how many particles belong to each subsystem. When the generation of a state is conditioned on the measurement of all its photons—and therefore its complete destruction—we talk about *postselected* states.

2.2.3 Beamsplitters and more

After they are created and before they are measured, photons are often transformed by beamsplitters and other optical elements. A lossless beamsplitter is characterized as a unitary operator

$$U_{BS} = \exp\left(\theta(e^{i\phi} a^\dagger b - e^{-i\phi} a b^\dagger)\right). \quad (2.51)$$

Using the Baker–Campbell–Hausdorff formula²⁰ one can relate the creation operators $a^\dagger$ and $b^\dagger$ as

$$U_{BS} a^\dagger U_{BS}^\dagger = \cos(\theta) a^\dagger - e^{i\phi} \sin(\theta) b^\dagger \equiv t a^\dagger - r b^\dagger, \quad (2.52)$$

$$U_{BS} b^\dagger U_{BS}^\dagger = e^{i\phi} \sin(\theta) a^\dagger + \cos(\theta) b^\dagger \equiv r a^\dagger + t b^\dagger. \quad (2.53)$$

where the coefficients $t = \cos \theta$ and $r = e^{i\phi} \sin \theta$ are, respectively, the *transmissivity* and *reflectivity* of the beamsplitter, such that $r^2 + t^2 = 1$. Acting on an arbitrary 2-mode Fock state, the unitary leads to

$$\begin{aligned} U_{BS} |P, Q\rangle &= U_{BS} \frac{(a^\dagger)^P (b^\dagger)^Q}{\sqrt{P!Q!}} |0, 0\rangle = \frac{(U_{BS} a^\dagger U_{BS}^\dagger)^P (U_{BS} b^\dagger U_{BS}^\dagger)^Q}{\sqrt{P!Q!}} U_{BS} |0, 0\rangle \\ &= \sum_{p=0}^P \sum_{q=0}^Q \sqrt{\binom{P}{p} \binom{Q}{q}} \binom{P-p+q}{q} \binom{Q-q+p}{p} \\ &\quad \times (-1)^p t^P r^{P+Q-p-q} |P-p+q, Q-q+p\rangle. \end{aligned} \quad (2.54)$$

The operator from equation 2.51 models several optical devices, such as waveplates or polarizing beamsplitters. In a waveplate, the modes a and b represent the two different polarization modes within the same optical path. The same applies for mode shifters. In the case of the polarizing beamsplitter, the action can be described as two beamsplitters, each acting on a different polarization mode while connecting 2 optical paths.

20: being $e^X Y e^{-X} = \sum_{n=0}^{\infty} [X, Y]_n / n!$, where $[X, Y]_{n+1} = [X, [X, Y]_n]$ and $[X, Y]_0 = Y$.

2.3 Experiments, graphs, and beyond

21: as we will see, the postselection renders annihilation operators irrelevant.

In this section, we introduce the path identity experiments and the different ways they can be computed and represented. SPDC sources are the workhorse of this section, and until the last part, we assume they are pumped very weakly. Thus, we approximate them as²¹ $S_2(\zeta) = \exp(\zeta^* ab - \zeta a^\dagger b^\dagger) \approx (1 - \zeta a^\dagger b^\dagger)$, where $|\zeta| = r \ll 1$. This perturbative approximation is often referred to as the *low-gain regime*. Additionally, the postselection conditions on the emission events will bound the total number of photons in the experiments. In the last part, we will challenge these assumptions and evaluate their implications.

These are the foundations upon which Pytheus and further work have been built, which will be presented in the next 2 chapters. While some concepts may reappear, this section offers an overview of the graph-based representation and the non-perturbative, circuit-like extension. Such a side-to-side comparison is not present in the papers.

2.3.1 Path identity and interference

In equation 2.2, we defined a general qubit in an arbitrary superposition of polarization states, $|H\rangle$ and $|V\rangle$, which lead to probabilistic outcomes when measuring their polarization. In this description, the superposition was between the local, intrinsic properties of a photon. In the famous *double-slit experiment*, the superposition is non-local: the particle's position is in a superposition of 2 distant locations, the two slits. This superposition leads to an interference pattern in the photon's position after the slits. Yet, if a detector is placed in one of the slits and the particle's position is revealed, the superposition and the resulting interference disappear. For the superposition to happen, no information about the particle's position must be accessible. No measurement, intentional or not, must take place. This non-local superposition can also occur between the photons' origin, as shown in 1991 by the Zou, Wang, and Mandel (ZWM) experiment [30, 31]. Their experiment uses two nonlinear SPDC crystals, pumped coherently and so weakly that at most one photon pair is created in total, yielding

$$\frac{1}{\sqrt{2}}(a^\dagger b^\dagger e^{i\delta} + c^\dagger d^\dagger)|\text{vac}\rangle, \quad (2.55)$$

where the modes a and b (c and d) are the optical paths of the photons of the first (second) crystal. The term $e^{i\delta}$ comes from a phase shifter in the optical path a , between the sources. The photons are emitted with equal probability and, except for their location, they are indistinguishable. Now, by aligning the optical paths of a and d , they merge into a single mode d , which leads to the state

$$\frac{1}{\sqrt{2}}(d^\dagger b^\dagger e^{i\delta} + c^\dagger d^\dagger)|\text{vac}\rangle = \frac{1}{\sqrt{2}}d^\dagger(b^\dagger e^{i\delta} + c^\dagger)|\text{vac}\rangle. \quad (2.56)$$

Assuming that only 2 photons are created, we now have a superposition between the photon in the mode b and the photon in the mode c . That is, a superposition between emission events in 2 different crystals. Moreover, their relative phase is given by the phase shifter with which they have never interacted. Inversely, if the phase δ is unknown, the value can be obtained by making the modes b and c interfere. To do so, one can redirect the optical paths b and c towards a 50:50 beamsplitter²², which

22: in the notation from equation 2.54, $b^\dagger \rightarrow tb^\dagger - rc^\dagger$, and $c^\dagger \rightarrow rb^\dagger + tc^\dagger$, where $t = r = 1/\sqrt{2}$.

leads to the final superposition (adding a global phase $e^{-i\delta/2}$)

$$d^\dagger \left(\cos(\delta/2)b^\dagger - i \sin(\delta/2)c^\dagger \right) |\text{vac}\rangle. \quad (2.57)$$

From this state, the probability of photodetection at the two spacial modes, b and c , reveals the value of δ . The optical path where the phase was applied is not even measured, and its photons can have different properties than those we detected. Two decades after its conception, variations of the experiment²³ found applications in metrology.

The indistinguishability by path identity, where the origin of the photons is unknown, can also be used to generate entanglement. Shortly after the ZWM experiment, Hardy proposed a similar setup to generate Bell states [33]. Again, two SPDC crystals produce a single photon pair in total, with equal probability. This time, the optical paths of both sources merge entirely, but instead of a phase shifter in one path, there is a mode shifter in both. We can assume the shifter converts the originally polarized photons of the SPDC sources into their orthogonal state, e.g., from $|H\rangle$ to $|V\rangle$. With this arrangement, the state leaving the second crystal is $(|HH\rangle + |VV\rangle)/\sqrt{2}$, a superposition of polarized states that emerges from a superposition of the origin of the photon pair. This technique has been further expanded by adding extra modes and optical paths, generating multiparty, high-dimensional entanglement [34].

23: the photons loss between the sources, by reflection or absorption, can also be measured [32]. To model these, the mode a does not merge perfectly into d , but instead $a^\dagger \rightarrow \sin \chi a^\dagger + e^{i\delta} \cos \chi d^\dagger$.

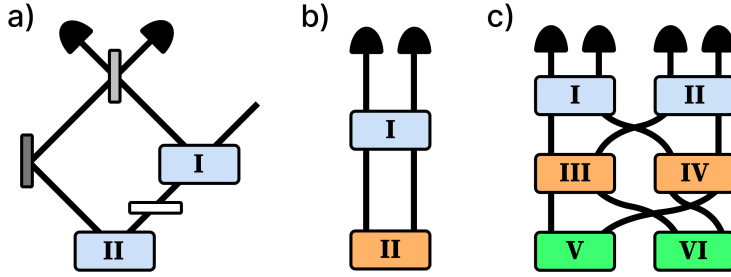


Figure 2.2: origins of path identity. a) the ZWM experiment [30, 31], leads to the interference described in equation 2.57, b) Hardy's experiment [33] generates a Bell state, and c) Krenn's experiment [34] produces a 3-dimensional GHZ state between 4 parties. These three experiments are based on quantum indistinguishability by path identity [32].

2.3.2 Superposition of perfect matchings

Fig. 2.2 shows two experiments to generate entangled states (b and c). One produces a 2-dimensional entanglement between 2 parties, a Bell state, and the other 3-dimensional entanglement between 4 parties, a GHZ state. To do so, the latter must produce one photon per optical path, i.e., two photon pairs. While 21 possible terms generate 4 photons, only 3 of them meet the postselection condition. The resulting state is [34, 35]

$$\begin{aligned} \mathcal{P}\Phi(\omega) |\text{vac}\rangle &= \mathcal{P}S_2(\omega_I)S_2(\omega_{II})S_2(\omega_{III})S_2(\omega_{IV})S_2(\omega_V)S_2(\omega_{VI}) |\text{vac}\rangle \\ &\approx \left(\omega_I a_0^\dagger b_0^\dagger \omega_{II} c_0^\dagger d_0^\dagger + \omega_{III} a_1^\dagger c_1^\dagger \omega_{IV} b_1^\dagger d_1^\dagger + \omega_V a_2^\dagger d_2^\dagger \omega_{VI} b_2^\dagger c_2^\dagger \right) |\text{vac}\rangle, \end{aligned} \quad (2.58)$$

where $\Phi(\omega)$ represents the set of SPDC sources acting on the vacuum, each with a squeezing factor $\omega_j \in \mathbb{C}$, where $|\omega_j| \ll 1$ and $j \in [I, \dots, VI]$ indicates the source. The postselection is encoded in $\mathcal{P}$, the measurement operator²⁴ of the threshold detectors, which project the state into the subspace where all optical paths contain a photon. That is, when all detectors click. With all sources equally pumped, the resulting state is

$$\frac{1}{\sqrt{3}} \left(a_0^\dagger b_0^\dagger c_0^\dagger d_0^\dagger + a_1^\dagger c_1^\dagger b_1^\dagger d_1^\dagger + a_2^\dagger d_2^\dagger b_2^\dagger c_2^\dagger \right) |\text{vac}\rangle = |\text{GHZ}\rangle_4^3. \quad (2.59)$$

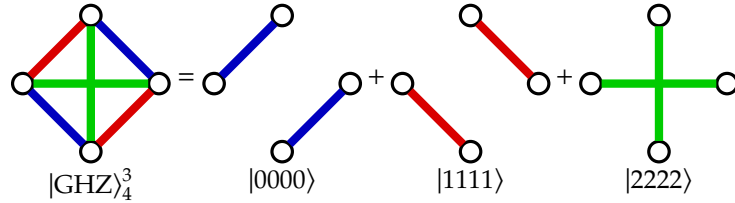
24: in the notation of equation 2.49, the projector reads as $\mathcal{P} = \otimes_\alpha P_{\geq 1}^\alpha$, where $\alpha \in \{a, b, c, d\}$ indicates the optical path.

Compared to the Bell state generation, in Fig. 2.2b, the sources in the GHZ experiment produce many more undesired terms. More differences arise if we add extra dimensions. The two-party experiment can be expanded to superpose an arbitrary number of modes. It only requires adding more photon-pair sources and mode shifters. Then, assuming a single photon pair is created, the resulting state is $\sum_k |kk\rangle$ (up to normalization). We can try to do the same in the GHZ case. We could connect two more SPDC sources, overlapping their optical paths with the previous crystals, and add mode shifters to gain a fourth dimension. However, the postselected state is then²⁵

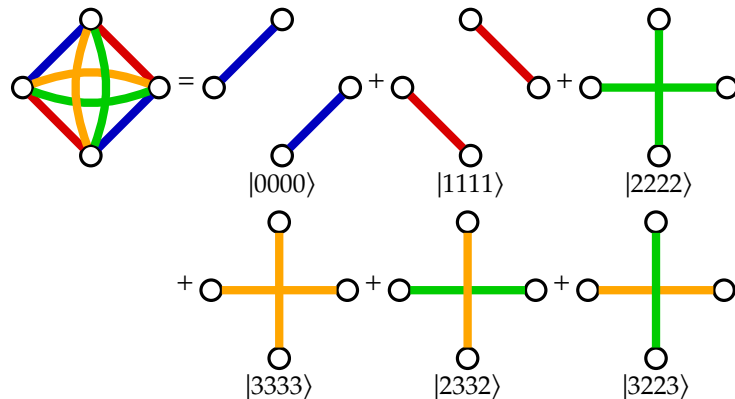
$$\frac{1}{\sqrt{6}} \left(\sqrt{3} |\text{GHZ}\rangle_4^3 + |3333\rangle + |3223\rangle + |2332\rangle \right) \neq |\text{GHZ}\rangle_4^4, \quad (2.60)$$

where extra terms are created that, despite fulfilling the postselection conditions, do not belong to the $|\text{GHZ}\rangle_4^4$. Moreover, since all terms are orthogonal to each other, they cannot be removed via destructive interference, at least not without additional elements.

To understand this phenomenon, it helps to visualize the path identity experiments as graphs. The photon sources, previously shown as boxes, can be represented as undirected edges²⁶ connecting the optical paths that reach the different photodetectors: the nodes. For clarity, the edge colors indicate the mode²⁷ of the created photons. Edges can have 2 colors, as crystals can emit photons in two different modes. In this representation, the superposed terms in equation 2.59 correspond to subgraphs where all nodes (detectors) have degree one. These are the *perfect matchings* of the graph. Fig. 2.2c is then shown as



Here, we assume all sources have the same pump and phase. That is, all edges have the same (real) weight. To obtain the state from equation 2.60 we add two extra sources to the graph, connecting them to the same nodes as the green edges. This is equivalent to adding two sources at the bottom of Fig. 2.2c, aligning their paths with the previous sources. In the graph representation, the new sources/edges lead to new perfect matchings



25: we assume the optical paths of the new sources merge with the previous, and that they all use the same pump.

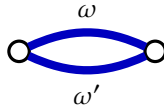
26: due to the postselection criteria—having one photon per detector—sources effectively commute. We challenge this approximation in the next section.

27: the mode shifter is merged with the source, acting as a single entity.

The last two terms, the undesired ones, combine photons with different modes, i.e., edges of different colors. Before, we had three disjoint perfect matchings, the maximum number for a 4 node graph. Yet, as we add more connections, some perfect matchings will share edges. This explains why the experiment from Fig. 2.2b can be expanded to arbitrary dimensions by simply adding more sources. Every edge forms its own perfect match in a two-node graph. In contrast, graphs with 6 or more nodes have only 2 disjoint perfect matchings. If there are more edges²⁸ than nodes, some of them will appear in several perfect matchings [36]. This is illustrated in Fig. 3.2, in the next chapter.

28: assuming that every edge belongs to at least one perfect matching.

While disjoint perfect matchings facilitate entanglement generation, they are not mandatory. High-dimensional entanglement between multiple parties can be reached with additional resources, including photon sources and ancillary detectors, i.e., optical paths [6, 19]. Usually, these designs exploit interference patterns between creation events, which leads to the cancellation of undesired terms. In the graph representation, interference is shown as two perfect matchings that produce the same state. The simplest case is a two-node graph with two edges of different weights and the same color. This leads to the state



$$= S_2(\omega)S_2(\omega')|\text{vac}\rangle \approx (\omega + \omega')|00\rangle, \quad (2.61)$$

where $\omega, \omega' \in \mathbb{C}$. Since both sources produce the same type of photons in the same optical path, the origin of the photon pair is indistinguishable. When $\omega = -\omega'$, this leads to the cancellation of the creation process, a phenomenon known as frustrated photon creation [37]. In general, interference may occur between an arbitrary number of emission events, or not occur at all, all within the same design. For example, the following graph encodes a path identity experiment that produces a 4-qubit state.

The weights $\omega_{j,k}^{m,n} \in \mathbb{C}$ indicate the squeezing parameter of a crystal emitting photons in mode m and n (colors) on the optical paths j and k (nodes), respectively. In this setup, $|0101\rangle$ emerges from 3 interfering²⁹ emission events, while two terms (perfect matchings) interfere for $|1111\rangle$, and for $|1010\rangle$. Assuming all weights have the same modulus, $\omega_{j,k}^{m,n} \equiv \omega e^{i\theta_{j,k}^{m,n}}$, one can, for example, add a relative phase π to $\omega_{2,3}^{1,1} = -\omega$. Then, if the other sources are in phase, $|1111\rangle$ is cancelled, but $|0111\rangle$, also produced by $\omega_{2,3}^{1,1}$, remains with a relative phase. In Chapter 3, these mechanisms are exploited to remove undesired emission events and reinforce the target ones, achieving high-dimensional entanglement among multiple particles. Moreover, new devices and postselection rules will be introduced, widening the applications of the graph representation.

29: equivalently, one can merge into a single weighted graph W every edge that contribute to a given state. For example, the 6 edges that contribute to $|0101\rangle$. The *hafnian* of the weighted adjacency matrix of W is the total amplitude of the state.

The experiments using the graph representation have an equivalent in path identity and can be implemented with existing optical hardware [17, 18, 38, 39]. While these designs could be envisioned in the circuit-like representation of path identity, with boxes and paths, the colored graphs are easier to visualize and interpret. This has proved very fruitful, especially when aiming for multiple related goals, such as GHZ or W states [8, 10]. In those cases, researchers can often identify patterns across solutions, extract underlying mechanisms, and extrapolate the findings [19, 21]. Current artificial intelligence can also exploit such patterns and find new designs [40, 41].

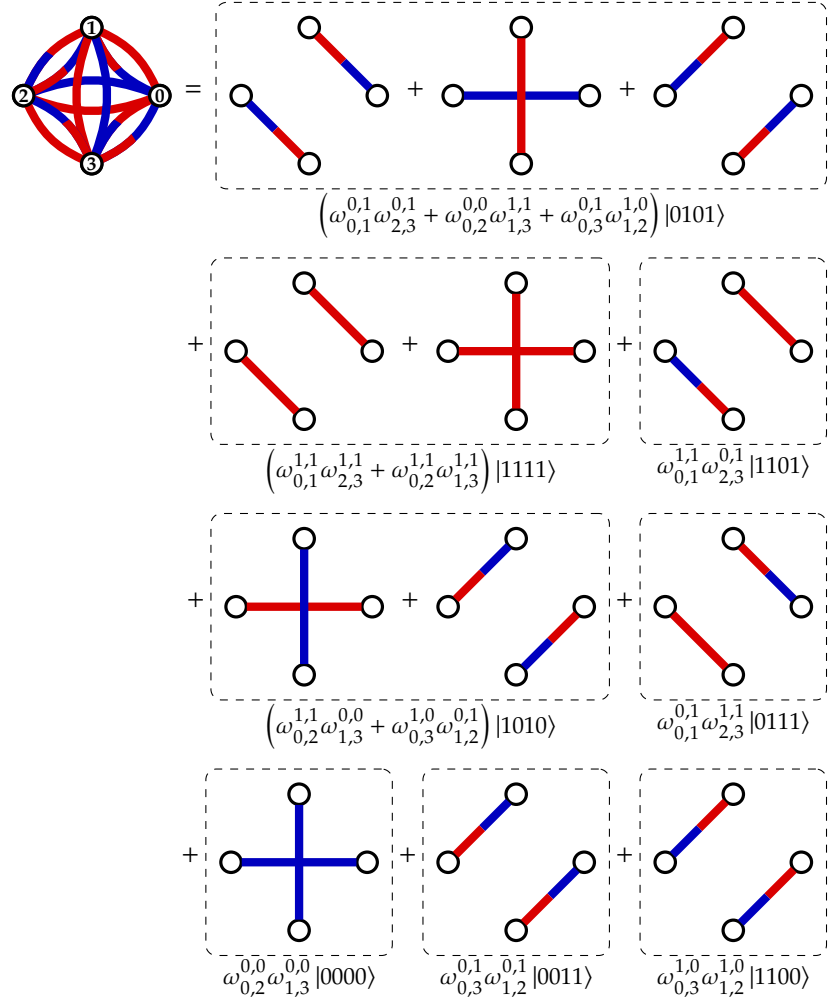


Figure 2.3: state computation in the graph representation. The product of the weights of the edges from the same perfect matching gives the amplitude of a state. Added to the other perfect matchings, they lead to the final superposition. The interference occurs between perfect matchings where the nodes are connected to edges with the same colors.

2.3.3 The non-perturbative regime

The graph representation was built upon a linear approximation of the squeeze operators, $S_2(\omega) \approx 1 - \omega a^\dagger b^\dagger$, where $|\omega| \ll 1$, the so-called, low-gain regime. Due to the low photon generation, emission events that occupied all optical paths, the postselected terms, were mostly those with as many photons as paths. Emission events that clicked all detectors with more photons than needed were just negligible. Hence, threshold detectors were sufficient to ensure that each path was occupied by at most one photon, effectively acting as PNR detectors. But what if we move beyond such a perturbative approximation and consider SPDC sources in their full³⁰ extent?

30: that is, as in section 2.2.1.

The most obvious consequence is the proliferation of creation events. Even if we only consider events in which all detectors click, perfect matchings are not enough. The total number of emitted photons may exceed the number of optical paths, and any source could emit multiple photon terms. In graph terms, we could consider those cases to be edge covers, i.e., subgraphs in which every node has degree one or more. Threshold detectors could not distinguish such events from perfect matchings, thereby degrading state fidelity. Moreover, as we consider higher-order terms, those with annihilation operators may also meet the postselection conditions. Then the order of the operators matters.

In the ZWM experiment [31], the precursor of path identity experi-

ments [32], the order of the optical elements was crucial. For the interference to occur, the phase shifter had to be located between 2 of the sources. Yet, for entanglement generation, the order of the sources was irrelevant, due to the low emission rates and postselection (see equation 2.59). Creator operators acting on different paths commute, and the same applies to repeated operators. However, due to the annihilation operators, SPDC sources do not commute, and the order of the operators strongly affects the generated states. This holds even if we use PNR detectors, which ensure that we get exactly one photon³¹ per path.

In a nutshell, when the laser pump increases, so does the number of relevant terms and the sources' sorting becomes paramount. This renders the weighted graphs insufficient³² to represent arbitrary experiments. Why would one abandon the comfort of the low-gain regime then? As shown in chapter 4, when weak laser pumps make photon emission rare, the probability of generating entanglement becomes extremely low. There is a tradeoff between the state's fidelity and how often we obtain it, and to explore it we need to compute high order creation events.

Computing endless amplitudes

Fig. 2.3 showed how to compute the state generated in a path identity experiment encoded as a graph. Perfect matchings contributing to the same term were grouped in boxes, and the resulting amplitude was the hafnian of the weighted adjacency matrix of a graph containing all contributing edges. Since the weights were complex, obtaining a single amplitude requires computing the hafnian of a symmetric complex matrix. That is a #P-complete problem [11], a very demanding³³ computation, and the graph representation is 'just' the perturbative approximation of path identity experiments. The challenge grows in Chapter 4, as we compute terms with an arbitrary number of photons. To do so, we use the operators described in section 2.2, defining them in finite Fock spaces with as many terms as we can compute. From the resulting states, truncated due to computational limits, we obtain the fidelity and success probability, i.e., how often we generate states that trigger the right detectors.

Computing so many terms is sure demanding, but even if the space must be truncated, this approach enables us to process arbitrary states. There exists, however, a more compact representation for the so-called *Gaussian states*, where N -mode states are encoded in $2N \times 2N$ symplectic³⁴ matrices. Coherent and squeezed states are examples of such states, and linear transformations on them, such as beamsplitters, also lead to Gaussian states [43–45]. Those are enough to model most of the transformations in Chapter 4, as they generally start with multimode vacuum³⁵. However, when we apply the projective measurements to several optical paths, the resulting states are no longer Gaussian. Then we need Fock basis amplitudes to compute the fidelity of the final state, as well as the probability of meeting the postselection criteria. Extracting the desired quantities from the symplectic matrices requires expensive computations such as hafnians and torontionians, diluting the benefits of the Gaussian representation.

31: for example, the terms $a^\dagger$, $aa^\dagger a^\dagger$, and $a^\dagger aa^\dagger$, they all produce one photon, but the amplitudes are different.

32: if sources act on different modes, as in Figure 2.2c, they do commute, so the weighted graph representation suffices.

33: the hafnian of $2n \times 2n$ complex matrices can be computed in $O(n^{3/2} 2^n)$ time [42]. That is a single amplitude from a maximum of d^{2n} , given a state of $2n$ qudits.

34: that is, a real $2N \times 2N$ real matrix M , such that $M\Omega M^T = \Omega$, being Ω a block diagonal matrix $\Omega = \bigotimes_{k=1}^N \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

35: in Chapter 3 we use a larger variety of initial states.

Digital discovery of 100 diverse quantum experiments with PyTheus

3.

This chapter adapts the homonymous paper [6].

After Mario Krenn showed how path identity experiments could be represented with colorful graphs [8–10, 34], we wanted a tool that would enable other scientists to use this representation and search for novel quantum experiments. Taking some examples from previous work [35], Sören Arlt and Carlos Ruiz Gonzalez, the main authors, created Pytheus, a Python library able to reproduce the previous findings and expand the applications [6].

To showcase the benefits and new functionalities of our software, our collaborators provided interesting and diverse quantum experiments. These are presented in the Appendix A. Jan Petermann searched for states that maximized different entanglement metrics, in Chapter A.2. Sharareh Sayyad suggested interesting states from the condensed matter literature, in Section A.5. Tareq Jaouni, under the supervision of Ebrahim Karimi, introduced different measurement schemes, in Section A.7. Nora Tischler proposed mixed states particularly relevant to the quantum formalism, in Section A.3. Xuemei Gu expanded the use of the graph representation to generate states in the Fock basis, in Section A.4. Mario Krenn showed how our framework could maximize other properties such as the assembly index, beyond standard quantum information tasks, in Section A.9. Many other experiments were found by Sören Arlt and Carlos Ruiz Gonzalez, reaching more than 100.

The writing was a coordinated effort from all the authors. All collaborators described their experiments, and the text was then edited as a single piece, incorporating feedback and avoiding duplication. The main authors led the software description.

Abstract

Photons are the physical system of choice for performing experimental tests of the foundations of quantum mechanics. Furthermore, photonic quantum technology is a main player in the second quantum revolution, promising the development of better sensors, secure communications, and quantum-enhanced computation. These endeavors require generating specific quantum states or efficiently performing quantum tasks. The design of the corresponding optical experiments was historically powered by human creativity but is recently being automated with advanced computer algorithms and artificial intelligence. While several computer-designed experiments have been experimentally realized, this approach has not yet been widely adopted by the broader photonic quantum optics community. The main roadblocks consist of most systems being closed-source, inefficient, or targeted to very specific use-cases that are difficult to generalize. Here, we overcome these problems with a highly-efficient, open-source digital discovery framework PyTheus, which can employ a wide range of experimental devices from modern quantum labs to solve various tasks. This includes the discovery of highly entangled quantum states, quantum measurement schemes, quantum communication protocols, multi-particle quantum gates, as well as the

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optimization of continuous and discrete properties of quantum experiments or quantum states. PyTheus produces interpretable designs for complex experimental problems which human researchers can often readily conceptualize. PyTheus is an example of a powerful framework that can lead to scientific discoveries—one of the core goals of artificial intelligence in science. We hope it will help accelerate the development of quantum optics and provide new ideas in quantum hardware and technology.

3.1 Introduction

Photons, the individual particles of light, have long been used as the core player for fundamental experiments and applications in quantum information science [46]. Photons do not easily interact with their environments; therefore, they can be distributed over large distances—which makes them a key resource for long-distance quantum communication [47, 48] and experiments that require strict Einstein locality conditions [49–51]. Using advanced measurement-based quantum computing schemes, photons are among the most promising candidates for future quantum computers [52]. Entanglement between two or more photons can be produced without a vacuum or cooling, and therefore many advanced experimental results can be achieved directly with table-top setups. Furthermore, the bosonic nature of photons allows for the generation of complex entangled quantum states of indistinguishable photons that are a key resource for quantum-enhanced measurements [53]. These potential applications have led to enormous technological advances in integrated chips for fast and precise control of photonic quantum states [54–57], high-quality single-photon sources [58–61], novel photon-pair sources [62], photon number resolving detectors [63, 64], and advanced high-quality multi-photon interference [65–68].

One question now is how to utilize these technologies to build up exciting new experiments for the foundations of quantum physics and practical quantum hardware.

Historically, the design of quantum experiments strongly relied on the intuition and creativity of human experts who leverage their experience and come up with blueprints of experiments. However, due to the unintuitive phenomena and enormous combinatorial space of the potential designs, it becomes extremely difficult for human researchers to discover more complex quantum setups. It might be possible that there are high-quality solutions to experimental design questions far outside of the region where humans' intuition fails. How could we possibly find such extraordinary solutions?

This question has sparked a strong interest in the automated discovery of quantum experiments with computers, overviewed in [1]. The invention of these tools for quantum optics experiments [5] have indeed overcome experimental limitations and allowed for new avenues in laboratories for entanglement research [69–72]. One crucial question is whether we can also learn something about physics from these tools. And indeed, several new concepts have been published that were purely discovered through automated design [5], such as a new general idea of entanglement structure [34], and generalized constructions of photonic quantum gates [73]. Those concepts were discovered by tedious analysis of the computer's solutions, which was time-consuming. The problem was that

the algorithms were powerful enough to find unknown solutions but had no incentive to present a simple, human-understandable form of it.

This was solved by the invention of Theseus [35], an efficient algorithm for the discovery of new quantum experiments that can readily be interpreted by humans. The key insight was a shift in the representation. Rather than describing quantum experiments as quantum optical components on an optical table, experiments are described as graphs of correlations between photons. This representation, which has been a derivative of a computer-discovered concept itself, was developed in [8–10]—and allows working with a much more natural representation, which can be translated back at any point to an experiment consisting of optical elements. (It should be noted that the representation is independent of photonic graph states for measurement-based quantum computing [74–76], and it is so far unknown how to translate among them.)

In this paper, we introduce PyTheus¹, a highly-efficient, open-source, automated design and discovery framework for quantum optics experiments. At the core, PyTheus uses a much extended graph-based representation of quantum optics, which allows us not only to represent entanglement and quantum gates, but lets us design quantum measurements, quantum communication protocols, optimize experimental properties, and discover quantum systems that involve single-photon sources, mixed states, and states entangled in the photon-number basis. Besides the advances of the scientific scope, we note that PyTheus is written in PYTHON, and therefore can readily be combined with machine learning frameworks such as TENSORFLOW and PYTORCH, and allows for immediate parallelization in computer clusters.

1: GitHub: <https://github.com/artificial-scientist-lab/PyTheus>

To showcase the applicability of PyTheus, we demonstrate the discovery of 100 previously unknown or advanced implementations of quantum optics experiments, ranging from exciting new systems for entanglement research to quantum states from condensed matter physics that are interesting for quantum simulation purposes, new ways of performing quantum communication tasks such as entanglement swapping, new quantum state measurements, and quantum gates. The experiments can involve both probabilistic photon sources and deterministic single-photon sources, and many of them are readily implementable in today's modern quantum optics labs. In the GitHub repository, we present the instructions for PyTheus that discover each of the examples. We hope that PyTheus's efficiency, generality, and low entry barrier kick-starts the application of computer-discovered quantum setups in experimental laboratories worldwide, and inspires new exciting computer-inspired ideas and directions for fundamentals and applications of photonic quantum physics research.

While the goal of this paper was to demonstrate the discovery capability of PyTheus, in several cases, it was impossible *not to see* clear generalizations and reasons why the solutions work. We show this in some cases below. One of the exceptionally interesting concepts we discovered was a new quantum multiphoton interference effect that can simulate probabilistic multi-pair sources just with photon pairs. We describe this new physics concept and its application in a parallel paper [19].

The article is structured in the following way: In section 3.2, we introduce the graph-based representation of quantum optics, which lies at the heart of PyTheus. In section 3.3, we introduce the idea of the computational PyTheus framework, which we then apply to the discovery of 100 new quantum experiments in section 3.4. In section 3.5, we explain some future(istic) ideas that might lie ahead of us.

3.1.1 Related work

The first automated and artificial-intelligence-driven design methods for new quantum experiments were introduced in 2016 (for a more detailed review on the topic see [1, 77]). One of them, *Melvin*, was focused on specific photonic quantum information tasks such as quantum state generation and quantum transformations, using discrete learning techniques [5]. The second one, *Tachikoma*, focused on the discovery of new experimental setups for quantum metrology tasks and used genetic algorithms for discrete optimization [78]. *Tachikoma* has been expanded to incorporate neural network surrogate models to speed up the search process for new quantum-enhanced measurements [79, 80]. At the same time, the ideas of *Melvin* have led to numerous implementations of experiments in various laboratories [69–72] and the extraction of new ideas and concepts in quantum physics [34, 73]. Automated design tools have helped to build new ways to perform quantum information tasks such as quantum cloning [81]. Compared to these tools, PyTheus does not work on the discrete search space. Discrete spaces are very challenging to navigate as gradients cannot be used. Rather, PyTheus uses domain knowledge in the form of a new physics-inspired representation that is entirely continuous.

These ideas have later been expanded by using reinforcement-learning [82, 83], for quantum communication [84, 85], recurrent neural networks [86], and deep generative models such as variational autoencoders [87] or logical AI [88]. Compared to these tools, PyTheus uses direct optimization on the outputs of a physical simulator and does not rely on learned simulators or strategies. That makes it significantly faster.

Various quantum physics groups and companies have also developed simulators and optimizers since. A main focus there is on the design of experimental settings for photonic quantum computing [89–92]. One remarkable simulator is *Strawberryfields* [90], which is focused on design and optimization tasks for continuous-variable photonic quantum computing and quantum machine learning tasks. Recent updates include auto-differentiation which significantly speeds up the rate of optimization. A related software package is [93], which allows for fast computation of tasks related to Gaussian boson sampling. Compared to these tools, the focus of PyTheus is different. PyTheus is built for discrete-variable quantum optics, and not targeted to photonic quantum computing (or boson sampling tasks). Furthermore, one main motivation is the interpretability of the discovered results, which is achieved via a topological optimization on the graph-based representation.

An alternative methodology that focuses not only on the design question but also on understanding the underlying physical concepts is *Theseus* [35]. There, the algorithm employs a graph-based representation to describe photonic experiments, and the final results are topologically simplified graphs that can be interpreted and conceptualized in a much more straightforward way than representations that work directly on the optimization of optical elements. Compared to *Theseus*, PyTheus expands this idea and applies it to many new situations, inaccessible before, such as the design of quantum measurement and communication setups via the Choi–Jamiołkowski isomorphism.

Related work has shown how quantum computers could overcome the enormous computational of designing quantum optical hardware [94]. PyTheus relies, for now, on classical computers, but recent hardware

advances might enable the execution of tools like PyTheus on quantum hardware [95].

3.2 Graphs and quantum experiments

The connection between quantum optical experiments and graph theory was discovered a few years ago [8–10] and has been further developed as a design algorithm Theseus for new quantum experiments [35]. In the graph-experiment representation, each colored weighted graph corresponds to a quantum experimental setup, and *vice versa*. Each edge and each vertex of the graph represent a correlated photon pair and a photon path, respectively. Its complex weight denotes the amplitude of the photon pair, and the edge color represents a photon's internal mode number for a given path, which corresponds to the photon's degree of freedom such as polarization [96], path [55, 57, 97], transverse spatial modes [98–101], time-bin [102] or frequency [62, 103]. This abstract graph representation allows us to have the full information of quantum optical experiments and has been used for discovering quantum states and transformations [35]. At first glance, it might seem that there is fundamentally no difference between the path degree of freedom (which is encoded as vertices) and the internal degrees of freedom of photons (which are encoded as colors of edges). However, this is only true information theoretically. Physically, the path degree of freedom is exceptional, because it allows to add spatially separation between photons and thereby perform non-locality experiments.

Here we significantly extend the bridge between graphs and experiments, which allows us to perform design and discovery tasks for quantum state generation (for pure and mixed states and on the photon number basis), quantum measurements, quantum communication protocols, and gates for quantum computing. In Table 3.1, we show the correspondence between graph theory and quantum experiments. The graph representation can be directly translated into different experimental implementations. In the remaining part of this section, we explain how these graphs encode quantum states, and how to translate them to experimental setups.

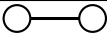
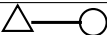
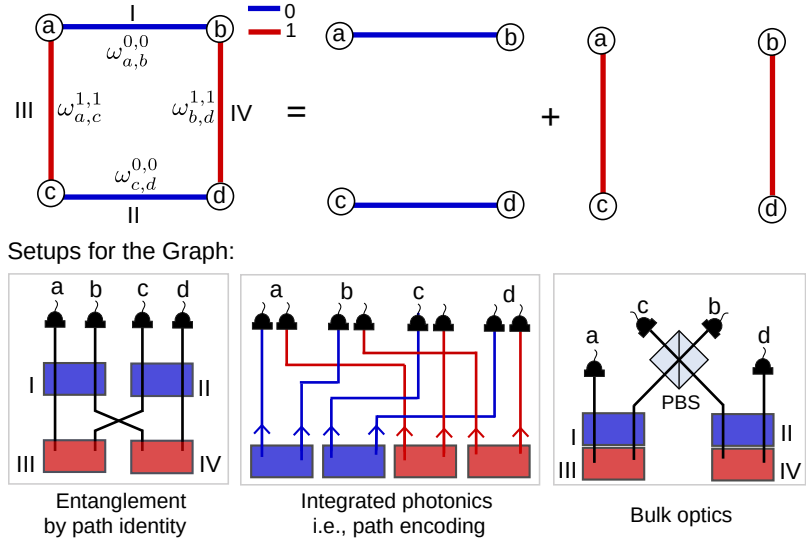
Graph theory		Experiment
color weighted graph		quantum experiment
vertex		path to photodetector
		heralded optical path*
		single photon source
		incoming photon
		ancillary photodetector
		number resolving detector
		environment interaction
edge	color	internal mode number
	weight $\in \mathbb{C}$	amplitude**
		negative amplitude
		correlated photon pair
		single photon path

Table 3.1: the correspondence between graph theory and quantum experiments. *For some experiments one must know the total amount of photons crossing a group of optical paths (see section A.4). This is clarified in the graph figures with a gray envelop. **Unless the contrary is specified, all weights are real values.

Figure 3.1: an abstract graph representation of quantum experiments producing GHZ states. There, edges and vertices represent correlated photon-pair and optical paths, respectively. The edge colors and weights ω are mode numbers and complex coefficients. The blue and red colors refer to the mode numbers 0 and 1, respectively. There are two perfect matchings (i.e., a subgraph that covers all vertices only once), and their coherent superposition leads to a quantum state $|\psi\rangle = \omega_{a,b}^{0,0} \omega_{c,d}^{0,0} |0000\rangle + \omega_{a,c}^{1,1} \omega_{b,d}^{1,1} |1111\rangle$. We can get a four-qubit GHZ state by setting all weights equal. The graph can be translated into several quantum setups with different technologies, such as entanglement by path identity or path encoding used in integrated photonics or standard bulk optics (e.g., using polarization encoding). The squares I-IV are probabilistic sources that create photon pairs. For clearer presentation, we draw the source in the color corresponding to the mode they contribute.



3.2.1 Quantum state generation

Probabilistic photon-pair sources

Probabilistic photon-pair sources, which are typically based on nonlinear processes such as spontaneous parametric down conversion (SPDC) and four-wave mixing (FWM) [27], are one of the most widespread resources to generate entangled and correlated pairs of photons. A range of photonic quantum experiments using probabilistic sources can be interpreted as a weighted colored graph [8–10, 35]. There, each vertex represents an optical path to a detector and each edge refers to the correlated photon pair produced by a probabilistic photon-pair source. The edge weight is the amplitude associated with the photons, and the edge color describes the photon's internal mode number (i.e., the degree of freedom of a photon). The connection between the graph and the corresponding quantum state is given by the weight function [35]

$$\Phi(\omega) = \sum_m \frac{1}{m!} \left(\sum_{e \in E(G)} \omega(e) x^\dagger(e) y^\dagger(e) + \text{h.c.} \right)^m, \quad (3.1)$$

where $E(G)$ is the set of edges of the graph. The quantum state is obtained by applying the weight function to the vacuum, i.e. $|\psi\rangle = \Phi(\omega) |\text{vac}\rangle$. The term h.c. stands for hermitian conjugate, which includes annihilation operators. As an example of states using four path (i.e., a , b , c , and d) with two-dimensional internal modes (i.e., 0 and 1) in Fig. 3.1, the $\Phi(\omega)$ is given as

$$\begin{aligned} \Phi(\omega) \approx \sum_N \frac{1}{N!} & (\omega_{a,b}^{0,0} a_0^\dagger b_0^\dagger + \omega_{b,d}^{1,1} b_1^\dagger d_1^\dagger \\ & + \omega_{c,d}^{0,0} c_0^\dagger d_0^\dagger + \omega_{a,c}^{1,1} c_1^\dagger a_1^\dagger + \text{h.c.})^N, \end{aligned} \quad (3.2)$$

where $\omega = (\omega_{a,b}^{0,0}, \omega_{b,d}^{1,1}, \omega_{c,d}^{0,0}, \omega_{a,c}^{1,1})$ is a list of edge weights $\omega_{x,y}^{i,j} \in \mathbb{C}$ and $|\omega_{x,y}^{i,j}|^2 < 1$, the superscript and subscript represent the mode number and the optical path, respectively. $x_k^\dagger$ is the creation operator of a photon

in path x with mode k . The pair-emission process is up to the N-order, and the probability of occurrence for lower-order events is higher than that of higher-order ones. In principle, the hermitian conjugate terms influence the final state. However, for the low pump regime, the effect of the annihilation terms is negligible in many cases. For example this is the case when the final state is conditioned on having a photon in every detector. In all of the examples that we present here, it is safe to neglect the annihilation operators.

An alternative method to compute these systems in a non-approximate way is to follow the Takagi decomposition (see [104], specifically Eq.27), which leads to reliable solutions also in the strong-pump regime. Naturally, this method is more expensive to compute. In our manuscript, we do not need to use this more involved method, as the low-pump approximation holds with high accuracy.

Experimentally, a common way to obtain a quantum state is to condition the experimental results on a simultaneous detection event in each detector, which is also called the n -fold coincidence detection. In our graph representation, this only happens when a subset of the edges contains each of the n vertices exactly once (see the subset of blue edges or red edges in Fig. 3.1), which is the so-called perfect matching of a graph. For the example in Fig. 3.1, we neglect the empty mode and higher-order terms $N > 2$ to post-select the quantum state. There are two perfect matchings (two blue edges and two red edges) in Fig. 3.1, which contribute to two quantum terms $|0000\rangle$ and $|1111\rangle$. The weight of a perfect matching is the product of all its edge weights. For each term in a quantum state, the weight is given by the sum of all weights of the perfect matchings that contribute to it. Therefore, the weights for quantum terms $|0000\rangle$ and $|1111\rangle$ are $\omega_{a,b}^{0,0} \omega_{c,d}^{0,0}$ and $\omega_{a,c}^{1,1} \omega_{b,d}^{1,1}$ respectively. In the end, a coherent superposition of the two perfect matchings in the graph leads to the post-selected quantum state, which is

$$|\psi\rangle \approx \omega_{a,b}^{0,0} \omega_{c,d}^{0,0} |0000\rangle + \omega_{a,c}^{1,1} \omega_{b,d}^{1,1} |1111\rangle. \quad (3.3)$$

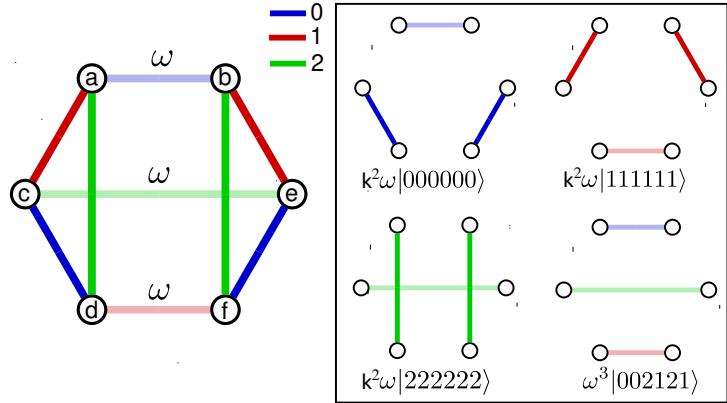
If we set all weights the same and normalize the state, we can then reach a four-particle Greenberger-Horne-Zeilinger (GHZ) state.

Translation to experiments. A colored weighted graph can directly be translated into several quantum optical experiments using different technologies [35], as shown in Fig. 3.1. The most straightforward way is applying the concept of entanglement by path identity [32, 34, 72], for which the graph representation has initially been designed [8]. In this scheme shown on the left side of Fig. 3.1, paths of indistinguishable photons produced from multiple crystals are overlapped, leading to a coherent superposition of possible origins for a photon. Equipped with the graph-experimental correspondence in Table 3.1, we now translate four edges and vertices into four photon-pair sources and optical paths. Here, we encode photon's internal mode number in polarization such that blue and red edge colors in the graph depict horizontal and vertical polarization, respectively. The four sources are set up in such a way that two sources (blue edges) produce photons with states $|00\rangle$ while the other two (red edges) produce photons with states $|11\rangle$. We then pump all sources coherently and consider that two of the four sources produce a photon pair in this example. Conditionally on perfect matching or post-selection, we have only two cases (two-photon pairs come either from the sources I and II or from the sources III and IV) that contribute to the final quantum state in equation 3.3. All other combinations of sources

do not lead to only one photon in each of the detectors. For example, a photon pair in crystals I and III leads to two photons in a , one photon in b , and c , but no photon in d ; therefore, this case is disregarded. If the mode numbers are indistinguishable, one can observe a new form of multi-photon interference, which has first been experimentally observed in [67, 68] and which forms the basis of many of the proposed experiments below.

Another promising technology is using integrated photonic, which usually uses path encoding [54–57], where each photon's one mode goes to one detector, as shown in the bottom middle-part of Fig. 3.1. In this example, each photon has two path modes depicted in blue and red colors, respectively. Therefore, one can consider that each vertex in Fig. 3.1 actually contains two sub-detectors that directly connect to the paths for the photon's two-mode numbers (i.e., blue and red paths). The vertex is activated when only one of the two sub-detectors clicks. In this way, perfect matching means the situations where four sub-detectors click in either blue paths or red paths, leading to the desired quantum state. Moreover, one can also use the standard bulk optics for generating photonic entanglement [105, 106]. When the edges are in the same color, which means the corresponding photons have the same mode number, one can use either probabilistic beam splitters or directly path-identified photon-pair sources to form the edge. If the edges have different colors (for example, two-mode numbers in Fig. 3.1), one can employ mode-dependent beam splitters such as polarizing beam splitters (PBSs) and transverse spatial mode sorters [46, 107] to create an edge. With polarization encoding in the example, we can insert a PBS between two cross-crystal sources [107], for the state generation. Yet, such a translation is not unique, and an expert may further simplify the proposed setup.

Figure 3.2: asymptotic solution for quantum experiments producing a three-dimensional six-particle GHZ state. All edges have weights k but those labelled as ω . The four perfect matchings of the graph lead to the quantum state $|\psi\rangle = k^2\omega(|000000\rangle + |111111\rangle + |222222\rangle) + \omega^3|002121\rangle$. To minimize the amplitude of the undesired term the weights must fulfill $k^2\omega > \omega^3$. Moreover, $k^2\omega > k^4$ to guarantee that $(N>3)$ -order terms from equation 3.2 are also negligible.



Asymptotic solutions. Until now, we know that each perfect matching can contribute to a term of the quantum state, and their coherent superposition gives the desired state. With the above graph-experimental link, one now expects to produce arbitrary multi-particle entanglement in high dimensions, e.g., a six-particle three-dimension GHZ state. A GHZ state appears when all perfect matchings are disjoint, i.e., every edge is used only in one perfect matching [8, 9]. However, it has been shown that it is impossible to construct a graph of more than four vertices with only three disjoint perfect matchings [8, 9]. Fig. 3.2 exemplifies this point. Three disjoint perfect matchings contribute to the desired quantum terms. However, there is another perfect matching that leads to an additional term that is not part of the GHZ state (the so-called Maverick term [8]). While we cannot fully erase the term, and therefore achieve a GHZ state

with perfect fidelity, we can reduce the weight of the undesired term. Yet, one must be cautious with $(N>3)$ -order terms from equation 3.2, since they also need to be smaller than the GHZ state terms we are targeting. These two constraints boil down to the inequalities $k^2 > \omega^2$ and $\omega > k^2$, respectively, which are both experimentally feasible. We call this and similar graphs *asymptotic solutions* (Fig. 3.2).

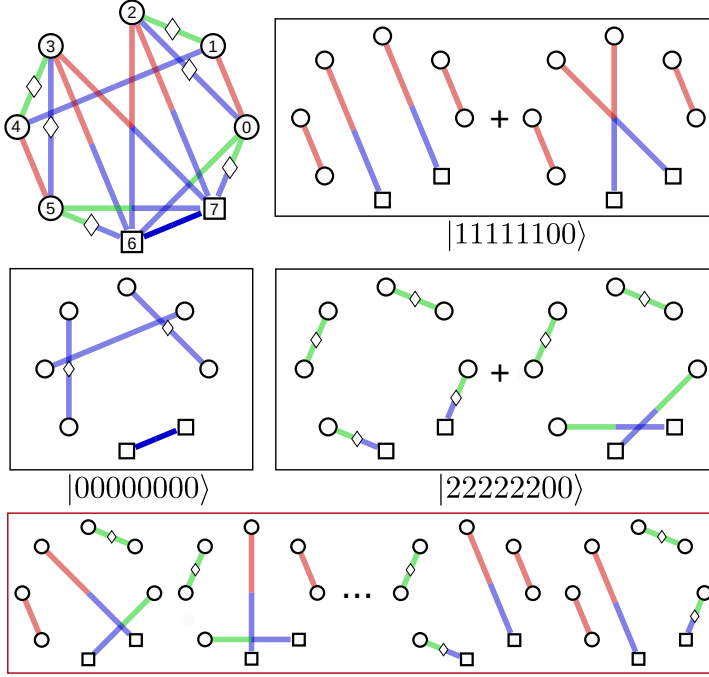


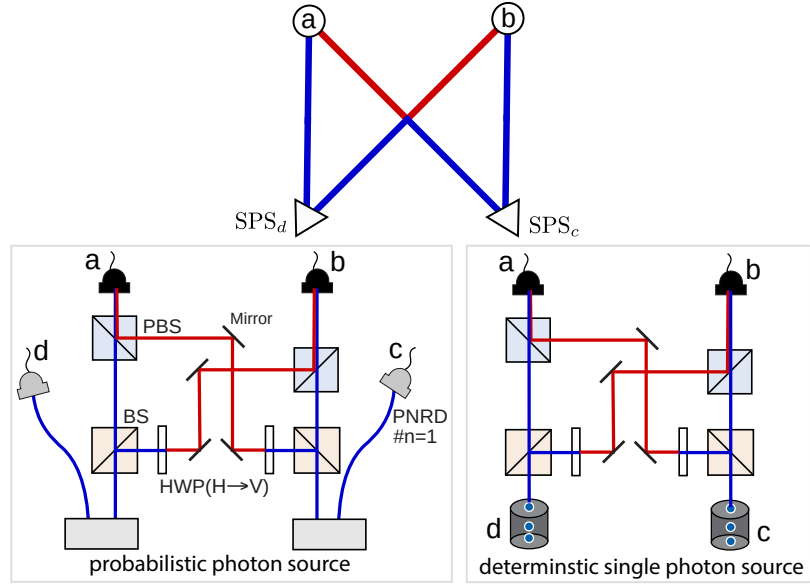
Figure 3.3: an abstract graph for producing a three-dimensional six-particle GHZ state with two ancillary particles. The diamonds along the edges represent that the edge weights are negative. Vertices 6 and 7 depicted by squares are for ancillas, which are in the same mode number (e.g., blue color for mode number 0). Five perfect matchings in black boxes contribute to the three terms in the desired GHZ state, while the rest of the perfect matchings are destructively canceled out.

Ancillary particles. Although there is a limitation to the state generation with standard linear optics and post-selection in the graph representation [9, 108], one can further employ ancillary particles to completely get rid of Maverick terms [19, 109]. For example, assisted with two ancillary photons, we can obtain the six-particle three-dimension GHZ state (details see Fig. 3.3). Additionally, to create quantum states with an odd number of particles, one can just use some odd numbers of photons produced from photon-pair sources as ancillary particles [10]. We present further examples in section 3.4.

Deterministic single-photon sources

In the above section 3.2.1, we have treated an edge in a graph as a probabilistic photon-pair source (in general, as the correlated photon pair) and a vertex for an optical path connecting to a single-photon (non)sensitive detector. Experimentally, one can construct a single-photon source that relies on a probabilistic photon-pair source, where one photon is detected heralds the presence of the other [63]. For example, the single-photon source in path SPS_d (SPS_c) is conditionally on the detection of only one photon in the path d (c) using a probabilistic photon-pair source and a photon-number-resolving detector, as shown in Fig. 3.4. In the graph representation, one can consider that there is an edge between the vertex SPS_d (SPS_c) and a virtual vertex which always needs to have exactly one incoming edge. For simplicity, we can just use an input vertex (e.g., SPS_d or SPS_c , without drawing its adjacent virtual vertex) for such a single-photon source based on a probabilistic photon-pair source, see Fig. 3.4. There are no connecting edges between two input vertices. The

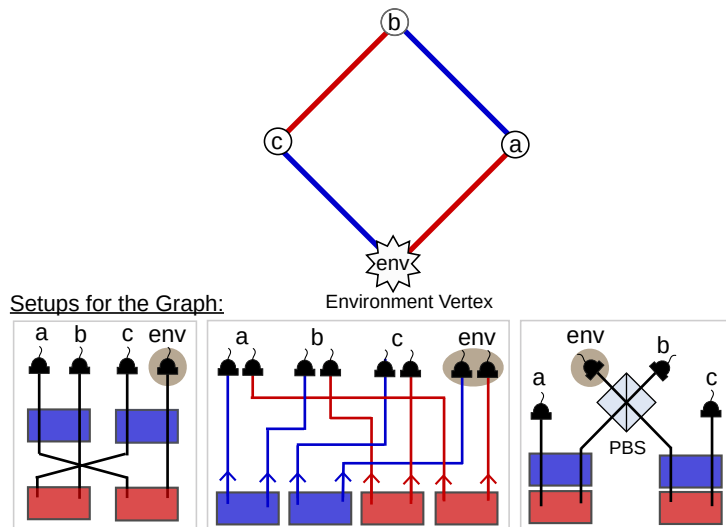
Figure 3.4: an abstract graph for quantum experiments producing a Bell state with single-photon sources. The vertices SPS_d and SPS_c described as triangles stand for the incoming path for single-photon sources while vertices a and b stand for the optical paths. No edges are allowed between input vertices SPS_d and SPS_c . The perfect matchings in this graph lead to a two-dimensional Bell state. The corresponding experiments are listed below. Experimentally, a single-photon source can be either a heralded single-photon source implemented with a photon-pair source and a photon-number-resolving detector (PNRD) or a deterministic single-photon source which can be an ideal quantum-dot-based source. In this experiment, the two photons arrive at the same time (and have the same wavelength, polarization, and spatial mode) such that they are indistinguishable.



edge connecting to an input vertex is reinterpreted as photons propagate towards a detector [35, 110].

Interestingly, this can also be explained in Klyshko's advanced-wave picture, described as "an intuitive treatment of two-photon correlation with the help of the concept of an effective field acting upon one of the two detectors and formed by parametric conversion of the advanced wave emitted by the second detector" [111, 112]. In other words, we can treat an edge as a quantum information transfer of one single system; thus, one can describe many abstract quantum information flows instead of just probabilistic photon-pair sources. In general, our input vertex representation is directly applicable to deterministic single-photon sources such as the ideal on-demand sources based on semiconductor quantum dots [58–61] or multiplexed single-photon sources [113]. Therefore, PyTheus can design experiments that use single-photon sources.

Figure 3.5: an abstract graph for producing mixed states. The detection paths were labeled a , b , c , and env , and internal modes described in blue and red are mode numbers 0 and 1. A mixed state can be generated by using one of the vertices as the environment (vertex env in this example). Experimentally this could be implemented by measurements that do not discriminate between the photon modes, for instance, detectors with a large active area. The corresponding setups for the graph are listed below.



Mixed states

So far, we have considered coherent creation processes, resulting in pure states. However, experimentally it is possible to introduce incoherence. Whenever distinguishing information about the quantum state escapes from the experiment to the environment, a partial trace of the density matrix is performed, producing a mixed state. This fact can be used systematically in the graph representation. To do so, we dedicate one of the vertices as the environment (environment vertex). We produce the full quantum state

$$|\psi\rangle = |\psi_0\rangle |0\rangle_{\text{env}} + |\psi_1\rangle |1\rangle_{\text{env}} + |\psi_2\rangle |2\rangle_{\text{env}} + \dots \quad (3.4)$$

Then, tracing out the environment contribution, we obtain

$$\rho = |\psi_0\rangle \langle \psi_0| + |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + \dots \quad (3.5)$$

We illustrate this procedure by a simple example in Fig. 3.5. From a graph that produces

$$|\psi\rangle = |000\rangle_{abc} |0\rangle_{\text{env}} + |111\rangle_{abc} |1\rangle_{\text{env}} , \quad (3.6)$$

we trace out the last photon obtaining

$$\rho_{000111} = |000\rangle \langle 000| + |111\rangle \langle 111|. \quad (3.7)$$

This procedure is experimentally achieved by using detectors that do not distinguish between the modes in vertex env. For example, if spatial modes are used, one can use a wide-area photon detector or multi-mode fibers in front of the detector. If the time degree of freedom is used to encode the mode numbers, one can integrate over the entire time such that one cannot distinguish between time-bins. Doing so, we introduce mixedness and recover the state in equation 3.7. Similarly to the translation in 3.2.1, we can also translate the graph for creating mixed states into different schemes, as shown in Fig. 3.5. Therefore, PyTheus can be used to design experiments for complex and interesting mixed states.

States entangled in the photon-Number basis

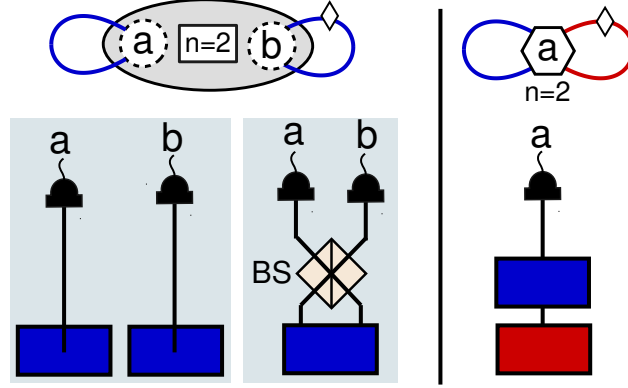
Fock states containing a fixed number of particles in a given spatial mode form a complete basis for many-body Hilbert spaces. The superposition of Fock states brings, among others, the well-known N00N states [115, 116] that promise many advantages such as the best-possible quantum-enhanced precision, super-sensitivity and super-resolution [53, 117]. The N -particle N00N state is (up to normalization)

$$|N00N\rangle_2^N := |N, 0\rangle_{a,b} + |0, N\rangle_{a,b} , \quad (3.8)$$

where the subscript number 2 means there are two paths (e.g., a and b), $|0\rangle$ indicates an unoccupied mode. The concept of the N00N state in equation 3.8 can be extended to a multi-mode case, where N particles are distributed in one of several optical paths [118, 119].

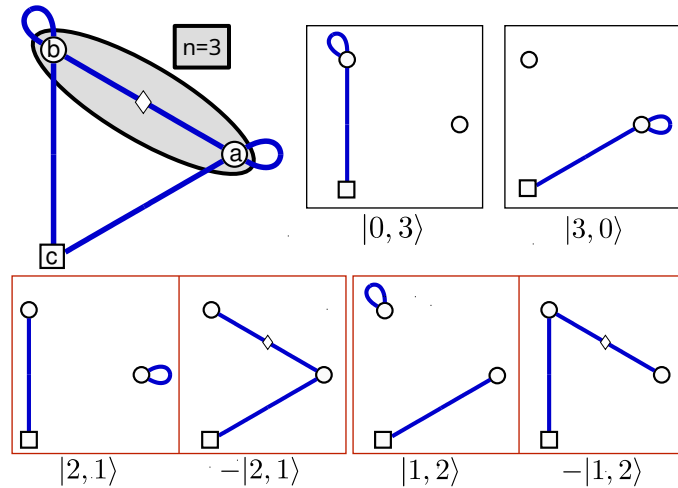
We know the total number of photons N going through a set of optical paths, but in contrast to the previous sections, we ignore how many photons occupy each of the paths. Therefore, perfect matchings are not

Figure 3.6: graphs for producing $|N00N\rangle_2^2$ states and the related experiments. On the left side, the graph with two blue edges (i.e., two self-loops) is for a path-entangled N00N state $(|2, 0\rangle_{a,b} - |0, 2\rangle_{a,b})/\sqrt{2}$. We translate this graph into several setups (below the graph). The setups using path identity or path encoding are the same in this case. With bulk optics, one can achieve the state with a Hong-Ou-Mandel interferometry [65, 114], where two identical photons enter a 50:50 beam splitter. One can also perform such a state in polarization, i.e., $(|2, 0\rangle_{H,V} - |0, 2\rangle_{V,H})/\sqrt{2}$ on the right side. The blue and red colors are, respectively, for horizontal and vertical polarization. The coherent superposition of blue and red self-loops gives the N00N state. Its related setup is described below the graph. There either two horizontally polarized photons (blue edge) or two vertically polarized photons (red edge) are in path a with a single-photon sensitive detector.



the only contributions to the final state that we must consider. Here, we need to compute all combinations of edges that lead to a total number of N photons for a given set of all optical paths. This includes combinations with repeated edges, which represent multiple photon pairs from the same source, and self-loops, in which a source produces a collinear photon pair [120, 121]. Moreover, for some contributions there can be vertices with degree zero, as long as the total number of photons—the total degree—is N . The conditioning on the total number of photons must be considered when employing the N00N states in metrology experiments, as described in previous work [53, 122].

Figure 3.7: graph for producing $|N00N\rangle_2^3$ state with an additional ancillary particle. The total photon number (including ancillary one) for each component in the state is four. There are six combinations of two edges that cover the ancillary vertex only once. Two of them (black box) contribute to the required terms $|3, 0\rangle$ and $|0, 3\rangle$, and the others $|2, 1\rangle$ and $|1, 2\rangle$ inside the red box cancel out.



As an example, we show a graph for the $|N00N\rangle_2^2$ state in Fig. 3.6. In this case, the total photon number is $N = 2$. All edges are in the same color, thus corresponding to the standard path-entangled N00N states. A self-loop edge indicates that there are two photons in its connected vertex. Thus the coherent superposition of two photons being in one of the two vertices (one of the self-loops produced a photon-pair, but we ignore it) leads to a coherent superposition of $|2, 0\rangle_{a,b}$ and $|0, 2\rangle_{a,b}$, i.e., a two-mode two-photon N00N state. One can now translate the

graph into quantum experiments, similarly, as we did in section 3.2.1. In this example, the setups using entanglement by path identity and path encoding are the same. With bulk optics, one can use probabilistic beam splitters that provide the mixing between the two input photons, to bunch two photons in one of the two optical paths a and b , which is the well-known Hong-Ou-Mandel (HOM) effect [65, 114]. Moreover, we can also generate such a N00N state in polarization or transverse spatial modes instead of the path; see Fig. 3.6 for details.

As for the previous states, ancillary paths can assist in the creation of states on the photon-number basis. Here we condition the final state on the existence of a total number of photons in the non-ancilla detectors but, again, the ancilla paths are reached by single photons. Fig. 3.7 shows an example in which the ancilla (vertex 3) receives only one photon, and the other two detectors get a total of three photons.

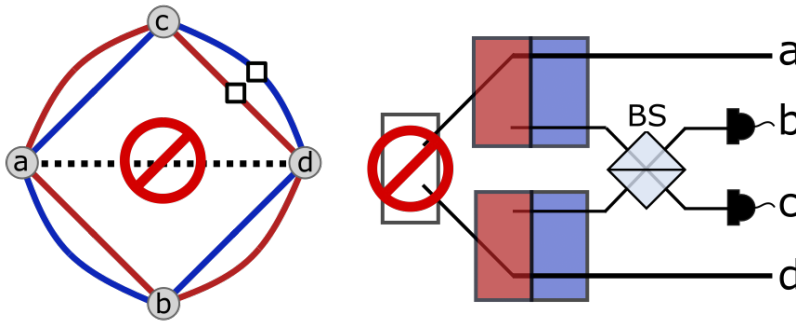


Figure 3.8: an abstract graph for entanglement swapping. On the right, an experimental setup for entanglement swapping is shown. The two particles a and d are entangled when two ancillary detectors click. A common source of particles a and d (as shown crossed out below the setup) would circumvent the point of entanglement swapping. The two particles are to be entangled without having interacted with each other. This restriction manifests in the graph corresponding to the experiment (shown on the left). An edge between the vertices a and d (dashed line) is not permitted.

3.2.2 Quantum communication

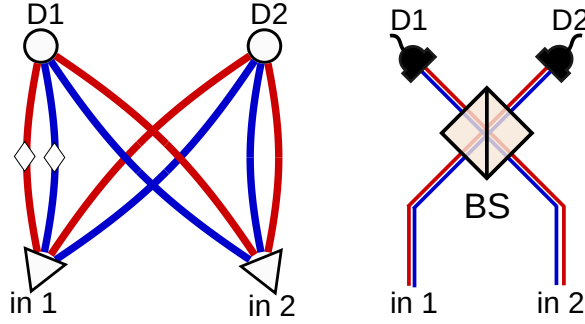
In future quantum networks that connect individual users with quantum resources and quantum computers, the distribution of quantum entanglement is essential. One key concept used in quantum communication in quantum networks is entanglement swapping [123, 124], where two qubits that never interacted can be entangled. This provides an important route for interesting tests of quantum foundations and plays an indispensable role in quantum technology such as quantum repeaters [125]. PyTheus allows us to explore new directions that could lead to solutions that require fewer resources than current techniques or are implemented in surprising ways. Many quantum network communication tasks have been experimentally implemented [124, 126–128], thus the new solutions by PyTheus can readily be implemented in laboratories.

We are interested in finding experimental setups for creating entanglement between particles that have not interacted or originated from a common source. With PyTheus this can be done for scenarios involving higher dimensions and multiple particles. Finding the corresponding graph works analogously to the state generation task shown in section 3.2.1, but comes with extra constraints on the graph. We set an entangled state (e.g. two particles in a Bell state) as a target for the optimization. The experiment corresponding to the resulting graph should create this state. To ensure that the two particles have no direct interaction, the graph must fulfill the following additional constraint. It should have no edge connecting the two corresponding vertices. Such an edge translated to a photon pair source would mean that the two photons could come out of the same source. This is shown in Fig. 3.8. Similarly, the connection of

two vertices to the same single-photon source vertex implies that they can not be space-like separated. Such constellations would not be valid for entanglement swapping, either. In the same way, as for incoming photons (single photon sources or input photons), the constraints are enforced by removing the edges from the starting graph.

Approaching these tasks with PyTheus makes it possible to explore new ways of distributing entanglement. We show the examples found by PyTheus in section 3.4.

Figure 3.9: an abstract graph for Bell state measurement. A measurement event happens if the photons **in1** and **in2** are in the Bell state $|\Psi^-\rangle$. Simultaneous clicks in the detectors **D1** and **D2** signify a successful Bell measurement. The corresponding setup is given for the graph, which is a typical Bell state measurement setup.



3.2.3 Quantum measurements

A graph may also be interpreted as a quantum measurement on an input quantum state. Such measurements are of utmost importance in many quantum communication tasks, for example, a Bell state measurement for quantum teleportation. The goal of measurements here is to distinguish different orthogonal incoming states, e.g., in the case when all detectors click simultaneously. Photons entering the experiment through input paths are represented analogously to single-photon sources introduced in section 3.2.1. An input photon corresponds to one vertex in the graph. An edge connecting an input vertex to a detector vertex corresponds to the photon traveling to that detector, following Klyshko's advanced-wave picture [111] as explained in section 3.2.1. The same constraints apply, excluding edges connecting two vertices belonging to input photons.

As an example, we show a graph in Fig. 3.9 for the well-known Bell state measurement (BSM) [129, 130], which plays an important role in many quantum information tasks. When the two detectors click, the incoming photons are projected to one Bell state $|\Psi^-\rangle$, which can be seen as the generation of a state given by a superposition of the two perfect matchings in the graph. To search for an experimental setup for measuring a particular state, we let PyTheus search for a graph that would produce the state under consideration of the topological constraints on the graph. The graph in Fig. 3.9 interpreted as state generation would produce the state $|\Psi^-\rangle$ under the condition that there is no connection between vertices **in1** and **in2**.

With a graph that 'produces' the state $|\varphi\rangle_{in}$, we can effectively perform a projective measurement ($P_0 = |\varphi\rangle\langle\varphi|$ and $P_1 = \mathbb{I} - |\varphi\rangle\langle\varphi|$) on the input photons. A coincidence of all detectors corresponds to the output 0 and any other pattern is interpreted as output 1.

The representation of measurements in terms of graphs constitutes an extension of the previous interpretation given in [35]. This allows us to use PyTheus for the discovery of measurement setups for any state and various constraints and conditions.

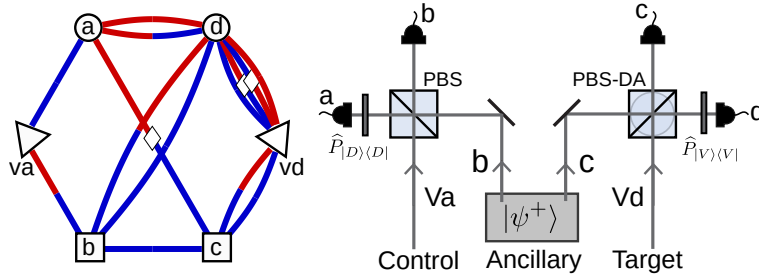


Figure 3.10: an abstract graph for a quantum CNOT gate shown on the left side. The right panels show the corresponding setup for a CNOT gate, which was realized by [131].

3.2.4 Quantum computation

Quantum gates are a crucial building block in quantum computation [24, 132]. A quantum gate performs a unitary transformation on an input state. These transformations can also be interpreted in terms of graphs. In these graphs, similar to measurements (described in subsection 3.2.3), the input photons are described by designated vertices. Likewise, we impose restrictions on which types of vertices can be connected by edges. Photons exiting the setup (output) are also represented by vertices, as in state generation. Additional ancillary photons can stem from probabilistic photon-pair sources and single-photon sources can also be involved. With PyTheus we can search for arbitrary quantum gates under a wide range of experimental conditions.

A target unitary quantum gate acting on an N -dimensional Hilbert space is specified by how each element of an orthonormal basis transforms. One canonical example in quantum gates is the controlled-NOT (CNOT) gate, which acts on two qubits and is described by the mapping:

$$\begin{aligned} |00\rangle &\rightarrow |00\rangle, \\ |01\rangle &\rightarrow |01\rangle, \\ |10\rangle &\rightarrow |11\rangle, \\ |11\rangle &\rightarrow |10\rangle. \end{aligned}$$

To find this gate, we let PyTheus search for a graph that would produce the state

$$|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|11\rangle + |11\rangle|10\rangle; \quad (3.9)$$

under consideration of the topological constraints on the graph (no connections between input photons). In Fig. 3.10, we show an experimental setup realized [131] together with its graph representation.

Photonic quantum gates fall into two main categories [133]. In a post-selected gate, the outgoing photons are detected directly after the gate. This ensures that there is exactly one photon in each path after the gate [134–141]. Because possibilities where two outgoing photons enter the same path are excluded, fewer experimental resources are required. However, this procedure destroys the outgoing state, and the particles can not be used further. The second category is heralded gates [131, 142–147], where only ancillary photons are detected, and the outgoing photons of the gate remain undetected. These gates are ultimately more useful since the output state can be used further. A graph for a heralded quantum gate is harder to discover and needs more experimental resources since the looser selection rules can lead to more unwanted terms. Both types of quantum gates (post-selected and heralded) can be represented with graphs and thereby be designed using PyTheus. Terms produced in post-selected experiments are represented by perfect matching, and

terms produced in heralded experiments are represented by collections of edges that cover the ancillary vertices.

The ability to perform design for quantum measurements and quantum gates relies on an idea that state generation and state propagation are closely related [35]. In the experimental quantum optics community, this idea is known as Klyshko's advanced-wave picture [111, 148], while in the quantum information science community it is often referred as Choi–Jamiołkowski isomorphism [149].

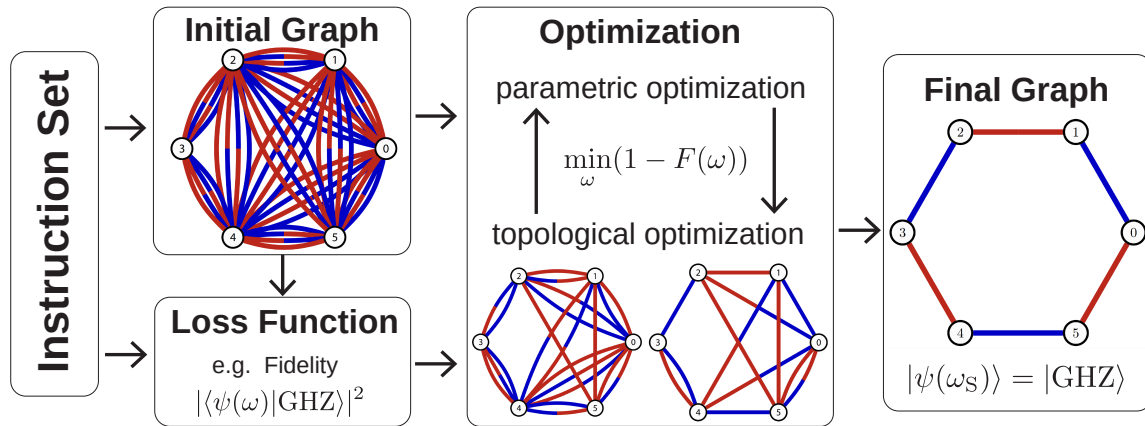


Figure 3.11: the PyTheus workflow. From the number of vertices and available dimensions, PyTheus builds the initial graph, which is fully connected except for vertex 3, which is only connected to its nearest neighbors. We impose this based on previous knowledge, but other constraints may arise depending on the available tools. Computing all the graph's perfect matchings, we find the more general state that the initial graph can produce: $|\psi(\omega)\rangle$. As the loss function, we choose the *Fidelity* of the state $|\psi\rangle$ with respect to the target: $|\text{GHZ}\rangle$. Aiming for an interpretable graph with perfect fidelity, we iterate the weight optimization with the edge removal until we find the simplest graph that produces $|\text{GHZ}\rangle = |\psi(\omega_S)\rangle$. The available resources, the loss function and further optimization details are specified in the *Instruction Set*.

3.3 The PyTheus library

Starting from a dense or fully connected graph, PyTheus uses gradient descent combined with topological optimization to find minimal graphs corresponding to some target quantum experiment. The PyTheus library greatly expands the range of applications of its predecessor, and it is significantly faster². The package is written in Python and is available on GitHub for further applications and development of the source code. The graph encoding is compatible with other tools such as *The Walrus*[93], a library to (among other applications) compute perfect matchings from the graphs' adjacency³ matrix.

PyTheus applications range from the creation of quantum states to the design of quantum communication protocols. While diverse, all these tasks are performed following the same steps, illustrated in Fig. 3.11. In this section, we explain the software workflow, which kind of loss functions we employ, and how to start using PyTheus.

Workflow overview. The *Instruction Set* file details *what* we want (e.g., a quantum state or a communication protocol) and some instructions about *how* to get it. The latter includes the entire allowed topology of the final solution, such as the type of photon sources/detectors and ancillary particles, how many of each we have, or which polarization modes can be used. It also contains the *Loss Function* to minimize (fidelity, count rate, or other metrics) as well as further optimization settings. In the example of Fig. 3.11, we want to obtain a post-selected six-particle,

2: For an eight-dimensional three-particle highly-entangled state (SRV(8, 5, 3)) PyTheus was almost five times faster than the current state-of-the-art method [35]. Examples of higher particle numbers were not readily available for the previous method, but we expect increasingly higher advantages for PyTheus.

3: The loops' weights in the adjacency matrix must be multiplied by 2.

two-dimensional GHZ state, $|\text{GHZ}\rangle_6^2 = |000000\rangle + |111111\rangle$. We employ 6 standard photodetectors and the photon pair sources described in section 3.2.1. In this example, we also specify further topological constraints of the final solution (thus of the initial graph)—here, vertex 3 must not have connections to vertex 0, 1 and 5.

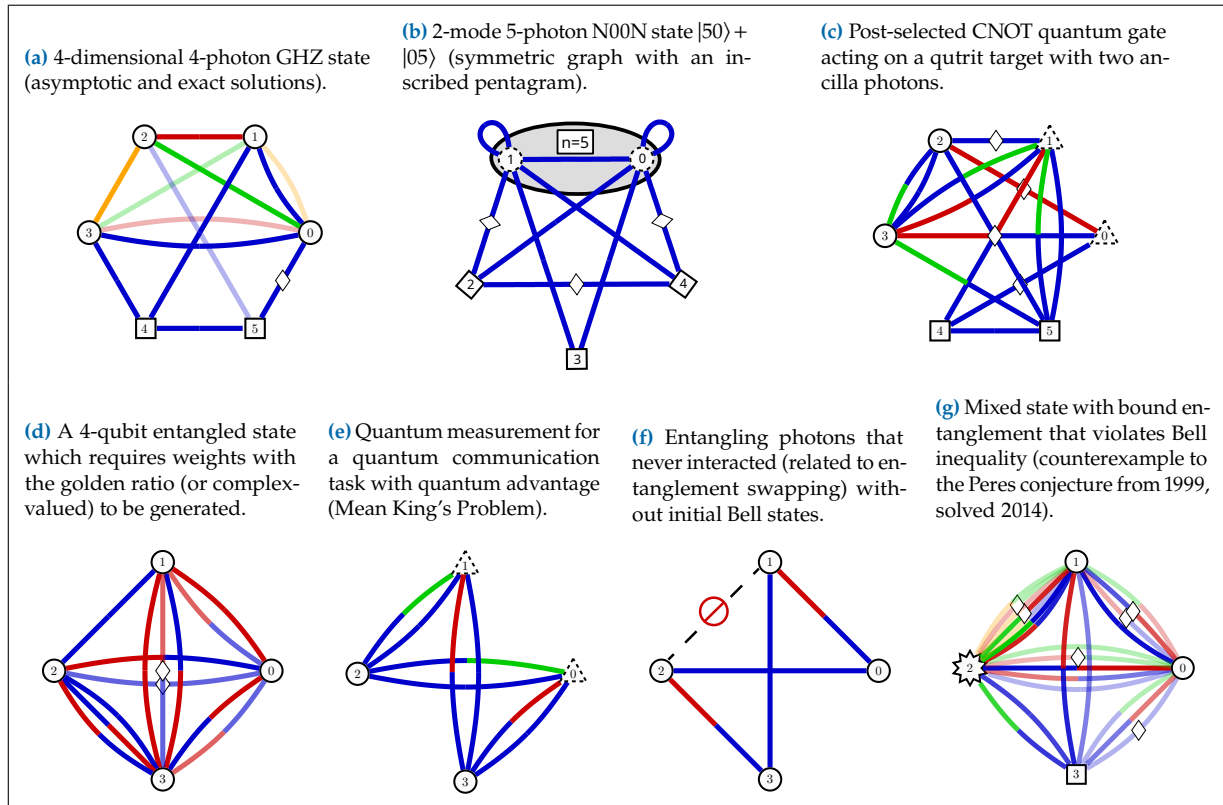
The experimentally available topology specified in the instruction set lead to the *Initial Graph*. This weighted graph represents all possible states/experiments that can be produced with the available resources. The number of vertices, the type of edges, or which vertices can be directly connected are some of the *Topological Constraints* that one can impose on the graph or that may follow from the available tools or from the task we want to implement—the sources described in section 3.2.1 and the protocols from section 3.2.2 are representing examples. On the initial graph of Fig. 3.11 the six vertices are connected with bicolored edges and based on previous knowledge, we only connect vertex 3 with its nearest neighbors.

From the initial graph, $\mathcal{G}(\omega)$, we can compute the more general state produced by the weight function $\Phi(\omega)$ acting on the vacuum (see equation 3.2). However, we do not consider the infinite terms produced by $\Phi(\omega)$, we only compute the creation events which fulfill certain *Conditional rules*. For example, in Fig. 3.11 the post-selection rule results in terms for which all detectors click. These creation events are represented by the graph's perfect matchings, and produce a coherent superposition of kets leading to the state $|\psi(\omega)\rangle$. We can specify more complex conditioning rules, which will be relevant, for example, for N00N states. There, we condition on states with a total final photon number in the output nodes. The conditioning rules are imposed by experimental circumstances 3.2.

Once we extract a general state $|\psi(\omega)\rangle$ from the initial graph, we can optimize it according to some *Loss Function*. As in Fig. 3.11 we seek a particular state, $|\text{GHZ}\rangle$, we maximize the *Fidelity* of our state accordingly: $F = |\langle \text{GHZ} | \psi(\omega) \rangle|^2$. To find a specific states we can also use the *Count Rate*, a metric that approximates how often our experiment produces such a state. Alternatively, we can maximize physical properties like entanglement rather than looking for a specific state. To perform quantum information tasks like measurements or communication protocols, we must translate each task into a state creation process—see sections 3.2.3, 3.2.2, and 3.2.4 for further details.

If the available experimental resources suffice, by optimizing the weights, we will find a graph that satisfies our needs. However, the optimal solution is not unique, and even if many of the weights vanish, we will likely obtain a very dense final graph, which will be hard to interpret. Therefore, to extract useful insights from our setups, we must simplify the graphs as much as possible, alternating the optimization of the weights' values with the removal of edges. This *Topological Optimization* will lead us to a simple graph, for which the removal of any additional edge would unacceptably raise the loss function's value.

For some graphs, like the one shown in Fig. 3.11, no smaller graph can produce the target state [8]. However, in most cases, it is challenging to ascertain whether there exists a simpler solution, especially when we choose to optimize for real weights instead of complex ones. With respect to these subtleties, following the workflow, we first generate graphs with abundant ancillary particles and reassess the results with the goal of reducing the number of resources. Using this procedure, we propose one hundred experiments to be realized in quantum optics.

Table 3.2: a collection of seven diverse highlights, each of which is interesting in its own right.

3.4 Hundred Experiments (in the appendix)

4: <https://github.com/artificial-scientist-lab/PyTheus>

The one hundred experiments conceived by PyTheus are Appendix A and in the group repository⁴. Table 3.2 shows seven discovery highlights.

- (a) A multi-photon entangled state that goes beyond the barrier of 3-dimensions [9]. It requires only two ancillas and has a small number of edges which makes it a very promising proposal for practical application, and observation of new properties at the foundation of quantum mechanics [150].
- (b) The generation of a two-mode N00N state with 5 photons. The associated graph, very symmetric, makes use of 3 ancillas.
- (c) A post-selected 3-photon control gate (or Toffoli) that does not require any ancilla photons.
- (d) A 4-photon qubit quantum state with equal coefficients $c = \frac{1}{\sqrt{7}}$, that requires the ratio between certain weights to be the golden ratio. Alternatively, a lower number of complex-weighted edges can also lead to such state.
- (e) A previously unknown quantum measurement scheme that allows the experimental implementation of a quantum communication protocol, the Mean King problem, proposed in 1987 [151].
- (f) A very surprising form of quantum entanglement swapping, which does not rely on the generation of two Bell pairs and a Bell state measurement.
- (g) Experimental setup of a 2-photon mixed entangled state that falsified the Peres conjecture [152]. The state is bound entangled (its entanglement cannot be distilled), however, it can be used to violate Bell's inequality [153, 154]. This could lead to an experimental falsification of the Peres conjecture.

3.5 Outlook

This article uses a list of examples to showcase the broad range of possibilities of digital discovery in quantum optics. The goal to design one hundred novel and intriguing experiments has led us to collect a diverse set of tasks for which experimental setups could be designed. To make the discoveries possible, we expanded the theoretical framework underlying the algorithm and produced faster, more versatile open-source software. We hope that this work can be an inspiration for experimental physicists in two ways: (1) to explore some of the experiments as they are proposed in this article (2) to see that PyTheus can be used to design experiments for a wide range of targets and for individually tailored specifications of experimental restrictions and resources. To explore particular examples that have not been covered in this article, the open-source library PyTheus can be used. Many more avenues leave room for significant further exploration, particularly within the optimization for targets other than fidelity, as well as more specific experimental constraints. For this, the PyTheus framework can readily be expanded in the future.

A number of extensions that would be interesting include the discovery of experiments that maximize success probability in terms of heralding efficiency or Bell state measurement efficiency, especially as experimentally available technologies became increasingly powerful [155]. The potential of PyTheus on suggesting experimental setups can pave the way to find experiments to simulate quantum states, which are nowadays merely realizable in extreme (thermodynamics) conditions, such as high pressure and low temperature [156]. An exciting extension would be the analysis whether we can extend state generation to encoding the dynamics of a quantum state into the weights of the graphs. Along with this line of research, it is theoretically tantalizing to explore the putative correlation between the complexity of time-evolved states [157] and the complexity of associated graphs as outputs of PyTheus.

As a final thought, we want to view our results through an additional lens. We have created a dataset of 100 hand-selected quantum experiments that are some aspects exceptional (which PyTheus has discovered). This dataset might be large enough for highly efficient artificial intelligence algorithms to bootstrap an intuition of what properties make an experiment interesting, and allow them to produce proposals for new, hopefully equally interesting quantum experiments. It would sure be exciting to investigate the physical properties of the proposals that the machine believes are *interesting*. In the best case, it could act as an inspiration for new ideas for the human scientist [14, 158].

Automated experimental design for high-probability entanglement generation

4.

This chapter adapts the homonymous paper [22].

Mario Krenn wanted to expand the Pytheus framework beyond the low-gain regime, where creation events are very rare, in order to achieve high fidelities. Xuemei Gu provided software to perform such computations, accounting for arbitrary terms in the Fock basis that were previously neglected [29]. Within our computational budget, limited by the high memory requirements, we choose significant and diverse states to generate, using different postselection conditions. They were all optimized and analyzed by Carlos Ruiz Gonzalez, who also took over the writing with valuable feedback from Mario Krenn and Xuemei Gu.

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Abstract

Entangled photons are widely used in quantum technologies. Many photonic experiments generate them with probabilistic photon-pair sources that can be modeled as squeeze operators. In practice, these sources are usually treated in the low-gain (perturbative) regime, keeping only the leading single-pair term and neglecting higher-order multi-pair emission events. In pursuit of fidelity, the probability of successful entanglement generation can become extremely small, a tradeoff often ignored. Here we develop an automated design algorithm for quantum experiments to optimize both fidelity and success probability while accounting for higher-order multi-pair emissions. Our discovery algorithm explores different design topologies subject to varying hardware constraints. It optimizes the source parameters to reduce undesired higher-order terms or even benefit from them. Our experiments outperform previous proposals for widely used states, including heralded Bell states, W states, and NOON states, paving the way for more efficient photonic technologies.

4.1 Introduction

Entanglement is a crucial resource in quantum technologies [26], enabling quantum computing [159–161], sensing [162, 163], and communications [164, 165]. Photonics is a promising platform for these advances [166, 167]: photons can travel long distances with low decoherence, often without cryogenic cooling [168, 169], and benefit from mature technologies such as optical fiber and integrated circuits [170–172]. To generate entangled photons, many setups combine probabilistic sources, such as spontaneous parametric down-conversion (SPDC), with linear elements and different detection schemes [6, 173].

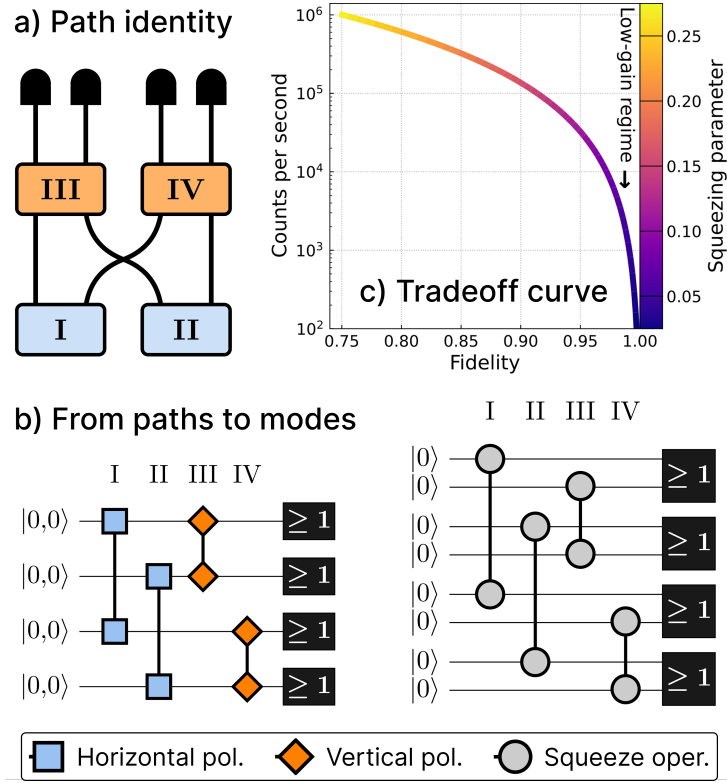
When pumped with strong fields, SPDC sources can produce a high number of photon pairs [174]. Yet, most experiments operate in the perturbative low-gain regime. The pump is so weak that multi-pair emission events are negligible, and each source is approximated by a single pair-creation term. Successful state generation is then identified by specific detector-click patterns [173]. However, as the reported success probability is conditional on those detection outcomes, a fundamental tradeoff is

overlooked. As the pump decreases to preserve fidelity, the probability of observing the required detection pattern becomes extremely small.

Here, we model SPDC sources as full squeeze operators [29, 175–177], revealing that the photon statistics neglected by the perturbative approximation are central for efficient entanglement generation. This turns experimental design into a joint optimization of two competing properties: fidelity and success probability. To improve the tradeoff, we engineer interference between higher-order contributions, reducing undesired emissions. Interestingly, we see that higher-order events can enhance the experiments' performance when using ancillary photons.

We first revisit the generation of GHZ states to illustrate how the trade-off emerges beyond the low-gain approximation. We then present our algorithm to optimize the experiments' parameters and configurations, maximizing the success probability for a range of target fidelities. Finally, we apply the methods to produce a 4-qubit W state, a heralded Bell state, and $N = 3$ and $N = 4$ N00N states, surpassing previous proposals [6, 10, 178]. These are widely used states with different entanglement structures, detection schemes, and photon statistics. By moving beyond the low-gain approximation, our work sheds new light in probabilistic entanglement generation.

Figure 4.1: generating a 4-qubit GHZ state. (a) We replicate an entanglement generation experiment with SPDC crystals and threshold detectors [7]. When all detectors click, the projected state is a superposition of horizontally and vertically polarized photons from the blue (I & II) and orange (III & IV) sources, respectively. With two Fock modes, we encode the photons as $|H\rangle = |1, 0\rangle$ and $|V\rangle = |0, 1\rangle$. (b) The SPDC crystals act as squeeze operators on 4 different optical paths, initialized at vacuum with 2 polarization modes: $|0, 0\rangle$. Splitting each path into 2 modes, we see the sources commute, as they act on different modes. (c) Increasing the common pump r , we transcend the low-gain approximation and reveal the tradeoff between fidelity and success probability. Here and in following figures, the shown counts per second assume the crystals are pumped by a laser with 10ns repetition rate.



4.2 Beyond the low-gain regime

The setup of Fig. 4.1, introduced in the low-gain regime [7], generates a 4-qubit GHZ state, $(|HHHH\rangle + |VVVV\rangle)/\sqrt{2}$, from four SPDC sources. Since the photodetectors ignore polarization, we cannot know which sources fired when each detector clicks. This indistinguishability of emission events is the basis of path identity experiments [17, 30–32, 38, 39, 72, 179]. In Fig. 4.1a, with a common pump $r \ll 1$ and phase $\theta = 0$, each source

can be approximated as $S_2(r, \theta) = \exp(r(e^{-i\theta}ab - e^{i\theta}a^\dagger b^\dagger)) \approx (1 + rab - ra^\dagger b^\dagger)$, where a ($a^\dagger$) and b ($b^\dagger$) are annihilation (creation) operators acting in two different modes. The resulting state is, approximately,

$$\begin{aligned} |\psi\rangle &= S_I S_{II} S_{III} S_{IV} |vac\rangle \\ &\approx (1 + rh_1 h_3 - rh_1^\dagger h_3^\dagger)(1 + rh_2 h_4 - rh_2^\dagger h_4^\dagger) \\ &\quad (1 + rv_1 v_2 - rv_1^\dagger v_2^\dagger)(1 + rv_3 v_4 - rv_3^\dagger v_4^\dagger) |vac\rangle, \end{aligned} \quad (4.1)$$

where $h_j^\dagger$ ($v_j^\dagger$) is the creation operator of horizontally (vertically) polarized photons in the j -th optical path. When the four threshold detectors click, the final state is

$$\begin{aligned} |\tilde{\psi}\rangle &= r^2(h_1^\dagger h_2^\dagger h_3^\dagger h_4^\dagger + v_1^\dagger v_2^\dagger v_3^\dagger v_4^\dagger) |vac\rangle \\ &\quad - r^3(h_1^\dagger h_2^\dagger h_3^\dagger h_4^\dagger v_1^\dagger v_2^\dagger + \dots) |vac\rangle + \dots \\ &\approx r^2(|HHHH\rangle + |VVVV\rangle). \end{aligned} \quad (4.2)$$

Hence, in the low-gain approximation, terms with extra photons are neglected, annihilation operators can be omitted, and the fidelity is assumed optimal.

What the approximation obscures is how unlikely successful detection is, since the dominant terms with unpopulated paths are left out. Beyond this low-gain regime, when the pump grows, so does the success probability. At the same time, as the weight of high-order terms increases, the fidelity drops. This tradeoff is shown in Fig. 4.1c, where the approximation from equation 4.2 is only a small part of the picture.

To recompute this experiment and find our new proposals, we model the SPDC sources as full squeeze operators in the Fock basis. We compute a (truncated) infinite series of amplitudes for arbitrary emission and absorption events [28, 29]. The two-mode SPDC is described as¹

1: detailed amplitudes in equation 2.46.

$$S_2(\zeta) |p, q\rangle = \sum_{k=0}^{\infty} \sum_{n=0}^{\min(p,q)} \mathcal{A}_{k,n}^{(2)}(p, q; \zeta) |p-n+k, q-n+k\rangle, \quad (4.3)$$

where $\zeta = re^{i\theta}$, and for the single-mode case,

$$S_1(\zeta) |p\rangle = \sum_{k=0}^{\infty} \sum_{n=0}^{\lfloor p/2 \rfloor} \mathcal{A}_{k,n}^{(1)}(p; \zeta) |p+2(k-n)\rangle. \quad (4.4)$$

4.3 Automated experimental design

We want to produce entangled states with high success probability via automated design of quantum experiments [1, 5]. We know from the baseline experiments that high fidelity can be reached, but not which probabilities to expect. Thus, we aim for a target fidelity, f_0 , minimizing

$$\begin{aligned} \mathcal{L}_\pi(\mathbf{r}, \boldsymbol{\theta}) &= -w_1 \log \mathcal{P} + w_2 \log ((\mathcal{F} - f_0)^2) \\ &\quad + w_3 \|\mathbf{r}\|_1 + w_4 \|\boldsymbol{\theta}\|_1 + w_5 \mathcal{E}, \end{aligned} \quad (4.5)$$

where $\mathcal{P}$ is the success probability, $\mathcal{F}$ the fidelity, $\mathbf{r} \equiv \{r_j\}$ and $\boldsymbol{\theta} \equiv \{\theta_j\}$ the squeezing parameters ($r_j e^{i\theta_j} = \zeta_j$), and π a fixed ordering of the sources. The remaining terms are optional: L1 regularization, $\|\cdot\|_1$, to enforce sparsity, and the truncation error, $\mathcal{E} = 1 - \text{Tr}(\rho)$, to penalize the

loss of amplitudes due to the cutoff on the Fock space. Sweeping the target fidelity f_0 from 75% to 99%, we find the Pareto front of the design. Since $\mathcal{P}$ may fluctuate by orders of magnitude, the weights, w_i , must be tuned for different detection schemes and target fidelities f_0 .

Each experiment starts from a multi-mode vacuum, on which we apply a series of squeeze operators in a given order $\pi: \{S_j(\zeta_j)\}_\pi$. The optical paths are then measured by threshold detectors, photon-number resolving (PNR) detectors, or left undetected. A successful detection is described as a projection M on the resulting state $\rho(\zeta, \pi)$, which gives the success probability: $\mathcal{P} = \text{Tr}(M\rho)$. To translate the success probability into expected counts per second, we assume the sources are pumped by a laser source with a repetition rate of 10ns. Finally, we trace out the ancillary modes from the state $\rho' = M\rho M^\dagger / \mathcal{P}$ and compute the fidelity: $\mathcal{F} = \langle \psi | \text{Tr}_{\text{anc}}(\rho') | \psi \rangle$.

To find the best designs, we randomly order the sources and optimize the parameters ζ . We repeat many times for different orderings and target fidelities. The best candidates for each f_0 are recomputed at a higher photon cutoff, to ensure $\mathcal{E} \rightarrow 0$, and pruned of redundant sources. This produces a pool of distinct topologies. Then, we reoptimize each of them starting from the previously optimized squeezing parameters, now with less sources, for the same target fidelity f_0 . Slowly varying f_0 across the desired range, we restart the optimizations from the previous optimal parameters. Selecting the best-performing topology, we obtain a set of experiments characterized by a high-dimensional curve $\zeta(f_0)$. This offers a simpler implementation than the initial zoo of unlike designs. Fig. 4.2 illustrates the procedure.

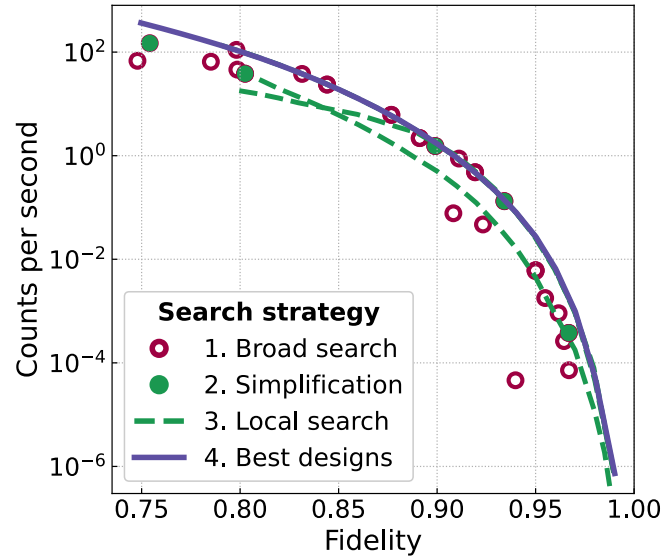


Figure 4.2: optimization strategy. The search takes four steps: 1) experiment optimization for random sortings and target fidelities f_0 (only a few shown), 2) simplification of the best candidates computed with a higher photon cutoff (convergence check), 3) optimization of previous solutions slowly varying f_0 , and 4) final choice of best-performing designs.

Compared to earlier work on automated experiment design [6, 35], here the source ordering is a search variable. In the low-gain regime, postselection leaves each path populated by at most one creation operator (see equation 4.2), so sources effectively commute and operators acting on the same modes merge: $(1 + \zeta_1 a^\dagger b^\dagger)(1 + \zeta_2 a^\dagger b^\dagger) \approx 1 + (\zeta_1 + \zeta_2) a^\dagger b^\dagger$. These simplifications enable graph-based representations of quantum experiments [8–10]. Otherwise, squeeze operators commute only when acting on different modes (as in Fig. 4.1b) or, on shared modes, when $\zeta_1 \zeta_2^* = \zeta_1^* \zeta_2$. Thus, the final state generally depends on the sources' ordering², turning experimental design into a joint search over a discrete ordering and a continuous parameter space.

2: See Supplemental Material at [URL will be inserted], for the source ordering effects.

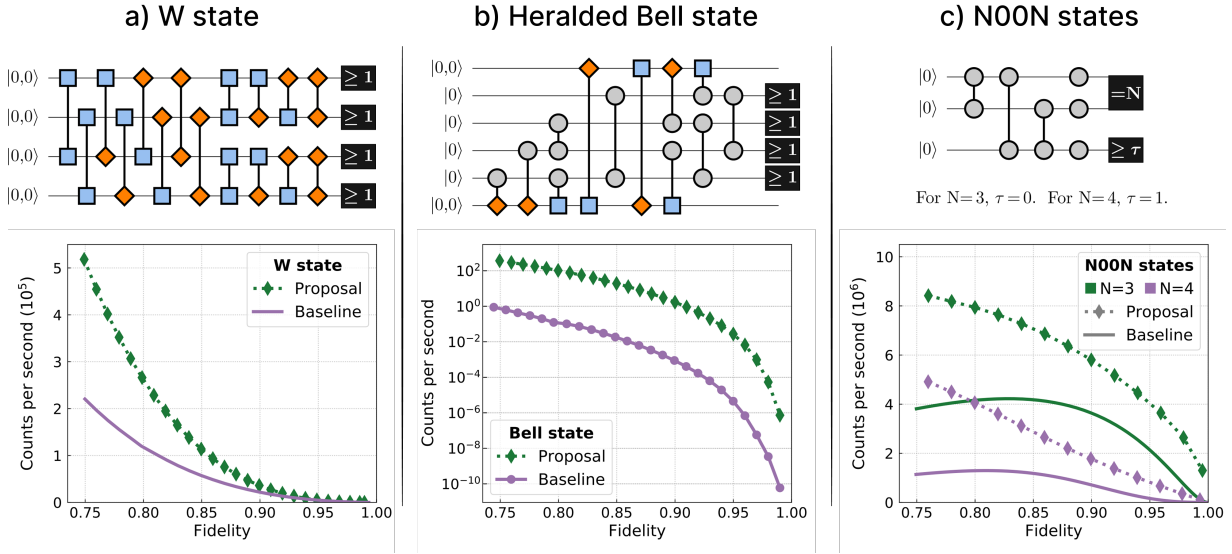


Figure 4.3: high probability entanglement by path identity. The presented experiments use SPDC crystals (squeeze operators) to generate entanglement. These path identity experiments are compared with previous designs, originally proposed for the low-gain regime, that neglected higher-order terms [6, 10, 178]. **a)** The first design produces a postselected W state of 4 qubits using threshold detectors. **b)** The second produces a heralded Bell state using 4 ancillary paths, each with a threshold detector. The entangled photons are not detected. **c)** The last one can generate two variants of N00N states using different parameters and postselection rules. For $N=3$, the threshold detector is not needed, i.e. $\tau = 0$, and the extra photons are lost. For $N=4$, we use a threshold detector and a single-mode squeezer acting on the ancilla mode.

4.4 Results

To showcase our approach, we present new setups to generate diverse forms of entanglement using different detection schemes: a postselected 4-qubit W state, a heralded Bell state with 4 ancillas, and 2-mode N00N states with 3 and 4 photons, both with an ancillary mode. We benchmark our designs against previous proposals [6, 10, 178], recomputed with higher pumps. For a fair comparison, we also optimized the source sorting when required, and adjusted the linear optical elements for the heralded Bell³. Fig. 4.3 shows the new designs and the performance improvement.

We start with a postselected 4-photon W state $|\psi\rangle = (|HHHV\rangle + |HHVH\rangle + |HVHH\rangle + |VHHH\rangle)/2$, an entangled state robust against photon loss and widely used in quantum information tasks [180, 181]. The baseline is an early work on path identity with a common squeezing factor for all sources [10], without interfering terms. Our design uses a different topology with additional sources, some of them producing pairs of vertical photons, and distinct phases that enable destructive interference. These features are deemed redundant in the low-gain regime, and yet they improve the fidelity-probability tradeoff at higher pumps.

Our second experiment generates a Bell state, $|\psi\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$, the most famous form of entanglement [25, 26]. The state's generation is conditioned on the detection of 4 ancillary modes that herald successful generation in the unmeasured paths [173]. The baseline combines a type-II SPDC source with several (polarizing) beam splitters and a half-wave plate [178, 182]. We fit the system into an 8-mode setup with 4 threshold detectors, like ours, replaced the linear optical elements with equivalent beam splitters, and optimized the parameters to obtain the curve shown in Fig. 4.3b. After the “translation” and despite the optimization efforts⁴, our setup outperformed the previous design by a wide margin.

³: See Supplemental Material at [URL will be inserted], for details on the baseline implementation.

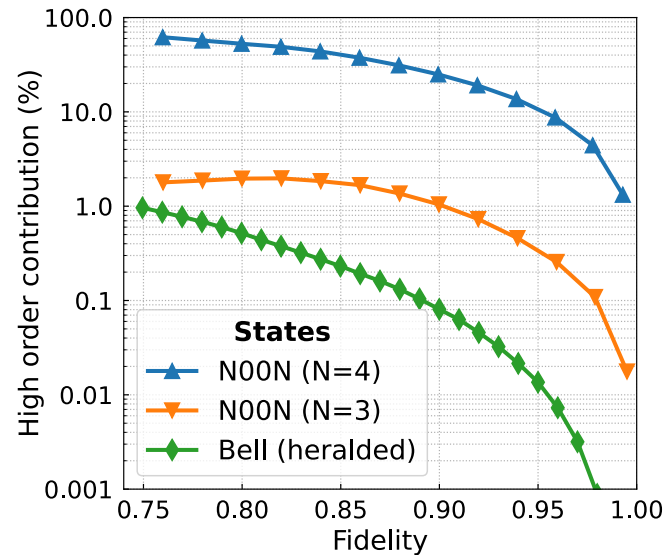
⁴: See Supplemental Material at [URL will be inserted], for details on the heralded Bell baseline implementation, which includes Ref. [183].

5: See Supplemental Material at [URL will be inserted], for the implementation of the 4-photon N00N baseline.

The two remaining experiments produce 2-mode N00N states $|\psi\rangle = (|N;0\rangle + |0;N\rangle)/\sqrt{2}$, with 3 and 4 photons, both using an ancillary mode. Such states are widely used in metrology [184, 185]. In contrast to the polarization-encoded states above, the entanglement here is between photon numbers in the two main modes, which a PNR detector pins to N . The constraints on the ancillary path are more lenient. For $N=3$, photon-pair creation guarantees an odd ancillary photon count, so no detector is needed. For $N=4$, a single threshold detector verifies that extra photons are in the ancillary path, at least two if the detector clicks. Two photons reaching a single threshold detector was not considered by the baseline. Yet, as it improves the performance, we applied the same trick for comparison⁵, and our designs still outperform it. For $N=3$, we use the same sources as the baseline but with different squeezing parameters. This led to new interference patterns providing a better performance.

Beyond the improvements shown in Fig. 4.3, the last 3 experiments share a common feature: their fidelity benefits from contributions with different photon numbers in the ancillary paths. As shown in Fig. 4.4, these effects are larger for the N00N states. For $N=4$, at 90% fidelity, more than a fifth comes from high photon-number emissions. Perturbative approaches overlook such terms, our framework turns them into a resource.

Figure 4.4: relative contribution to the fidelity of high-order terms. When using ancilla paths, creation events with a higher photon-number can also lead to the target (reduced) state. Yet, lower-order terms are still predominant.



4.5 Discussion

By computing emission events with an arbitrary number of photons, we designed efficient experiments for entanglement generation. Our algorithm combined the Fock-basis description of SPDC sources with different detection schemes. Aided by automatic differentiation, we optimized path identity setups to maximize fidelity and success probability. We surpassed previous proposals to produce a postselected W state, a heralded Bell state, and two N00N states. Moreover, when using ancillary paths, the fidelity benefited from multi-photon events, an unexplored resource in the low-gain regime.

The generated states were diverse and broadly applied. Yet, as the Hilbert space grew exponentially with the number of modes, we modeled relatively small systems. While the Gaussian formalism enables a more

compact representation of our experiments [43, 44], symplectic transformations cannot model arbitrary Fock states, and the extraction of Fock-state amplitudes via Hafnians is computationally demanding [186–189]. Scaling the approach to larger systems for arbitrary initial states may require new simulation techniques.

Our algorithm currently optimizes squeezing parameters after randomizing the orderings of the non-commuting SPDC sources. While symmetries reduce the count, we still face a large discrete search space and must resort to expensive optimizations without a guarantee of global optimality. Theoretical intuition or rigorous design rules for the experiments' topology are much needed. Related questions were explored in the graph-based representation of quantum experiments [190], showing how graph theory can guide experimental design [191–193]. It would be helpful to develop analogous principles for the non-commuting, high-gain regime.

Finally, while our proposals rely primarily on SPDC sources, linear optical elements can be included, as shown in the Bell state baseline. Including such elements as part of the optimization, alongside new non-linear tools, would expand the accessible design space. Furthermore, different hardware platforms also impose their own constraints. To name a few: photon losses, detector inefficiencies, or structural limits in integrated photonic circuits. Encoding these into the design process can foster the adoption of new experimental solutions.

Part III.

Gravitational Wave Detectors

Interferometric gravitational wave detection

5.

This chapter offers a brief introduction to gravitational waves and how to detect them. What are they? Where do they come from? And why do we care? All this will become clear. Moreover, by the end of the chapter, you will be familiar with the main advances that enabled the first (direct) detection of gravitational waves in 2015 [194]. We also review the remaining challenges, and how they are addressed. Finally, we introduce an expressive ansatz for interferometric gravitational wave detectors.

5.1 Gravitational waves . . . 57

5.2 Advanced detectors . . . 61

5.1 Gravitational waves

In this section we introduce the generation and propagation of gravitational waves, modeled as a perturbation in an almost flat spacetime. Afterwards, we try to detect them using a Michelson interferometer.

5.1.1 Linearized gravity and the TT gauge

In the space vacuum, far from any source, gravitational waves are a small perturbation in an otherwise flat spacetime. Therefore, the resulting metric tensor is

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad (5.1)$$

where $\eta_{\mu\nu}$ is the *Minkowski metric*, i.e. $\text{diag}(-1, 1, 1, 1)$, and $|h_{\mu\nu}| \ll 1$. This is the so-called *linearized gravity*, where all nonlinear terms $O(h^2)$ are neglected. To find $h_{\mu\nu}$, we must solve the *Einstein equation* for the vacuum

$$R_{\mu\nu} - \frac{1}{2}g^{\alpha\beta}R_{\alpha\beta}g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} = 0, \quad (5.2)$$

where $R_{\mu\nu}$ is the *Ricci curvature*, G is the gravitational constant, c is the speed of light in vacuum, and $T_{\mu\nu}$ is the *stress-energy tensor*—as we are in vacuum, $T_{\mu\nu} = 0$. The curvature $R_{\mu\nu}$ is defined as

$$R_{\mu\nu} = \partial_\alpha \Gamma_{\mu\nu}^\alpha - \partial_\nu \Gamma_{\mu\alpha}^\alpha + \Gamma_{\alpha\beta}^\alpha \Gamma_{\mu\nu}^\beta - \Gamma_{\mu\beta}^\alpha \Gamma_{\alpha\nu}^\beta, \quad (5.3)$$

where $\Gamma_{\mu\nu}^\sigma$ are the *Christoffel symbols*,

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2}g^{\sigma\alpha} (\partial_\mu g_{\nu\alpha} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu}). \quad (5.4)$$

Since $\eta_{\mu\nu}$ is a constant, and only linear terms on $h_{\mu\nu}$ remain, we obtain

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2}\eta^{\sigma\alpha} (\partial_\mu h_{\nu\alpha} + \partial_\nu h_{\mu\alpha} - \partial_\alpha h_{\mu\nu}), \quad \Gamma_{\sigma\mu}^\sigma = \frac{1}{2}\partial_\mu h, \quad (5.5)$$

where $h_\sigma^\sigma = h$ is the trace of $h_{\mu\nu}$. Inserting these definitions into equation 5.3, the last two terms become negligible, and the resulting Ricci tensor and scalar are

$$R_{\mu\nu} = \frac{1}{2} (\partial^\alpha \partial_\mu h_{\nu\alpha} + \partial^\alpha \partial_\nu h_{\mu\alpha} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h), \quad (5.6)$$

$$R = g^{\alpha\beta} R_{\alpha\beta} \approx \eta^{\alpha\beta} R_{\alpha\beta} = \partial^\alpha \partial^\beta h_{\alpha\beta} - \square h \quad (5.7)$$

where $\square = \partial^\alpha \partial_\alpha = \nabla - \partial^2/c^2 \partial t^2$ is the d'Alembert operator. Replacing the terms in the Einstein equation 5.2, we obtain

$$\begin{aligned} 0 &= R_{\mu\nu} - \frac{1}{2} g^{\alpha\beta} R_{\alpha\beta} g_{\mu\nu} \approx R_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} R \\ &= \frac{1}{2} (\partial^\alpha \partial_\mu h_{\nu\alpha} + \partial^\alpha \partial_\nu h_{\mu\alpha} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h - \eta_{\mu\nu} \partial^\alpha \partial^\beta h_{\alpha\beta} + \eta_{\mu\nu} \square h) \\ &= \frac{1}{2} (\partial^\alpha \partial_\mu \bar{h}_{\nu\alpha} + \partial^\alpha \partial_\nu \bar{h}_{\mu\alpha} - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \partial^\alpha \partial^\beta \bar{h}_{\alpha\beta}). \end{aligned} \quad (5.8)$$

Notice that in the final (simpler) expression, we used the *trace reversed* perturbation, $\bar{h}_{\mu\nu}$, such that $h_{\mu\nu} = \bar{h}_{\mu\nu} - \eta_{\mu\nu} \bar{h}/2$ and $\bar{h} = -h$. We can simplify the expression further via *gauge transformation*, in particular we use the *Transverse-Traceless* gauge, or TT gauge. First, we impose the *Lorenz* gauge, where $\partial^\mu \bar{h}_{\mu\nu} = 0$, leading to the wave equation

$$\square \bar{h}_{\mu\nu} = 0. \quad (5.9)$$

Moreover, assuming the wave propagates through the z axis, i.e., $\bar{h}_{\mu\nu} = \bar{h}_{\mu\nu}(t - z/c)$, then $\bar{h}_{\mu 0} = -\bar{h}_{\mu 3}$, and with the synchronous gauge, $h_{\mu 0} = 0$. Hence, only 4 components of the perturbation $\bar{h}_{\mu\nu}$ are different from zero, the spatial components perpendicular to the propagation direction. This is the *transverse* part of the gauge. Since the gauge is also *traceless*, the trace reversed perturbation is the same than the original, $\bar{h}_{\mu\nu}^{TT} = h_{\mu\nu}^{TT}$, which leads to the final solution,

$$h_{\mu\nu}^{TT} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (5.10)$$

where $h_+ = h_+(t - z/c)$ and $h_\times = h_\times(t - z/c)$ are the two polarization modes of a plane wave of frequency ω traveling through the z axis at the speed of light, c . The effects of gravitational waves are often illustrated with still test particles, distributed in a ring perpendicular to the wave propagation. The displacement of these test particles is given by

$$\delta x^\mu = \frac{1}{2} h_{\mu\nu}^{TT} x^\nu, \quad (5.11)$$

where $x^\mu \in (x^1, x^2) \equiv (x, y)$. Taking the solution from equation 5.10, the displacement in the x and y coordinates is, respectively,

$$\delta x = \frac{1}{2} (x h_+ + y h_\times), \quad (5.12)$$

$$\delta y = \frac{1}{2} (x h_\times - y h_+). \quad (5.13)$$

To illustrate the effect of the two polarizations, Fig. 5.1 shows the effect of waves with amplitude $|h| = 0.5$ on a ring of test particles. Notice that the amplitude of a gravitational wave, or *strain*, is a dimensionless quantity that refers to how much the distance gets squeezed. Of course, our cartoonish figure is a wild exaggeration, as the first gravitational waves ever detected had a peak strain of 10^{-21} [194]. Paraphrasing the Royal Swedish Academy of Sciences, which awarded the discovery with the Nobel Prize in 2017, detecting such a perturbation is like measuring a distance of ten lightyears with a precision of a hair's width [195].

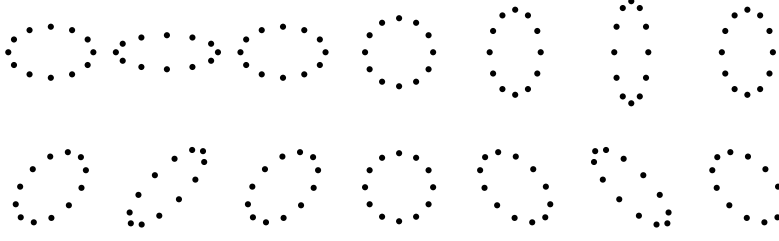


Figure 5.1: gravitational waves with different polarizations—plus above, cross below—acting on a ring of test particles.

5.1.2 Sources of gravitational waves

Using the perturbative treatment, we just described gravitational planar waves propagating in the vacuum. Two polarization modes, h_+ and $h_\times$, define the perturbations, but their amplitude and frequency are determined by their origin. To model the generation of gravitational waves, we cannot assume an empty space, $T_{\mu\nu} \neq 0$. Thus, equation 5.8 becomes

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \quad (5.14)$$

again imposing the Lorenz gauge. The new wave equation can be solved via Green's function, leading to

$$\begin{aligned} \bar{h}_{\mu\nu}(t, \mathbf{x}) &= \frac{4G}{c^4} \int \frac{T_{\mu\nu}(t - |\mathbf{x} - \mathbf{x}'|/c, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ &\approx \frac{4G}{c^4 r} \int T_{\mu\nu}(t - (r - \mathbf{n} \cdot \mathbf{x}')/c, \mathbf{x}') d^3x', \end{aligned} \quad (5.15)$$

where $\mathbf{x} = |\mathbf{x}|\mathbf{n}$ is the position of the observer with respect to the origin (wherever the source is), at a distance $r = |\mathbf{x}| \gg |\mathbf{x}'|$. Expanding the solution around $t' = t - r/c \gg \mathbf{n} \cdot \mathbf{x}'$, we obtain

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) \approx \frac{4G}{c^4 r} \left(\int T_{\mu\nu} d^3x' + \frac{\partial}{\partial t} \int T_{\mu\nu} \mathbf{n} \cdot \mathbf{x}' d^3x' + \frac{\partial^2}{c^2 \partial t^2} \dots \right). \quad (5.16)$$

Notice that each time derivative yields a term proportional to the velocity of the source, v . Hence, assuming the source's movement is non-relativistic, $v/c \ll 1$, we neglect all but the first term. The leading integral can be split in 3 contributions¹:

$$\text{Mass} \quad \int T^{00} d^3x' = Mc^2, \quad (5.17)$$

$$\text{Moment} \quad \int T^{0j} d^3x' = cP^j, \quad (5.18)$$

$$\text{Quadrupole} \quad \int T^{jk} d^3x' = \frac{1}{2} \ddot{Q}^{jk}, \quad (5.19)$$

where $j, k \in [1, 2, 3]$. The first term integrates the total mass of the system from the density, $\rho = c^2 T^{00}$, while the second integrates the momentum density, cT^{0j} . These integrals are constant, non-radiating terms², with no effect in the final gravitational waves. That leaves us with the leading quadrupole contribution, where $Q^{jk} = \int \rho x'^j x'^k d^3x'$. Finally, to recover the TT gauge solution, the perturbation that we actually measure, we take the transverse-traceless projection of the quadrupole [197, 198],

$$Q_{jk}^{TT} = \left(P_j^l P_k^m - \frac{1}{2} P_{jk} P^{lm} \right) Q_{lm}, \quad (5.20)$$

1: $\partial_j \partial_k T^{jk} = \partial_0^2 T^{00}$ in the linearized approximation, leading to the quadrupole expression [196] where $\ddot{Q} = \partial^2 Q / \partial t^2$.

2: while T^{00} also encodes the kinetic energy, we approximate it as the rest mass since $v \ll c$. Similarly, we neglect the reduction on the mass due to the wave radiation. These approximations are further discussed here [197].

3: as the wave propagates across the z axis, $n_1 = n_2 = 0$ and $n_3 = 1$.

where $P_{jk} = \delta_{jk} - n_j n_k$, being δ_{jk} the Kronecker delta, and n_j the unit vector³ between the source and the observer. Since $\bar{h}_{jk}^{TT} = h_{jk}^{TT}$, we obtain the quadrupole formula [197]

$$h_{jk}^{TT}(t, \mathbf{x}) = \frac{2G}{c^4 r} \ddot{Q}_{jk}^{TT}(t - r/c). \quad (5.21)$$

This expression connects the gravitational waves from equation 5.10 with the properties of the system that produces them. For example, a binary system with a circular orbit, where the bodies move at non-relativistic speed, produces waves of amplitude [198]

$$|h| = \frac{4G^2}{c^4} \frac{m_1 m_2}{r d}, \quad (5.22)$$

where r is the distance to the source, d is the distance between the moving bodies, and m_i their mass. The gravitational wave frequency is twice the orbital angular frequency of the binary system, $\omega = 2\Omega$, given by Kepler's third law: $\Omega^2 = G(m_1 + m_2)/d^3$.

The double pulsar PSR J0737–3039, a good example of such a system [199], produces waves of amplitude $|h| \sim 10^{-22}$. For comparison, this perturbation is 10 times weaker than the first wave ever detected, in 2015 [194], which was produced by the merging of two black holes. Modeling mergers is beyond the quadrupole formula, which assumed weak fields produced by bodies moving at non-relativistic speeds. Instead, modeling merging events requires numerical relativity [200], which is beyond this thesis' scope.

5.1.3 Naive interferometric detection

Gravitational waves carry information about distant cosmological events. Yet, to access the information, we must detect extremely feeble perturbations. In our first attempt to do so, we use a *Michelson interferometer*.

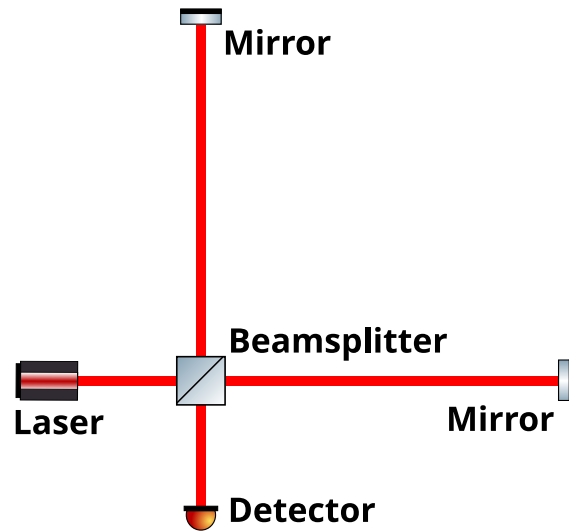


Figure 5.2: Michelson interferometer. In our Michelson interferometer, operated near the dark fringe, the lengths of the vertical and horizontal arms are almost identical.

In a Michelson interferometer, shown in Fig. 5.2, the laser beam is split by a beamsplitter into two perpendicular interferometer arms. At the ends of the arms, at distances L_x and L_y from the beamsplitter, mirrors reflect the light, which the beamsplitter then recombines. The light leaving the

beamsplitter through the bottom path reaches a photodetector, and the measured power is proportional to [197]

$$|E_0|^2 \sin^2(k_L(L_y - L_x)) \equiv P_0 \sin^2(\phi_0), \quad (5.23)$$

where $k_L = \omega_L/c$ is the wavenumber of the laser light, and E_0 its electric field. Now, if we introduce a perturbation $h_+(t) = h_0 \cos(\omega t)$, the optical path in the x-arm becomes⁴

$$\begin{aligned} L'_x(t) &= 2L_x + \frac{c}{2} \int_{t-2L_x/c}^t h_+(t') dt' \\ &= 2L_x \left(1 + \frac{ch_0}{4L_x} \int_{t-2L_x/c}^t \cos(\omega t') dt' \right) \\ &= 2L_x + \frac{h_0 c}{\omega} \sin(\omega L_x/c) \cos(\omega(t - L_x/c)), \end{aligned} \quad (5.24)$$

and similarly for the y-arm

$$L'_y(t) = 2L_y - \frac{h_0 c}{\omega} \sin(\omega L_y/c) \cos(\omega(t - L_y/c)). \quad (5.25)$$

With this, the difference between optical paths is

$$L'_y(t) - L'_x(t) = \frac{2}{k_L} (\phi_0 + \phi(t)) \quad (5.26)$$

where the second term, the phase added by the gravitational wave, is

$$\phi(t) \equiv -h_0 \frac{\omega L}{\omega} \sin(\omega L/c) \cos(\omega(t - L/c)), \quad (5.27)$$

with $L_x \approx L_y \equiv L$. Therefore, under the effect of a gravitational wave, the power at the detector becomes $P_0 \sin^2(\phi_0 + \phi(t))$. To further simplify the expression, one could set exactly $L_x = L_y$. Yet, if we keep a constant phase $\phi_0 \neq 0$, since $\phi(t) \ll 1$, we obtain

$$P_0 \sin^2(\phi_0 + \phi) \approx P_0 (\sin^2 \phi_0 + \sin(2\phi_0) \phi). \quad (5.28)$$

This makes the power response linear⁵ to the phase $\phi(t)$, and thus to the wave amplitude. The constant power term is then suppressed by low pass filter. Yet, the Michelson interferometer is severely limited. Looking at equation 5.27, we see how the signal peaks at $\omega L/c = \pi/2$. The waves from the black hole merger detected in 2015 reached their maximum amplitude, $h = 10^{-21}$, with a frequency of $\omega/2\pi = 150\text{Hz}$ [194]. The corresponding optimal arm length is 500km, totally infeasible and far beyond the 4km arms used by the LIGO detector. Moreover, a Michelson interferometer with 4km arms would capture around a 1/80 of the optimal⁶ signal. Elongating the optical path within a "short" arm is one of the main advances of LIGO's detector, and the first we discuss.

4: light propagation is described by the space-time interval $ds^2 = 0$, which under a perturbation $h_+(t) \ll 1$ is: $ds^2 = -c^2 dt^2 + (1 + h_+) dx^2 + (1 - h_+) dy^2 + dz^2$. Thus, light in x-arm propagates (to the right) as $dx \approx (1 - h_+/2)cdt$, and similar for the y-arm (to the top): $dy \approx (1 + h_+/2)cdt$.

5: if $\phi_0 = 0$, the response to $\phi(t)$ would be quadratic.

6: what is the weakest signal we can detect? This question will be addressed in section 5.2.2.

5.2 Advanced detectors

We have modeled gravitational waves and deduced from their properties what the cosmological events that produced them were like. However, our first detector proposal, a Michelson interferometer, required unreasonable proportions. In this section, we explore the advances that enable the LIGO collaboration to detect gravitational waves with setups of a few

kilometers [194], rather than hundreds. We then review the main sources of noise and how to mitigate them. Finally, we introduce the quasi-universal interferometer, or UIFO [12], an expressive experimental ansatz for designing new detectors.

5.2.1 Mirrors everywhere

In the Michelson interferometer, gravitational waves induced an additional phase in the electric field reaching the detector (see equation 5.27). This led to a detectable change in the power, but to maximize the signal, the interferometer's arms had to be hundreds of kilometers long. Fortunately, a more practical solution exists. By placing a second mirror at the beginning of each arm, light bounces back and forth between the mirrors before going back to the beam splitter. Hence, each *cavity* elongates the effective optical path length, increasing the phase difference between the light coming from each arm.

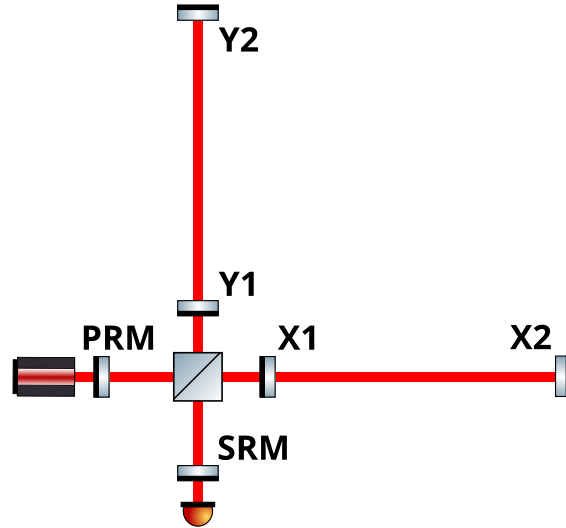


Figure 5.3: simplified LIGO detector. The depicted interferometer uses cavities in both arms, as well as recycling and signal power mirrors.

Focusing on the horizontal cavity from Fig. 5.3, we assume a stable light source that reaches the cavity mirror (X1), at $x = 0$, with an electric field⁷

$$E_{\text{in}}(t) = E_0 e^{-i\omega_L t}. \quad (5.29)$$

Assuming a reflection coefficient r_1 , the reflected field is

$$E_{\text{out}}^{(0)} = r_1 E_{\text{in}} = r_1 E_0 e^{-i\omega_L t}. \quad (5.30)$$

However, $E_{\text{in}}^{(0)}$ is not the only field leaving the mirror. If the laser was running for long enough, at a previous time $t - 2L/c$, a field $t_1 E_0 e^{-i\omega_L(t-\tau)}$ crossed the first mirror, being L the cavity's length. Then, at time $t - L/c$, it reflected in the second mirror, at $x = L$, returning as $-r_2 t_1 E_0 e^{-i\omega_L(t-L/c)}$, and leaving the first mirror as⁸ $E_{\text{out}}^{(1)} = -r_2 t_1^2 E_0 e^{-i\omega_L(t-2L/c)}$, at the same time than $E_{\text{out}}^{(0)}$ reflected. These two fields are then added, but there are more of them. A field that crossed the first mirror at time $t - 4L/c$ reached the second mirror at time $t - 3L/c$, bounced back and reached the first mirror at $t - 2L/c$. That is, when the previous field was entering. Then the field reflected in the first mirror, again in the second, and left the cavity as $E_{\text{out}}^{(2)} = -r_1 r_2^2 t_1^2 E_0 e^{-i\omega_L(t-4L/c)}$. Iterating this, we can go back to arbitrary

7: for the same E_0 , the field's amplitude was half in equation 5.23, as it was split in 2 arms.

8: we define real reflection and transmission coefficients, and a negative phase is added to the field reflected inside the cavity. This is what we saw in Section 2.2.3. However, now we consider the mirror's loss $\epsilon \ll 1$, such that $r^2 + t^2 = 1 - \epsilon$.

times $t - 2nL/c$, leading to an output field (as usual $k_L = \omega_L/c$)

$$E_{\text{out}}^{(n)} = -E_0 e^{-i\omega_L t} r_1^{n-1} r_2^n t_1^2 e^{i2k_L n}. \quad (5.31)$$

Finally, adding the terms $E_{\text{out}}^{(n)}$ and replacing $t_1^2 = 1 - \epsilon_1 - r_1^2$, we obtain⁹

$$E_{\text{out}}(t) = \sum_{n=0}^{\infty} E_{\text{out}}^{(n)} = E_0 e^{-i\omega_L t} \frac{r_1 - r_2(1 - \epsilon_1)e^{i2k_L L}}{1 - r_1 r_2 e^{i2k_L L}}. \quad (5.32)$$

9: $E_{\text{in}}^{(0)}$ is the maverick term. It does not cross the first mirror nor bounce inside the cavity, so no negative phase is added.

Now, as we did for the Michelson interferometer, we introduce a perturbation $h_+ = h_0 \cos(\omega t)$. As before, the field $E_{\text{out}}^{(1)}$, which went back and forth inside the cavity, will be added to $E_{\text{out}}^{(0)} = r_1 E_0 e^{-i\omega_L t}$, which reflected in the first mirror at time t . However, we no longer assume that $E_{\text{out}}^{(1)}$ entered the cavity at time $t - 2L/c$, but at

$$t_0^{(1)} = t - \frac{2L}{c} - \frac{h_0}{\omega} \sin(\omega L/c) \cos(\omega(t - L/c)) \equiv t - \frac{2L}{c} - \frac{\delta\phi(t)}{\omega_L}. \quad (5.33)$$

This is like in equation 5.24, where we described light propagating in one of the Michelson interferometer's arms. Similarly, the term $E_{\text{out}}^{(2)}$, added to the previous two fields, did not enter the cavity at time $t - 4L/c$ but at

$$\begin{aligned} t_0^{(2)} &= t_0^{(1)} - \frac{2L}{c} - \frac{\delta\phi(t_0^{(1)})}{\omega_L} \\ &\approx t - \frac{4L}{c} - \frac{1}{\omega_L} (\delta\phi(t) + \delta\phi(t - 2L/c)), \end{aligned} \quad (5.34)$$

where we approximate $\delta\phi(t_0^{(1)}) = \delta\phi(t - 2L/c - \delta\phi(t)/\omega_L) \approx \delta\phi(t - 2L/c)$. Finally, the general case is

$$t_0^{(n)} = t - \frac{2L}{c}n - \frac{1}{\omega_L} \Phi_n(t) \quad \text{where } \Phi_n(t) \equiv \sum_{k=0}^{n-1} \delta\phi(t - 2kL/c), \quad (5.35)$$

and replacing $t - 2nL/c$ by $t_0^{(n)}$ in equation 5.31 leads to

$$E_{\text{out}}^{(n)} = -E_0 e^{-i\omega_L t} r_1^{n-1} r_2^n t_1^2 e^{i2k_L n} e^{i\Phi_n(t)}. \quad (5.36)$$

Now each term has $E_{\text{out}}^{(n)}$ an extra phase¹⁰ $e^{i\Phi_n(t)} \approx 1 + i\Phi_n(t)$. Accordingly, the sum of all fields leaving the cavity's left mirror is

10: again, we are leveraging that $h_0 \ll 1$ to simplify the equations.

$$\begin{aligned} E_{\text{out}}(t) &= E_0 e^{-i\omega_L t} \left(\frac{r_1 - r_2(1 - \epsilon_1)e^{i2k_L L}}{1 - r_1 r_2 e^{i2k_L L}} \right. \\ &\quad \left. - i t_1^2 r_2 e^{i2k_L L} \sum_{n=1}^{\infty} (r_1 r_2 e^{i2k_L L})^{n-1} \Phi_n \right). \end{aligned} \quad (5.37)$$

We already saw the first term in equation 5.32, so we focus on the extra field contribution, the sum. For convenience, we define $u \equiv r_1 r_2 e^{i2k_L L}$, $\varphi \equiv \omega(t - L/c)$, and $\theta \equiv \omega L/c$, which leads to

$$\begin{aligned} \sum_{n=1}^{\infty} u^{n-1} \Phi_n &= h_0 \frac{\omega_L}{\omega} \sin \theta \sum_{n=1}^{\infty} u^{n-1} \sum_{m=0}^{n-1} \cos(\varphi - 2\theta m) \\ &= h_0 \frac{\omega_L}{\omega} \sin \theta \frac{\cos \varphi - u \cos(\varphi + 2\theta)}{(1 - u)(1 - 2u \cos(2\theta) + u^2)}. \end{aligned} \quad (5.38)$$

Notice that both terms in equation 5.37 are proportional to $1/(1-u)$, whose magnitude reach a maximum for $e^{i2k_L L} = 1$, i.e., in *resonance*. That is how we want our cavities to operate, since it leads to the maximum signal. Having imposed resonance, we rewrite the two terms of the output field, the *carrier* and the *signal*, as $E_{\text{out}}(t) = E_0 e^{-i\omega_L t} (C - iS(t))$, where

$$C \equiv \frac{r_1 - r_2(1 - \epsilon_1)}{1 - r_1 r_2}, \quad (5.39)$$

$$S \equiv h_0 t_1^2 r_2 \frac{\omega_L}{\omega} \frac{\sin(\theta)(\cos(\omega t - \theta) - r_1 r_2 \cos(\omega t + \theta))}{(1 - r_1 r_2)(1 - 2r_1 r_2 \cos(2\theta) + (r_1 r_2)^2)}. \quad (5.40)$$

We are very close to our final goal: translating the effect of a gravitational wave into a measurable signal. To do so, we add the fields that exit the cavities and leave the beamsplitter through the bottom path. Assuming the signal h_+ was aligned with the arms, these fields are¹¹

$$E_x = -\frac{1}{2} E_0 e^{-i(\omega_L t - 2k_L l_x)} (C - iS), \quad (5.41)$$

$$E_y = +\frac{1}{2} E_0 e^{-i(\omega_L t - 2k_L l_y)} (C + iS). \quad (5.42)$$

Here, we account for an extra phase $2k_L l_x$ due to the distance between the beamsplitter and the cavities. The lengths are almost identical, just as the arms of the Michelson interferometer, but the difference leads to a linear response of the signal, as in equation 5.28. Notice that only the gap between the beamsplitter and each mirror differs, not the cavity length. For the cavities to be in resonance, their length must be fixed based on the light's frequency. We also assume the mirrors have the same properties in both cavities. All together, the field leaving the beamsplitter is

$$E(t) = E_x + E_y = iE_0 e^{-i(\omega_L t - 2k_L l)} (\sin(\phi_0)C + \cos(\phi_0)S), \quad (5.43)$$

where $2l = l_x + l_y$ and $\phi_0 = k_L(l_y - l_x) \ll 1$. Finally, since S is linear to $h_0 \ll 1$, the final power is approximately¹²

$$|E_0|^2 |\sin(\phi_0)C + \cos(\phi_0)S|^2 \approx P_0 (|\sin(\phi_0)C|^2 + \sin(2\phi_0)CS). \quad (5.44)$$

This result is almost identical to equation 5.28, but instead of the signal $\phi(t)$, we have a cavity term $C \approx -1$ and the new signal term $S(t)$, which for $\theta = \omega L/c \ll 1$ is¹³

$$S \approx h_0 t_1^2 r_2 \omega_L \frac{\omega_0 \cos(\omega t) + \omega \sin(\omega t)}{\omega_0^2 + \omega^2} = \frac{h_0 t_1^2 r_2 \omega_L}{\sqrt{\omega_0^2 + \omega^2}} \cos(\omega t - \delta), \quad (5.45)$$

where $\omega_0 \equiv (1 - r_1 r_2)c/2L$ and $\tan \delta \equiv \omega/\omega_0$. The signal peaks at $\omega \rightarrow 0$, just like $\phi(t) \approx -h_0 \omega_L L \cos(\omega t)/c$, but with a gain¹⁴

$$\left| \frac{S}{\phi_0} \right| = \frac{t_1^2 r_2}{(1 - r_1 r_2)^2} \approx \frac{2}{1 - r_1}, \quad (5.46)$$

where $r_1 \lesssim r_2 \approx 1$, and the mirrors are close to lossless. Hence, for a $r_1 = 0.99$, the signal is enhanced by 2 orders of magnitude [201].

11: the beamsplitter is 50:50, thus the $1/2$. The reflected term from the x-arm gains a negative phase.

12: in general, $|a + b|^2 = |a|^2 + |b|^2 + 2\text{Re}(a^*b)$, but for now $C, S \in \mathbb{R}$.

13: $a \cos x + b \sin x = \Delta \cos(x - \delta)$, being $\Delta \equiv \sqrt{a^2 + b^2}$ and $\tan \delta \equiv b/a$.

14: this is proportional to the cavity's *fineness*, $\mathcal{F} = \pi \sqrt{r_1 r_2} / (1 - r_1 r_2) \approx \pi / (1 - r_1)$. Roughly speaking, the fineness indicates how well the cavity stores the field, and its proportional to the average time that a photon stays within the cavity.

Recycling mirrors

The benefits of the cavities are clear, but we must still discuss two of the mirrors in the LIGO design: the power and signal *recycling mirrors* shown in Fig. 5.3. Like the cavities, they "store" the laser field for longer times, but rather than elongating the effective optical path, they enhance the signal readout in different ways [202, 203].

The role of the power recycling mirror (PRM) is almost self-explanatory. We are operating the interferometer very close to the dark fringe, with $\phi_0 \ll 1$. Therefore, most of the field that reaches the beamsplitter returns to the source. From the point of view of the laser, the beamsplitter and the cavities act together as a high reflectivity mirror¹⁵ with $r_{\text{int}}^2 = 1 - \Lambda \approx 1$, being Λ the fraction of the field that does not return to the source. To recycle the otherwise wasted field, we place a mirror between the laser and the beamsplitter, with reflectivity r_p and transmissivity t_p . The PRM and the "effective" mirror of the interferometer create a small cavity, building up a field¹⁶

$$E_p = t_p E_{\text{in}} \sum_{n=0}^{\infty} (r_p r_{\text{int}} e^{i\phi_p})^n = \frac{t_p E_{\text{in}}}{1 - r_p r_{\text{int}} e^{i\phi_p}}, \quad (5.47)$$

where E_{in} is the field entering the cavity from the laser, $\phi_p \equiv 2k_L l_p$, and l_p is the gap between the beamsplitter and recycling mirror. In resonance, this leads to a power gain of

$$\left| \frac{E_p}{E_{\text{in}}} \right|^2 = \frac{t_p^2}{(1 - r_p r_{\text{int}})^2} = \frac{t_p^2}{(1 - r_p + r_p \Lambda)^2}, \quad (5.48)$$

which reaches a maximum for¹⁷ $t_p^2 = \Lambda$. Using an almost lossless recycling mirror, $t_p^2 \approx 1 - r_p^2$, the gain is approximately $\Lambda^{-1} \gg 1$. Thus, with the same light source, the power P_0 in equation 5.44 is significantly enhanced.

Finally, we study the signal recycling mirror (SRM). Like the PRM, it also creates a cavity with the "effective" interferometer mirror. However, the effective reflectivity now depends on the frequency of the incoming field. To understand this, we must consider what occurs when the field returns to the beamsplitter, and before that, it is convenient to tidy up equation 5.43. That is, the field that leaves the beamsplitter heading to the detector for the first time¹⁸

$$E(t) = E'_0 \sin \phi_0 C e^{-i\omega_L t} \quad \text{carrier sideband} \quad (5.49)$$

$$+ E'_0 \cos \phi_0 S_0 \frac{e^{i\theta} e^{-i(\omega_L + \omega)t}}{1 - r_1 r_2 e^{i2\theta}} \quad \text{upper sideband} \quad (5.50)$$

$$+ E'_0 \cos \phi_0 S_0 \frac{e^{-i\theta} e^{-i(\omega_L - \omega)t}}{1 - r_1 r_2 e^{-i2\theta}} \quad \text{lower sideband} \quad (5.51)$$

For clarity, we defined the constants $S_0 \equiv h_0 t_1^2 r_2 \omega_L \sin \theta / (2\omega(1 - r_1 r_2))$ and $E'_0 \equiv iE_0 e^{i2k_L l}$. There is nothing new in the expression, but it explicitly shows that the wave reaching the detector has three main field contributions, each oscillating at a different frequency.

Similar to the PRM, we take the field coming from the beamsplitter, $E(t)$, as the input field, E^{in} . However, now we compute the cavity field E_s by separately addressing the three components of $E(t) = E_{\text{carrier}} + E_{\text{upper}} + E_{\text{lower}}$. Starting with the original upper sideband field, $E_u^{(0)}$, the field bounces in the PRM, with a phase -1, and returns to the beamsplitter,

15: as almost no field goes towards the detector, what returns to the laser is, approximately, $\cos \phi_0 C E_{\text{in}} \approx -E_{\text{in}}$. Thus, $r_{\text{int}} \approx \cos \phi_0 C$.

16: this is very similar to the field produced in the arm cavities, in equation 5.32. However, since we are inside the cavity, there is only one transmission coefficient, and reflections always occur on the same side of the mirrors.

17: also called *impedance matching* [204].

18: to split the term S from equation 5.45, we massage the numerator, $2(\cos(\omega t - \theta) - r_1 r_2 \cos(\omega t + \theta)) = e^{-i(\omega_L + \omega)t} (1 - r_1 r_2 e^{i2\theta}) + e^{-i(\omega_L - \omega)t} (1 - r_1 r_2 e^{-i2\theta})$, and the denominator $1 - 2r_1 r_2 \cos(2\theta) + (r_1 r_2)^2 = |1 - r_1 r_2 e^{i2\theta}|^2$.

where is split. Then, as with the cavity described in equation 5.32, several optical paths are possible, and we must add them all. In the shortest path, the light bounces back in the first mirror and returns. Recombining the field in both arms, we obtain $E_{u1}^{(0)} = r_s e^{i\phi_s} \cos \phi_0 r_1 E_u^{(\text{in})}$, where again $\phi_0 = k_L(l_y - l_x) \ll 1$, $\phi_s \equiv 2k_L l_s$, and l_s is the distance between the beamsplitter and the recycling mirror. The index $u1$ indicates the single bounce in the SRM. For the terms $E_{u1}^{(n)}$, when light goes through the cavity $n \geq 1$ times, $E_{u1}^{(n)} = -r_s e^{i\phi_s} \cos \phi_0 r_1^{n-1} r_2^n t_1^2 e^{i2n\theta} E_u^{(\text{in})}$, almost identical to equation 5.31. Adding all terms we obtain¹⁹

19: the effect of gravitational waves, shown in equation 5.37, is neglected here, as it leads to higher order terms on h_0 .

$$\begin{aligned} E_{u1} &= r_s e^{i\phi_s} \cos \phi_0 E_u^{(\text{in})} \sum_{n=0}^{\infty} E_{u1}^{(n)} \\ &= r_s e^{i\phi_s} \cos \phi_0 \frac{r_1 - r_2(1 - \epsilon_1)e^{i2\theta}}{1 - r_1 r_2 e^{i2\theta}} E_u^{(\text{in})} \equiv r_{\text{int}}^{(u)} r_s e^{i\phi_s} E_u^{(\text{in})}. \end{aligned} \quad (5.52)$$

This defines the effective reflection acting on the upper band field, with a field's frequency $\omega_L + \omega$. Similarly, for an arbitrary frequency $\Omega = m\omega_L + \delta\omega$, with $m \in \mathbb{N}$, the effective cavity reflectivity is

$$r_{\text{int}}(\Omega) = \cos \phi_0 \frac{r_1 - r_2(1 - \epsilon_1)e^{i2\delta\theta}}{1 - r_1 r_2 e^{i2\delta\theta}}, \quad (5.53)$$

where $\delta\theta = \delta\omega L/c$. In other words, since the cavity is in resonance, $e^{i2L\omega_L/c} = 1$, then $\Omega \equiv \Omega \bmod \omega_L$. Accordingly, the carrier field keeps the same phase, no matter how many times it goes through the cavity, and $r_{\text{int}}(\omega_L) = \cos \phi_0 C$. With the same procedure as in equation 5.47, using $\sum_{n=0}^{\infty} (r_{\text{int}} r_s e^{i\phi_s})^n = 1/(1 - r_{\text{int}} r_s e^{i\phi_s})$, we obtain the carrier field

$$E_c^{(\text{out})} = \frac{t_s E_c^{(\text{in})}}{1 - C \cos \phi_0 r_s e^{i\phi_s}} = \frac{t_s E_0' \sin \phi_0 C e^{-i\omega_L t}}{1 - C \cos \phi_0 r_s e^{i\phi_s}}, \quad (5.54)$$

and the upper (+) and lower (-) sideband fields leaving the cavity towards the detector

$$\begin{aligned} E_{u/l}^{(\text{out})} &= \frac{t_s E_{u/l}^{(\text{in})} (1 - r_1 r_2 e^{\pm i2\theta})}{(1 - r_1 r_2 e^{\pm i2\theta}) - \cos \phi_0 (r_1 - r_2(1 - \epsilon_1)e^{\pm i2\theta}) r_s e^{i\phi_s}} \\ &= \frac{E_0' \cos \phi_0 S_0 t_s e^{\pm i\theta} e^{-i(\omega_L \pm \omega)t}}{1 - e^{i\phi_s} r_1 r_s \cos \phi_0 - (r_1 r_2 - e^{i\phi_s}(1 - \epsilon_1) r_2 r_s \cos \phi_0) e^{\pm i2\theta}}, \end{aligned} \quad (5.55)$$

The effect of signal recycling is not as obvious as for the power recycling. There is no global gain in the amplitude field, but looking at the denominator of equation 5.55, we see

$$\begin{aligned} &1 - e^{i\phi_s} r_1 r_s \cos \phi_0 - (r_1 r_2 - e^{i\phi_s}(1 - \epsilon_1) r_2 r_s \cos \phi_0) e^{\pm i2\theta} \\ &\equiv A_s - B_s e^{\pm i2\theta} \approx A_s - B_s (1 \pm i2\omega L/c) \\ &= B_s \frac{2L}{c} \left(\frac{c}{2L} \left(\frac{A_s}{B_s} - 1 \right) \mp i\omega \right) \equiv B_s \frac{2L}{c} (\omega_s \mp i\omega), \end{aligned} \quad (5.56)$$

20: if cavities end with almost perfectly reflecting mirrors, i.e., $r_2 \approx 1$, then

$$\omega_s \approx \frac{c(1 - r_1)}{2L} \frac{1 + r_s e^{i\phi_s}}{r_1 - r_s e^{i\phi_s}},$$

where $c(1 - r_1)/2L \approx \omega_0$, the frequency we saw in equation 5.45.

where $\theta \equiv \omega L/c \ll 1$. This simplifies the sideband expressions to²⁰

$$E_{u/l}^{(\text{out})} = E_0' \cos \phi_0 S_0 \frac{t_s c}{2LB_s} \frac{e^{\pm i\theta} e^{-i(\omega_L \pm \omega)t}}{\omega_s \mp i\omega}, \quad (5.57)$$

which leads to the total signal field

$$E_s^{(\text{out})} = E'_0 \cos \phi_0 S_0 \frac{ct_s e^{-i\omega_L t}}{LB_s} \frac{\omega_s \cos(\omega t) + \omega \sin(\omega t)}{\omega_s^2 + \omega^2}. \quad (5.58)$$

This resembles the pole²¹ we observed in equation 5.45, but ω_s can be tuned with the signal recycling cavity. Hence, the signal field no longer peaks at $\omega = 0$ but wherever we choose, with an adjustable²² width.

21: if $r_2 \approx 1 \Rightarrow \omega_s \approx \omega_0(1 + r_s e^{i\phi_s})/(r_1 - r_s e^{i\phi_s})$.

22: the integral of the field amplitude for different wave frequencies stays constant [205].

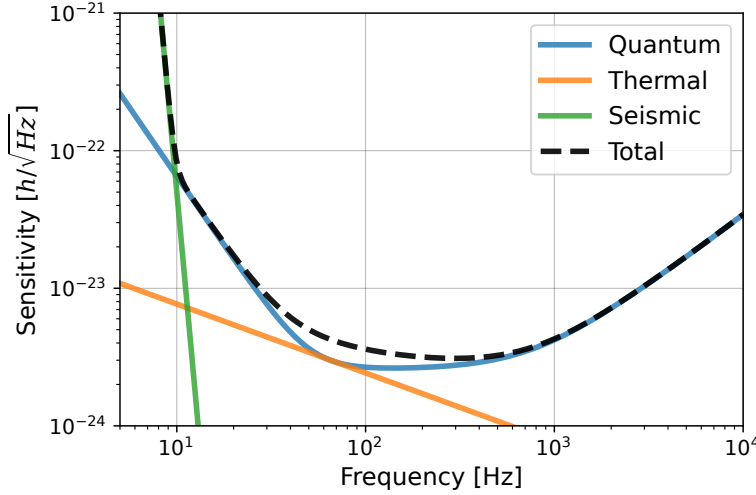


Figure 5.4: the noise budget. Simplified calculation of the total strain sensitivity from the Finesse documentation [206].

5.2.2 Strain sensitivity and the noise budget

Hitherto, we have been through several iterations of gravitational wave detectors. Starting with a Michelson interferometer, we used cavities to enhance the phase difference, and thus the signal field, induced by a gravitational wave. Then we add power and signal recycling mirrors. The former increased the laser's effective power, and the second allowed us to tune the field response to different frequencies. These advances lead to a power response linear to the strain

$$2\text{Re}(E_c^* E_s) \equiv h_0 H(\omega), \quad (5.59)$$

where $H(\omega)$, the so-called *transfer function*, characterizes the response of the interferometer to different gravitational wave frequencies [207]. We want the magnitude of $H(\omega)$ to be as large as possible, longer²³ cavity arms, higher light powers, the more the better. Unfortunately, not only are we limited by space constraints, unbounded light power leads to unbounded damage, which can render optical components useless. Moreover, even with ideal hardware, different noise sources limit the waves we can detect. The *noise budget*, the collection of noises that affect the gravitational wave detection, divided by the amplitude of the transfer function, leads to the *strain sensitivity*

$$\mathcal{S}(\omega) = \left| \frac{N(\omega)}{H(\omega)} \right|. \quad (5.60)$$

That is, the smallest wave's amplitude h_0 we can discern from the noise $N(\omega)$ given the transfer function $H(\omega)$. Therefore, the lower $\mathcal{S}$ is, the more sensitive our detector is. In this section, we review the main noise sources that limit the performance of interferometric detectors.

23: in equation 5.58, $S_0 \propto L$ and $\omega_s \propto 1/L$.

For a design with all the cavities we have described, the final sensitivity is shown in Fig. 5.4. Among the depicted noises, the quantum one is arguably the most fundamental. Technical advances and space interferometers can mitigate thermal and seismic noise, but the quantum nature of photons persists. From the two types of quantum noise, we start with the *shot noise*, which originates from the fluctuations in the number of photons reaching the detector. We start with the signal-to-noise ratio

$$\frac{S}{N} = \frac{\Delta P_{\text{gw}}}{\Delta P_{\text{shot}}}, \quad (5.61)$$

i.e., the ratio between the power response to the gravitational wave—the signal from equation 5.59—and the fluctuations in the overall power reaching the detector. While the signal S must not be confused with the strain sensitivity, they are closely related [197]

$$\mathcal{S}_{\text{shot}} = h_0 \sqrt{\tau} \frac{N}{S}, \quad (5.62)$$

where τ is the detection time. As shown in Chapter 2, the number of photons in laser light follows the Poisson distribution, where the average number of photons, N , is also the variance. Accordingly, $\Delta P_{\text{shot}} = \hbar\omega_L \sqrt{N}/\tau = \sqrt{\hbar\omega_L P_{\text{det}}/\tau}$, and since the signal is linear to $h_0 \ll 1$, most power at the detector comes from the carrier field, $P_{\text{det}} \approx |E_c^{(\text{out})}|^2$. All together, this leads to a strain

$$\mathcal{S}_{\text{shot}} = h_0 \sqrt{\tau} \frac{\Delta P_{\text{shot}}}{\Delta P_{\text{gw}}} = \frac{h_0 \sqrt{\hbar\omega_L} |E_c|}{2\text{Re}(E_c^* E_s)}. \quad (5.63)$$

Without unraveling the strain expression, we see it is inversely proportional to the signal field (see equation 5.58). Hence, for a given interferometer configuration with a fixed ϕ_s , the strain increases for larger wave frequencies and decreases with higher laser power.

The second part of quantum noise is *radiation pressure*. Since photons carry momentum, they apply a force on the mirrors that reflect them. As a result, the mirrors may shake or even move, creating an effective phase difference as stochastic as the photon generation that produces it. In the simple Michelson interferometer, with only one mirrors at each arm, the noise strain sensitivity due to radiation pressure is [197]

$$\mathcal{S}_{\text{rad}}^{(\text{Mich})} = \frac{4\sqrt{2\hbar\omega_L P}}{ML\omega^2 c}, \quad (5.64)$$

where M is the mass of the mirrors, and P is the laser power that reach them. Here the light reflects only once in the arm mirror, so power is given by the laser field. However, the cavities amplify the field, which reflects between the mirrors an average of $2\mathcal{F}/c$, where $\mathcal{F}$ is the cavity's finesse²⁴. The resulting power boost leads to

$$\mathcal{S}_{\text{rad}}^{(\text{cav})} \approx \frac{8\mathcal{F}\sqrt{2\hbar\omega_L P}}{ML\omega^2 c^2}. \quad (5.65)$$

Further derivations could introduce the effect of recycling cavities. However, here we already see that higher laser power increases the radiation pressure strain and that it dominates for low-frequency gravitational waves. This is the opposite of the shot strain, and thus, there is a trade-off between the two. For the interferometers we have introduced, these two

24: this is discussed next to equation 5.46, showing $\mathcal{F} \approx \pi/(1 - r_1)$.

terms lead to the final quantum strain sensitivity

$$\mathcal{S}_Q^2 = \mathcal{S}_{\text{shot}}^2 + \mathcal{S}_{\text{rad}}^2, \quad (5.66)$$

in which the terms are assumed independent. This changes if we introduce squeezed²⁵ light [208], used by the LIGO collaboration since 2019 [209]. As stated by the Heisenberg principle, the uncertainties of phase and amplitude are inversely correlated, and squeezed light allows us to tune their relation. One can reduce the amplitude uncertainty, and hence the radiation pressure noise, while increasing the phase uncertainty, which raises the shot noise. For frequency independent squeezing, there is no benefit. The reduction of one noise is canceled by the increase in the other. On the contrary, by adapting the squeezing to the gravitational wave frequencies, we can reduce the dominant noise: the shot at high, and the radiation pressure at lower ones. This trick leads to correlations between the noises that the quadrature sum neglects. Nonetheless, equation 5.66 suffices for the detectors described in this and the next chapter, as squeezed light is not used.

25: introduced in [Chapter 2](#).

Finally, we arrive at the main classical noise sources: seismic and thermal noise. In terrestrial detectors, seismic noise dominates at low frequencies [210], being referred to as the seismic "wall" [211]. The noise arises from the continuous movement of Earth's ground, which the detector must be isolated from. This is achieved by hanging the mirrors with systems of multiple pendulums [212, 213]. While engineering advances could reduce this noise arbitrarily, a more radical solution is planned: move the detector to the space [214]. Ironically, the suspension system is partially responsible for the thermal noise, our second classical menace. The pendulum system presents Brownian motion [215] that propagates to the mirrors, which are also affected by their own intrinsic thermal noise [216]. Cooling systems and material science are crucial for reducing these fluctuations, but optimizing the light beam shape can also reduce thermal noise [217]. As with seismic noise, thermal strain sensitivity is higher at low frequencies.

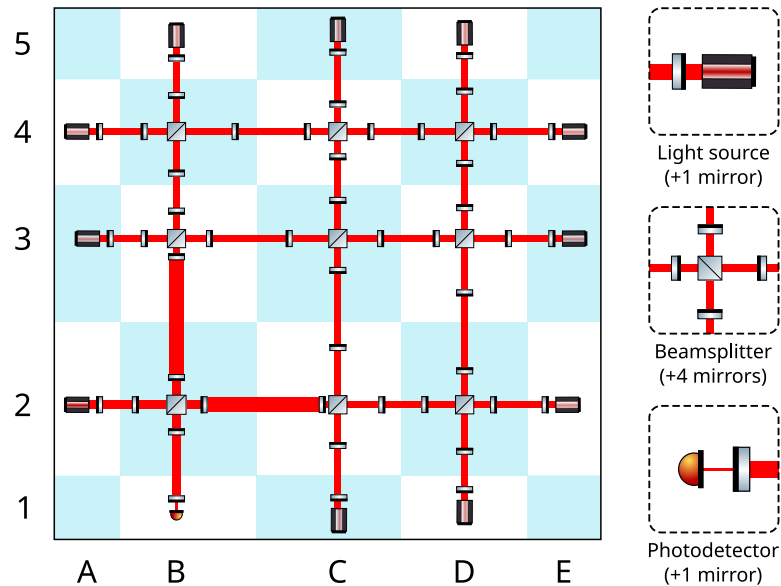
To conclude this brief review of the noise budget, is worth to mention the different impact that the quantum and classic noise have in the interferometer designing process. In the quasi-universal interferometer, our ansatz of gravitational wave detectors, the tunable parameters are the mirrors' and light properties, and the distribution of the optical components. By optimizing these elements, we reduce quantum strain sensitivity, our main target. In contrast, there is not much to do against classical noise. Seismic and thermal noise pose a tremendous engineering challenge, tackled by the combined efforts of several disciplines which keep improving the hardware. We model the noises based on their efforts, assuming the best technology available. The seismic noise is modeled as a fixed wall, and for thermal noise we focus in ensuring that the optical power remains within the limits the mirrors and the detector can withstand. We do not consider beam optimization.

5.2.3 The quasi-universal interferometer

The use of optical cavities showed multiple benefits. At the interferometer arms, they elongated the optical path, increasing the phase difference induced by the gravitational waves. On top, the recycling cavities raised the effective power, and even tuned the response to the wave signal. Pushing forward with the motto *mirrors everywhere*, our ansatz for interferometric

detectors consists on a set of beamsplitters distributed in a rectangular grid, with mirrors in between: the *quasi-universal interferometer* or UIFO. Surrounding this grid, we find an outer layer of mirrors, and then, at the end of each optical path, a light source pointing towards the grid. One of the light sources is replaced by a detector.

Figure 5.5: a quasi-universal interferometer. 3×3 UIFO with aligned beamsplitters and laser light sources. The thickness of the beams is a rough indicator of the power contained within different cavities. For the designs introduced so far, a 1×1 grid with a single beamsplitter suffices.



Looking at Fig. 5.5, we see how within this template, for the right set of parameters, one finds the initial Michelson interferometer and the upgrades from Section 5.2.1. Such parameters include the reflectivity of mirrors and beamsplitters, the distances between²⁶ them, and the power of each laser. Each combination of parameters—including discrete choices like the alignment of beamsplitters and detector location—leads to a quantum noise that we minimize across a certain frequency range.

Powered by the Finesse simulator [206], the UIFO template provided high-quality designs for different frequency ranges [12]. However, the computations were expensive, and as the designs are extremely sensitive, the search took millions of CPU-hours. The goal of next chapter is to accelerate the search by reducing the use of the Finesse, in favor of a neural surrogate trained with the data of different interferometers. Our goal is not to replace the original simulator completely, but to learn a simplified physics representation which can guide us effectively through the challenging landscape of UIFO configurations.

26: while beamsplitters must keep the rectangular grid, the mirrors between them can be located rather freely.

Neural surrogates for designing gravitational wave detectors

6.

This chapter adapts the homonymous paper [23].

As a continuation of the quasi-universal interferometer [12], Mario Krenn, Rana X Adhikari, and Yehonathan Drori considered surrogating the Finesse simulator to accelerate the search for new gravitational wave detectors. Carlos Ruiz Gonzalez took over those attempts, but the highly sensitive designs were hard to surrogate. Guided by Johannes Brandstetter, Sebastian Lehner, and Arturs Berzins, of the Machine Learning Institute at Johannes Kepler University, we understood the limitations of previous approaches and surpassed them. Helped by these ML experts and the advice of Sören Arlt, Carlos Ruiz Gonzalez implemented neural models able to surrogate the highly sensitive setups. The rest of the optimization algorithm combined the previously developed Urania with some new insights from Sebastian Lehner. Yehonathan Drori offered his expertise in gravitational wave detection to ensure the feasibility of the designs. As with the programming, the paper writing and the posterior review process were driven by Carlos Ruiz Gonzalez, with valuable advice and corrections from the other authors.

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Abstract

Physics simulators are essential in science and engineering, enabling the analysis, control, and design of complex systems. In experimental sciences, they are increasingly used to automate experimental design, often via combinatorial search and optimization. However, as the setups grow more complex, the computational cost of traditional, CPU-based simulators becomes a major limitation. Here, we show how neural surrogate models can significantly reduce reliance on such slow simulators while preserving accuracy. Taking the design of interferometric gravitational wave detectors as a representative example, we train a neural network to surrogate the gravitational wave physics simulator Finesse, which was developed by the LIGO community. Despite that small changes in physical parameters can change the output by orders of magnitudes, the model rapidly predicts the quality and feasibility of candidate designs, allowing an efficient exploration of large design spaces. Our algorithm loops between training the surrogate, inverse designing new experiments, and verifying their properties with the slow simulator for further training. Assisted by auto-differentiation and GPU parallelism, our method proposes high-quality experiments much faster than direct optimization. Within hours, our algorithm's findings outperform designs that the original optimizer takes five days to reach. Beyond detecting gravitational waves, our surrogate approach can be adapted to other interferometric systems. Moreover, our active learning approach is broadly applicable to other domains where simulator bottlenecks hinder optimization and discovery.

6.1 Introduction

Computer simulations are ubiquitous in all branches of science and engineering, providing predictions and a deeper understanding of complex interacting systems [218–220]. In experimental sciences, they are also used to optimize, control, and even design experimental setups. This goes beyond tuning a few parameters; novel algorithms have (partially) automated the conception of experiments, leading to new and often counterintuitive designs [1]. In physics alone, there are successful examples in quantum physics [5, 6, 78, 79, 83, 221, 222], particle physics [223–225], nano-optics [3, 226, 227], microscopy [228, 229] and gravitational wave physics [12], to name a few.

The inverse design of experiments is extremely computationally intensive [230], especially when combinatorial search is required. Therefore, the simulator speed is paramount to navigate the vast space of experimental designs. Fortunately, these high computational demands are mitigated by hardware improvements, such as GPUs [231, 232], and high-performance tools, like auto-differentiation or just-in-time compilation. These advances can also benefit slower CPU-based software, which can be surrogated with neural networks [233–240].

In this work, we leverage fast neural surrogate models to design interferometric gravitational wave detectors (GWDs). These sensitive experiments, able to detect distant cosmological events, can be simulated with the CPU-based software Finesse [206]. This open-source simulator was developed by the LIGO collaboration, the first to detect gravitational waves in 2015 [194], and is the de facto standard for GWD design. In recent work [12], Finesse was used to automate the search for new interferometric detectors by optimizing a highly parametrized experimental ansatz [12]. However, as the Finesse simulator is neither differentiable nor GPU accelerated, gradient-based optimization becomes increasingly inefficient when the system grows.

To accelerate the design process, we present SPROUT, short of *Surrogate Predictions for Reiterative Optimization with Update Training*. Our multi-step approach trains a surrogate model to predict the quality and feasibility of different GWDs, encoded as the parameters of a UIFO. First, we train the model with randomly parametrized designs to then generate new data that, once verified, is used for further training. Iterating the design and training (see Fig. 6.2), we produce successive generations of models that efficiently explore the design space and improve the solutions. This iterative optimization process, known as online optimization or active learning [241, 242], is a successful strategy when simulations (or experiments) are expensive [243–246].

Our approach does not replace the original simulator, we still use it to verify the designs generated by the faster GPU-based differentiable surrogates. However, by rapidly exploring the space of possible designs, even if it is an approximation, we reduce the overall usage of the CPU software and still obtain high-quality designs. Further, since the verifications of the GPU-based optimization are independent, we can parallelize the simulator calls arbitrarily, beyond what is possible with sequential gradient-based optimization. Finally, as the size of the GWD increases, the growing number of tunable parameters and longer simulation times make numerical gradient approximations prohibitively slow. Neural surrogates excel in such a scenario, as backpropagation and the vast parallelization capabilities of GPUs allow us to optimize thousands of experiments at breakneck speed. We obtain high-quality designs in a

fraction of the time, and, remarkably, even when we train the models with randomly parametrized designs, they find solutions far beyond anything we trained them with

Beyond gravitational wave detection, our models can be readily adapted to other interferometric systems, from nanometrology to optical atomic clocks [247, 248]. The goal just switches from strain to phase sensitivity, and local constraints to prevent optical damage remain relevant. More fundamentally, surrogating non-differentiable simulators via active learning provides a transferable methodology for general experiment design, enabling efficient navigation of high-dimensional parameter spaces in any domain where complex, modular systems are bottlenecked by traditional CPU simulation.

6.1.1 Related works

Surrogates for inverse design. When the computational cost of traditional simulators becomes unbearable, (neural) surrogate models are efficient, often differentiable, alternatives [249]. They are widely used for inverse design tasks in multiple fields, including engineering [250–252], chemistry [2, 4, 253–255], or physics [239, 240, 256–258], being particularly popular in photonics [233–236, 259, 260]. Most of these surrogates predict one or a few values that summarize the designs’ quality. In contrast, we predict an array of properties that indicate the quality as well as the hardware requirements to build the design. These predicted properties are incredibly responsive to the input parameters, and can fluctuate up to 10 orders of magnitude when tuning a single experimental parameter (see Fig. 6.1). We did not find such sensitive systems in the surrogate literature and took inspiration from other domains [261] to face the challenge (details on Section 6.2).

Neural networks in physics. Recent advances in machine learning are especially aimed at the physical sciences. Physics-informed neural networks (PINN) [262, 263], or neural operators [264, 265], are notable examples, widely used as PDE surrogates in fluid dynamics [266, 267], among other domains [268, 269]. However, these novel approaches are not suited to our task: predicting the experiment’s sensitivity and the light powers applied to the elements, based on the design parameters. For the PINNs, we lack simple physical laws to constrain the outputs. On the other hand, neural operators surrogate functions which can be evaluated at any discretization of a continuous input space. Here we have a finite set of input properties, from a finite number of optical parameters, and a fixed number of output properties. Thus, these tools offer us no advantage.

Pool-based active learning. Supervised learning requires costly labeled data. Thus, active learning strategies must carefully select which data to label, from a pool of unlabeled data points [241]. There are theoretical results on the strengths and limitations of several approaches [270–273], but these assumptions rarely hold when training deep neural networks. Therefore, the field relies mostly on heuristics and empirical evidence for particular applications [274–277]. Bayesian methods are also popular for small systems [278–281], but they do not scale well. In this work, we verify the designs obtained after a noisy optimization process of the surrogate simulator, pushing the model to generalize beyond its training dataset. We repeat this procedure for 5 rounds (see Fig. 6.2a). Our strategy resembles previous work on active learning [282, 283] and on adversarial training [284], where gradient-based optimization

generates adversarial examples that are mislabeled by neural [285, 286]. Retraining the networks with such examples increases their robustness and improves their predictions in high-quality design spaces.

6.2 Methods

This section describes how to design GWD using the *quasi-Universal InterFerOmeter* (UIFO), a highly overparameterized experimental setup introduced in previous work [12], which is used as a highly expressive ansatz. We also outline the surrogate models' architecture and training strategy, the building blocks of the SPROUT algorithm.

6.2.1 Gravitational wave detectors

Gravitational waves are perturbations of the spacetime caused by the motion of masses. They were predicted by Einstein's general relativity in 1916 [287] and directly observed for the first time in 2015 by the LIGO collaboration [194]. Feeble as they are, we can only detect the waves originating from massively energetic astrophysical phenomena, such as supernovae or the merging of black holes. To do so, experimentalists use some of the most sensitive devices ever created by humans – gravitational wave detectors – massive facilities whose components require extremely precise tuning. Yet, as shown in previous work [12], the space of GWD experiments is mostly unexplored. Many possible designs, including the LIGO detector, can be described within the UIFO framework, a rectangular lattice of parametrized optical elements. Using an expressive ansatz is a powerful tool to design new experiments, also in other fields [6, 229, 288], as it relaxes the optimization problem from (partially) discrete to purely continuous. The price we pay is to optimize larger systems of tunable elements, likely redundant, which contain previously known baselines and new designs able to surpass them.

As shown in Fig. 6.1, for a given choice of UIFO parameters, the detector will have a certain strain sensitivity for each possible frequency of a gravitational wave, which we refer to as *sensitivity curve*. The lower the sensitivity, the smaller space fluctuations it can detect. Additionally, it is necessary to limit the optical power in each component, otherwise their performance degrades. The same applies to the photodetector, which is orders of magnitude more sensitive. These and other constraints are summarized in Table 6.1.

The sensitivity and power values define the quality and viability of the designs and can be computed with Finesse, the CPU-based simulator we want to surrogate. As shown in Fig. 6.1iii), the quality of the interferometer can fluctuate by multiple orders of magnitude when tuning even a single parameter. Furthermore, some of the plotted sensitivity curves could not be implemented with current hardware, as the light power at several components surpasses the aforementioned limitations.

In this project, we focus on the continuous optimization of the parametrized ansatz UIFO and how the benefit of the surrogates increases with the size of the GWD. Therefore, to facilitate comparison between designs of different sizes, we fix all discrete design variables: the grid size, the beam splitters' orientations, and the photodetector's location. The reduced size of our setups also facilitates the experiments. For more practical applications, we would emulate larger designs with

different configurations that include even more sensitive designs than the LIGO detector. Details on the UIFO parametrization in Appendix C.1.

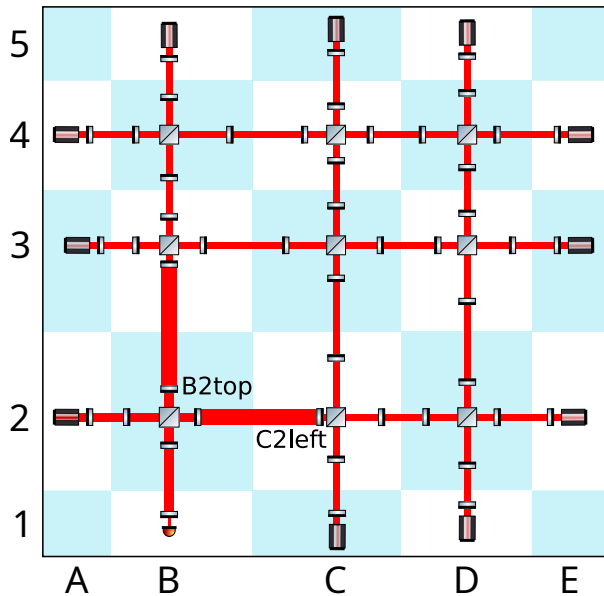
6.2.2 Design of experiments

Whether we simulate the UIFO with Finesse or rely on a neural surrogate, the strategy to design novel GWD is the same. Having fixed the discrete design choices, we optimize the UIFO's continuous parameters, our template for GWD, to minimize the geometric mean of the strain sensitivity.

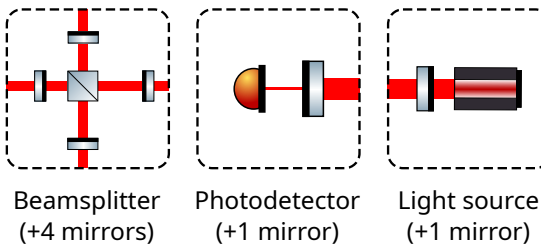
Furthermore, to guarantee the feasibility of the designs, they must meet the hardware constraints described in Table 6.1, which can be divided into two types. First, we have the maximum powers that the detectors, mirrors, and beam splitters can withstand. These are output constraints that are calculated by the simulator. We enforce them by adding penalties to the loss, using the differentiable function

$$\text{Penalty} \equiv (\alpha + \beta(x - x_{\text{th}}))\sigma(\gamma(x - x_{\text{th}})), \quad (6.1)$$

i) UIFO gravitational wave detector



ii) Cell Types



iii) Highly responsive solutions

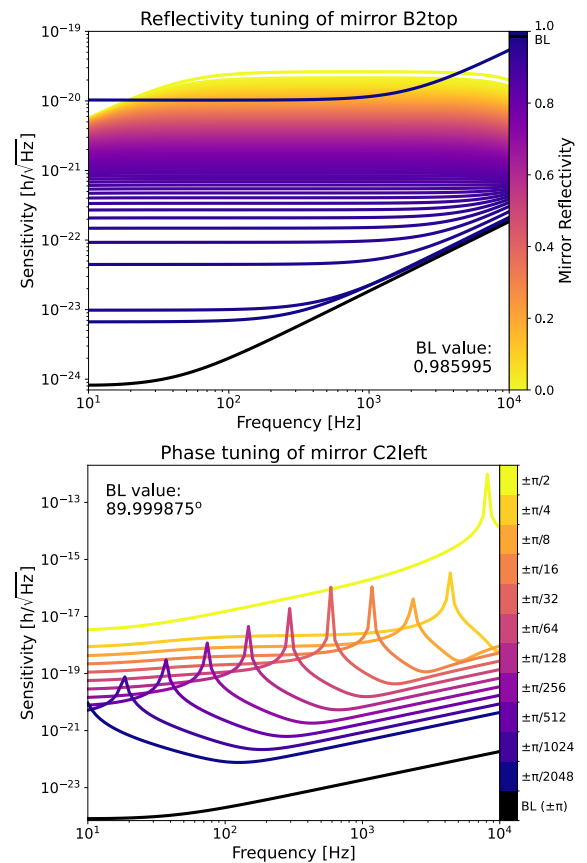


Figure 6.1: A highly expressive ansatz: quasi-Universal InterFerOmeter (UIFO). A large variety of gravitational wave detectors can be built by aligning optical elements in an irregular rectangular grid. This expressive parametrized template is referred to as quasi-Universal InterFerOmeter, or UIFO. i) The 3x3 UIFO, with 169 parameters, has 9 beamsplitter cells surrounded by light source cells and a single photodetector. By tuning the parameters of a small subset of the grid's components, we can recreate a simplified version of the GWD used by the LIGO collaboration. ii) In the depicted design and throughout this paper, we use beamsplitters, mirrors, photodetectors, and lasers. iii) The quality of a GWD is given by the smallest spatial deformation that it can detect for each gravitational wave frequency, represented by a sensitivity curve. As shown in the top plot, these curves can fluctuate in several orders of magnitude in response to a single parameter, like the reflectivity of the top mirror in cell B2. The change is even more drastic when tuning the left mirror in cell C2.

Table 6.1: ranges of the physical parameters of the UIFO designs. The loss/-transmission of the optical elements can be bounded before running the simulations. Also the optical path length, which ensures that the design remains within the 4km that LIGO design measures. The powers at the optical elements, computed by the simulator, are enforced via penalties. The lower bound for the optical loss, 5ppm, is a rough estimate of what is possible with current technology. We assume the lowest possible loss for all components, as larger values worsen the performance. Details in Appendix C.1.

Parameter	Min	Max
Optical path length	1m	4km
Optical loss per element	5ppm	
Optical transmission	15ppm	
Reflected optical power		3.5MW
Transmitted optical power		2kW
Power in photodetector		10mW

which approximates a step function with a step size α and a final slope of β , for $x > x_{\text{th}}$. The stiffness of the step is indicated with γ , σ is a sigmoid function, x is the (log) optical power applied to a given optical element, and x_{th} is the maximum power that the element can tolerate. The values of x_{th} depend on the optical element and whether the power is reflected or transmitted (see Table 6.1). We set the values of $\{\alpha, \beta, \gamma\}$ to $\{10, 1, 1000\}$.

In contrast, the constraints on element properties are enforced by restricting the input parameter space, for example, by limiting the reflectivity and losses of the mirrors and beam splitters. To guarantee spatial consistency, the distances between rows and columns of beamsplitters are described as fractions of a total larger distance, which must remain below 4 km. Finally, the positions of the different mirrors are also described by fractions (see Appendix C.1).

The final loss is the mean of the log sensitivity for N frequency values f_i (i.e. the log of the geometric mean) plus the described penalties,

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \log(S(f_i)) + \text{Penalties}. \quad (6.2)$$

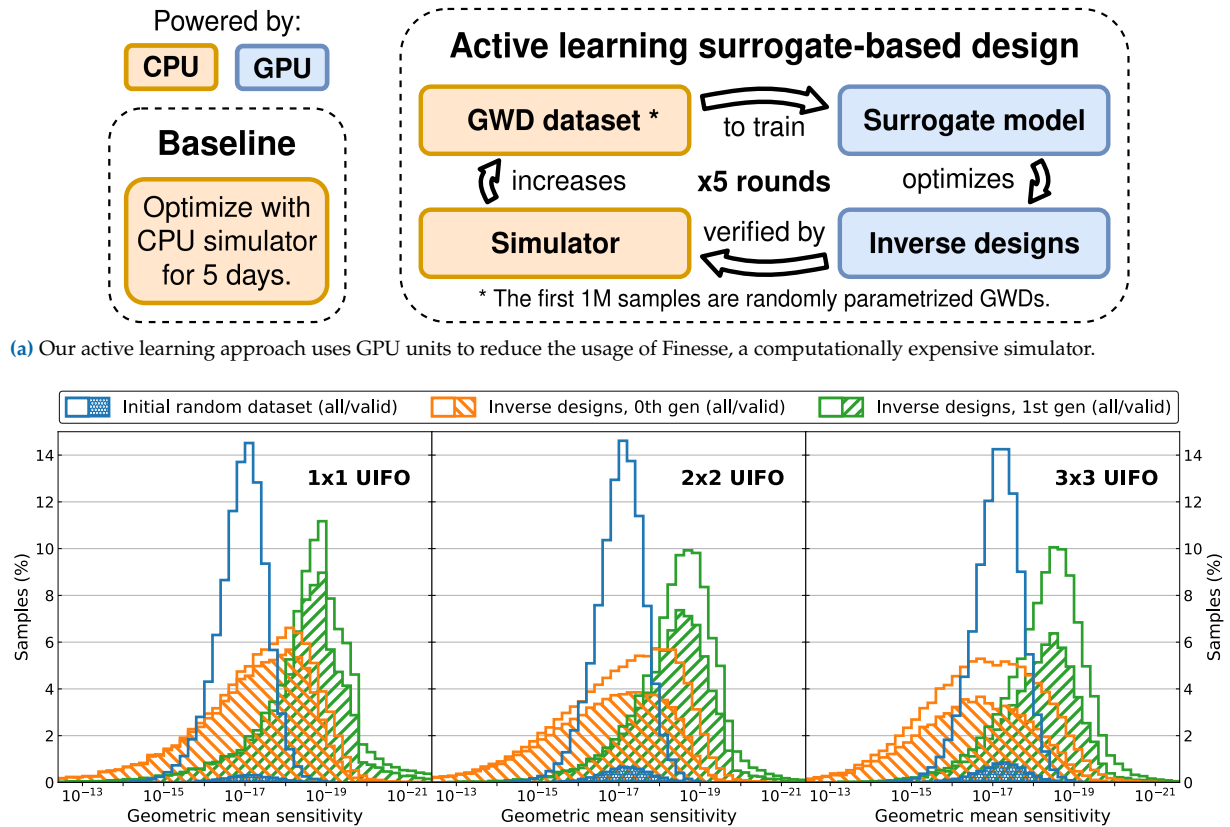
The set of frequencies are 101 logarithmically spaced values between 1Hz and 10kHz. To minimize the loss, we use the Adam optimizer [289] with learning rate of $\lambda = 0.1$. Moreover, we add annealed Gaussian noise, $\mathcal{N}(0, 2\epsilon^t \lambda)$, to the input update, a well-known strategy to optimize highly non-convex functions [290–292]. With $\epsilon < 1$, the noise decreases exponentially at every step t .

6.2.3 Surrogate models

In order to find new GWD designs, our neural networks must be able to simulate them, at least approximately. We want our surrogates to predict, based on the input parameters of the UIFO, the final sensitivity curve of the designs (their quality) and the light powers that are applied to each optical component (their feasibility). To process the grid-like structure of the UIFO template, we divide the interferometer into overlapping subgraphs, each with a different set of parametrized elements, and turn them into patches, which are then processed by a transformer encoder and mapped into the output space. Our architecture is heavily inspired by models such as Vision Transformers [293], MLP-Mixers [294], and other works that structure the input data into patches/tokens [295]. It has three main parts:

1. the patch extraction from the GWD and their embedding into tokens,
2. the processing of the tokens with a standard transformer encoder of 8 layers, and
3. the decoding of the processed tokens into the final output of sensitivities and powers.

Turning GWD into patches. To transform a GWD into a set of patches, we treat the UIFO grid as an attributed graph, i.e. a graph whose nodes and edges have several weights. The nodes are the beam splitters, the light sources, and the detectors, each with a different set of associated parameters. Between the nodes, we always find two mirrors, whose properties and relative position in the optical paths are the attributes of the edges that connect the nodes. Grouping the parameters of an edge and the two adjacent nodes, we define a patch. Since the nodes have different numbers of attributes, we pad the patches with zeroes up to length 12. As shown in Fig. 6.3, the patches overlap in the beam splitters, as their respective nodes have degree 4.



(b) The designs' quality (i.e. low sensitivity) surpasses the model's training data. Only valid samples meet hardware constraints.

Experiment size	1x1 (29 parameters)		2x2 (85 parameters)		3x3 (169 parameters)	
Design strategy	Simulator	Surrogate	Simulator	Surrogate	Simulator	Surrogate
Processor unit	CPU only	CPU+GPU	CPU only	CPU+GPU	CPU only	CPU+GPU
Time until $1e-20$	1h10'	14'+40'	6h05'	41'+1h09'	32h15'	1h23'+2h01'
Simulation calls to reach $1e-20$	5.5M	2.2M	9.6M	2.2M	25.6M	2.2M

(c) For the same quality, the surrogate-based design (+ random sampling) takes much less time than only using the simulator.

Figure 6.2: Iterative inverse design with surrogate models.

To the padded patches, we add a positional embedding (PE) and apply the trainable Fourier feature map, that is, a linear layer followed by a sine activation [261]. This mapping helps to overcome the spectral bias of neural networks towards low-frequency functions [296, 297], which can limit the capacity of our models to predict the properties of high-quality outliers. The projection of the Fourier map increases the embedding dimension of the patches to 192. It is followed by a linear transformation.

Transformer architecture. After adding a positional embedding to the tokens, we pass them to the transformer encoder [298]. We use eight standard transformer layers with the pre-layer normalization convention [299], and a multilayer perceptron (MLP) with hidden dimension 384, twice the size of the tokens. We chose the transformer architecture, as it has shown great success processing multiple types of data for a wide variety of tasks [300–308].

Towards the final output. After the transformer encoder, we apply a layer norm and reduce the tokens' size with a linear projection, followed by a GELU. Concatenating the tokens, we obtain a final vector that, after a norm layer and an MLP, takes the shape of the output. We enforce normalization by applying a sigmoid to the output: the array of sensitivities and the light powers.

A summary of the architecture is shown in Fig. 6.3. The implementation details are described in Appendix C.3, and all the code can be found in the repository¹.

The training datasets consist of pairs of input parameters and output sensitivities and powers. The input parameters are linearly normalized between 0 and 1, adjusting the rescaling to the different types of physical parameters. On the other hand, as the values of sensitivities and optical powers range across several orders of magnitude, we take their logarithm value for training. Details on data preprocessing in Appendix C.2.

6.2.4 Active learning strategy

To train the models, we use the Adam optimizer with weight decay and a plateau learning rate schedule. In the first training phase, the model learns from 1 million randomly parametrized designs.² With the trained models we run the Adam optimization process described in section 6.2.2, adding annealed noise with $\epsilon = 0.999$. Running 200,000 optimization trajectories for 16,384 steps on multiple GPU nodes, we store 5 designs for each trajectory³ producing one million new training samples each round (+10% for testing). After verifying the designs – in parallel – with the CPU-simulator Finesse, we use the old and the new samples, 2 million in total, to retrain the model. The total amount of training samples increases by 1 million at every training round, which are added to the training datasets in groups of 50K to stabilize training [309]. The neural networks are initialized with the weights of the previous training phase.

It is noteworthy that the quality of the first training dataset, randomly produced, is always very low. Less than 5% are valid, as most of them apply excessive light power in at least one optical component, usually the fragile detector. Yet, the first trained model produces valid designs much more sensitive than any sample it was trained with (see Fig. 6.2). That happens for all system sizes. After further training, the model produces increasingly better samples by learning from its previous designs, a virtuous cycle of self-improvement (see Fig. 6.4).

1: <https://github.com/artificial-scientist-lab/SPROUT>

2: Additional 10% are produced for testing.

3: We store the designs at (approximately) logarithmically spaced steps: 6, 48, 337, 2352, and 16384.

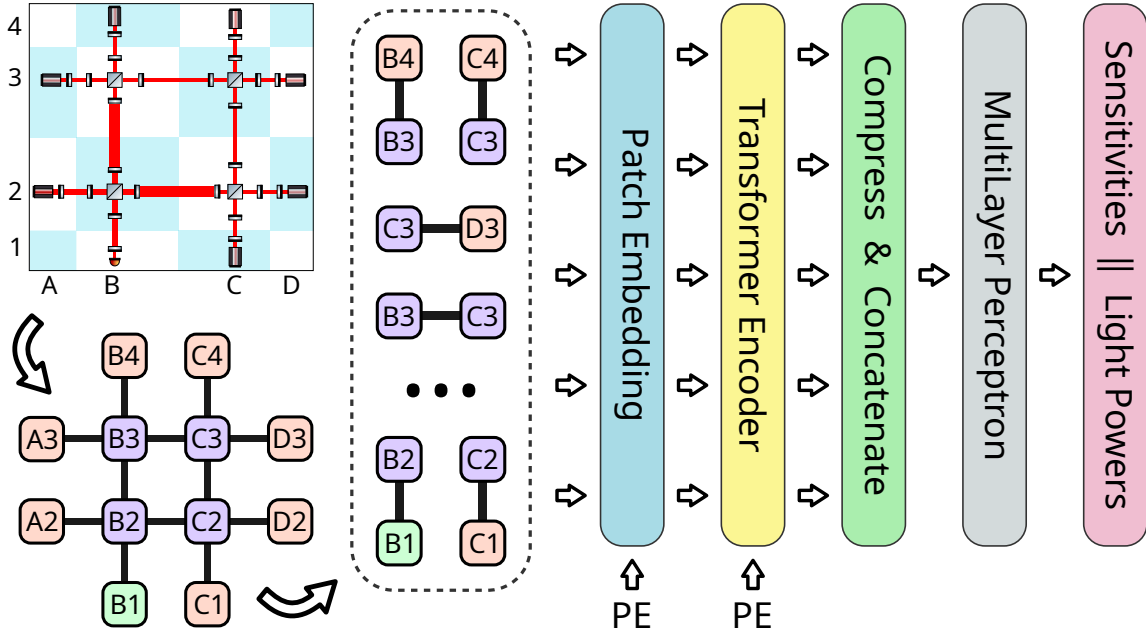


Figure 6.3: Overview of the surrogate model's architecture. After extracting the patches from the grid-based GWD, we transform them into (larger) token representations, which are then processed by a standard transformer encoder. To turn the tokens into the final output, we compress them with a learned projection and concatenate them. The resulting array, larger than the output by design, goes through a final MLP and, after a sigmoid, is turned into the output: the values of the sensitivity curve concatenated with the light powers at each optical element.

6.3 Experiments

To benchmark our surrogate-based algorithm, SPROUT, we compare it with the previous baseline: direct optimization with Finesse, the CPU-based simulator. Table 6.2 summarizes the most relevant feats of both approaches, while Fig. 6.4 shows the evolution of the surrogates' designs across several iterations of (re-) training and designing, comparing them with the CPU-powered optimization process. We also detail the resources employed for both approaches.

6.3.1 CPU-based optimization baseline

The original approach we aim to surpass is gradient-based optimization with the CPU-simulator Finesse. Therefore, we optimize 256 randomly parameterized UIFO samples with the Adam algorithm, adding annealed noise of $\epsilon = 0.999$, as with the surrogates. The main difference from surrogate optimization is the need to compute the gradients with finite differences, a much slower procedure than automatic differentiation. We ran the optimization for 5 days, and only the smallest UIFO with 29 parameters reached more than 10000 optimization steps.

The sensitivities of the final designs, optimized with Finesse, are shown in Fig. 6.4 for the 3 UIFO sizes along with the surrogate models' designs. The evolution of the CPU-based optimization over the 5 days is shown in Fig. C.3.

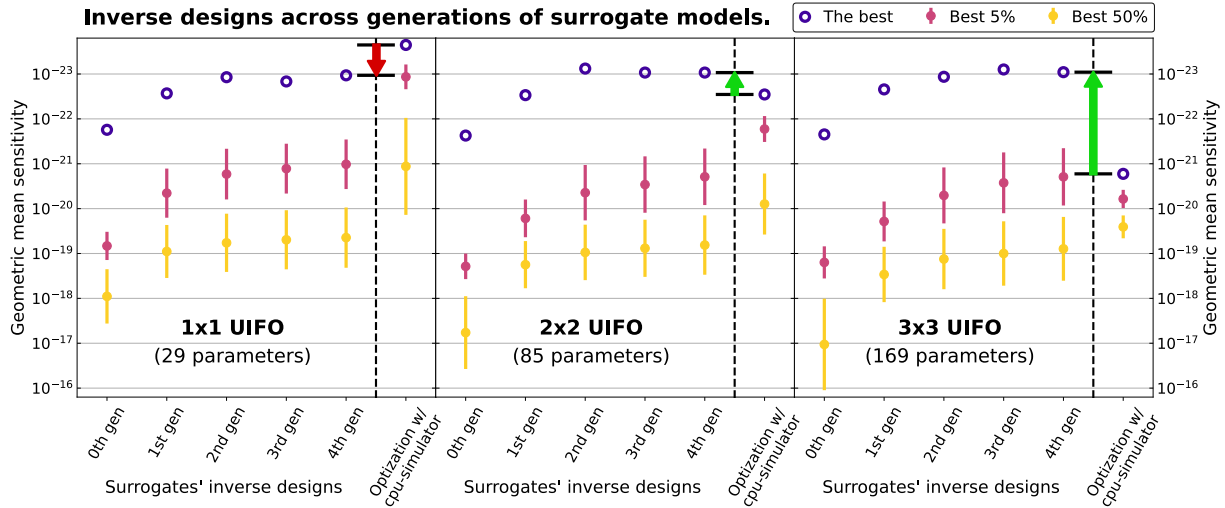


Figure 6.4: Evolution of the inverse designs’ quality after (re)training the surrogate model and comparison with direct optimization. After training the surrogate model with random data we use it to produce hundreds of thousands of novel solutions via gradient-based optimization. Once the designs are verified, they are added to the previous training dataset, to finetune the model. At every generation of finetuned models the training data adds 1 million of new samples. The plotted sensitivities, which improve after every generation, come from the last step of the optimization process of the 200K inverse-designed samples. The optimizations with Finesse, detailed in Fig. C.3, were limited to 256 samples per grid size and ran for 5 days. The benefits of the surrogate models are specially prominent in the larger systems.

6.3.2 The SPROUT algorithm

SPROUT overcomes the limitations of the CPU simulator by delegating part of the task to powerful and parallelizable GPU nodes. At each round, we train the model on a GPU node from *AMD Instinct MI300A APU*, and for inverse design, we use 11 such nodes. We optimize the designs in batches of 20000, for all UIFO sizes, minimizing the loss from equation 6.2, as shown in Section 6.2.4.

Storing five optimized samples for every optimization trajectory, we have 1 million of new input samples for training (+10% for testing). To verify them, we run Finesse in parallel jobs across 256 CPU cores from several *Xeon Gold 6130*. That is the same hardware used for the optimization with Finesse described in Section 6.3.1.

To ensure the reliability of our approach, we applied SPROUT twice for each size of the UIFO. Therefore, the models are initially trained with the same random dataset but, due to the different random seed, they present minor differences in the training process that also affect the design (see Appendix C.4). Nonetheless, the overall quality is usually very similar. The inverse designs shown in Fig. 6.4 are the aggregated data of both experiments, while the times in Table 6.2 are averaged from the two processes.

6.3.3 Results

As shown in Table 6.2, the advantage of the surrogate models for inverse design increases with the size of the experiment. For the largest UIFO, the surrogate model designs improve the sensitivities by more than an order of magnitude. The increasingly better designs presented in Fig. 6.4 show how the models are able to generalize beyond the data they are trained on. This is particularly noticeable at the first iterations (see Fig. 6.2b).

The key to the surrogates' performance is to utilize the GPU and the differentiability of the neural nets, enabling the simulation of many more experiments in a fraction of the time required by the CPU-based simulator Finesse. To give a loose comparison, for the larger 3x3 UIFO, an optimization step takes less than 0.3 seconds on a *AMD Instinct MI300A APU* GPU core for 20,000 samples. A single evaluation with Finesse takes around 1.2 seconds on a *Xeon Gold 6130* CPU node. Needing 169 additional simulator calls to compute the gradient, every optimization step can take more than 3 minutes. Therefore, as the size of the optimized GWD grows, so do the benefits of the surrogate models.

Table 6.2: Time, quality, and resource comparison of the 2 design strategies for 3 sizes of UIFO. Our surrogate-based approach drastically reduced the simulator calls to achieve the same design quality. The CPU-powered optimization ran for 5 days (fixed) in 8 units of *Xeon Gold 6130*, with 32 CPU cores each. With the same CPU resources, the verification of the 6.6M samples (5 rounds of inverse design + starting random dataset) takes only a few hours. The additional GPU time comes from the training time, which increased at every round with the size of the dataset (see Table C.2), together with the inverse design, which took (approximately) 16, 40, and 85 minutes, for the 3 respective sizes. The inverse design was distributed across 11 GPU cores from multiple *AMD Instinct MI300A APU* units, each optimizing 20K samples in parallel. Details on the computational resources in Appendix C.4.

UIFO grid size	1x1 (29 parameters)		2x2 (85 parameters)		3x3 (169 parameters)	
Design tool	Simulator	Surrogate	Simulator	Surrogate	Simulator	Surrogate
Simulator calls	560.6M	6.6M	189.3M	6.6M	95.2M	6.6M
Final samples	256	1.1M	256	1.1M	256	1.1M
Steps / sample	73001	16384	8600	16384	2189	16384
Best sensitivity	<u>2.25e-24</u>	1.07e-23	2.85e-21	<u>7.56e-24</u>	1.68e-21	<u>7.92e-24</u>
Time spent	CPU	CPU+ GPU	CPU	CPU+ GPU	CPU	CPU+ GPU
Until 1e-20	1h10'	14'+40'	6h05'	41'+1h09'	32h15'	1h23'+2h01'
Until 1e-22	5h09'	28'+1h44'	90h13'	1h23'+2h56'	Never	2h46'+4h37'
Total	5 days	1h24'+7h36'	5 days	4h10'+11h27'	5 days	8h18'+18h6'

6.4 Discussion

We introduced SPROUT, a surrogate-based approach to accelerate the inverse design of gravitational wave detectors. Our algorithm navigates the space of experimental designs by iteratively training neural surrogates, optimizing designs, and selectively verifying candidates with Finesse, the non-differentiable, CPU-based simulator that we are surrogating. Tested with a limited set of experimental tools, the surrogates exhibited outstanding generalization capabilities, as they designed experiments of much higher quality than those used for training. A surprising feat given the drastic fluctuations in the GWD's properties when modifying the input parameters.

By exploiting differentiability and GPU-based parallelization, the benefits of our neural network approach rise with the system size, as finite-difference methods become increasingly expensive. This approach transcends our immediate application, GW detection, as any multi-element interferometer could adopt SPROUT by generalizing the optimization target from strain to phase sensitivity, and still require high-power optical constraints. Furthermore, our methods can tackle bottlenecks in other fields. In nanophotonics, for example, our patch-based tokenization would map naturally to grid-like structures, while our use of Fourier features effectively could capture sharp spatial transitions. More broadly, the approach is domain-agnostic: applying it to a new problem requires no adaptation of the architecture or training procedure, only knowledge of the input and output data types of the target simulator. The framework offers a robust blueprint for sample-efficient exploration of complex,

modular design spaces, producing designs that reach well beyond the initial training distribution (Fig. 6.4).

With the positive results of our algorithm, there is still room for improvement. First, our inverse designs use a small set of optical elements and a few simple topologies. Constraining the task, we can compare the performance of different methods in a well-known arena, but we are limited to rediscovery and prone to stagnation. To reach SOTA detectors, we need other components and topologies. That leads us to the second point: our neural networks could be more flexible. We trained our neural networks on fixed discrete choices, but better models could pick different components, optimize the topology of experimental setups, or even predict the detector's sensitivity for arbitrary wave frequencies. To meet these ambitious goals, our architectures must map between different input and output spaces, and we need richer training data that encodes every possible design choice. These modifications might lead to general changes in the SPROUT algorithm, our third point. As the search space grows in dimensions and complexity, we might need more efficient training and exploration strategies. Some options include limiting the training time, intelligent sample weighting to favor specific designs' qualities, or implementing curriculum training with increasingly complex designs. Ensemble methods can foster exploration by aggregating predictions across diverse surrogate models, and diffusion-based design could be a powerful asset to explore different topologies.

To conclude, SPROUT is a promising tool for inverse design, whose principles and benefits apply beyond the search for GWD. Having showcased its potential in a constrained task, this work paves the way for further advances that lead to novel and more powerful experiments.

Part IV.

Conclusion

Conclusions and future directions

7.

7.1 Summary

7.1 Summary 85

7.2 Outlook 85

At the start of this document, we went through three stages of automated experimental design. We set up the ansätze, defined our goals, and discussed the importance of understanding our solutions. In these closing remarks, we will succinctly review and summarize the papers of this thesis through the lens of our automated design framework.

In the first project, the development of the Pytheus in [Chapter 3](#), the graph representation was expanded to novel applications. The nodes and edges were reinterpreted so that the graphs could represent new types of transformations and states. As the code was written in Python, we made the most of our cluster and found 100 novel designs. The purpose was to show a diverse range of experiments that could spark interest across multiple fields. Thus, we were less interested in extrapolating the solutions than in providing software that anyone could easily use. Even then, the graph interpretation was so intuitive that families of solutions emerged, as we could not overlook the patterns of some of the solutions gathered in [Appendix A](#).

Taking a step back towards the original path identity, [Chapter 4](#) explored terms omitted by the graph representation. Those are the emission events that postselection filters out, as well as the higher-order terms usually deemed negligible. From this broader view, in which the graphs were just a corner of the picture, we could extract new, often counterintuitive solutions that found better tradeoffs between state fidelity and generation probability. The abundance of interfering creation terms complicates computation, exploration, and understanding. Nonetheless, we outperformed several baseline designs, and the solutions we found lay on a high-dimensional curve that mapped onto the Pareto front. Instead of a zoo of disjoint solutions, small parameter variations could guide you across the best trade-offs, facilitating implementation.

Finally, in [Chapter 6](#), we presented neural surrogates that learned to model quasi-universal interferometers and navigate the challenging space of gravitational wave detectors. The training samples were obtained from the Finesse simulator using an active learning approach, in which the model's design suggestions were used to improve predictions. With occasional checks against the ground-truth simulations, we explored the space of UIFO designs at neck-breaking speed. We expect this work to be a guideline for circumventing the limitations of slow (sometimes legacy) simulators and exploiting the latest advances in parallelization.

7.2 Outlook

One of the main purposes of our ansätze was to avoid making discrete choices, thereby turning the search for new designs mostly into continuous optimization. However, as some discrete choices still had to be made, we better address them properly. There are many methods and heuristics for discrete optimization we could explore [[80](#), [224](#), [310](#)], and

recent advances in combinatorial optimization hold great promise for our ansätze [311, 312]. This is one of the clearest avenues for future research, even though, in the context of the Pytheus graph-based representation, it has been explored to some extent [40, 41].

Another active field of research we could benefit from, closely related to our surrogate models, is representation learning [313, 314]. The ability to learn the meaningful features of a physical system enabled our surrogates to effectively navigate the space interferometric detectors. But there is more to gain from a good representation. We may be able to extract insights from it [255, 315, 316], underlying patterns and mechanisms which facilitate further designs and discoveries. Moreover, these tools may help us to find families of homotopic designs [317, 318], i.e., different solutions connected by continuous changes¹ in their parameters, as those we found in Chapter 4. We could go a step further and ask for solutions not only to be homotopic, but also for the tunable parameters to meet certain properties. For example, in the case of interferometers, we rather tune the distance between mirrors than their reflectivity, as only the former can be changed in situ.

As the thesis approaches its end, the reader may wonder why we have not explored more of the aforementioned directions. While time constraints have played a role, it was often the case that imperfect methods could take us very far. Accordingly, we often preferred to tackle a new design task—find new ansätze, see which physics we should add...—than arbitrarily hone our search strategies. Therefore, this is the final, and perhaps more exciting continuation of this work: now that *you know my methods*, let's see where else we² can apply them.

1: we iterated several optimizations in which the loss function was gradually tuned, allowing us to turn a Pareto optimal solution into another via continuous parameter changes. This was inspired by adiabatic processes.

2: in my case, the author behind the *we*, I plan to do so in engineering design.

Part V.

Appendix

100 experiments from Chapter 3

A.

In the following, we introduce one hundred experiments conceived by PyTheus. A broad catalog of designs that, hopefully, will contribute to multiple branches of quantum optics, including quantum computing, communications, and quantum sensing. For each proposed experimental setup which is new and has not been described in any theoretical or experimental paper, we mark the corresponding graph with a number in a box counting from 1 to 100. The weights of each graph can be found in the GitHub repository, together with the instruction sets used to search them.

A.1 Generation of Entangled States

In this part, we propose ways to generate entangled states, which play an important role not only in our understanding of entanglement and the non-local nature of quantum mechanics but also in many quantum information applications.

GHZ States

Bell's theorem shows that Einstein-Podolsky-Rosen's propositions about local realism [319] are inconsistent when we apply them to quantum systems of two particles, revealed by the violation of the Bell inequality with quantum mechanical statistical correlations [25]. In the late 1980s, Greenberger, Horne, and Zeilinger took a step further from two to three particles. Interestingly, with a tripartite quantum system, local realism can be violated by quantum mechanics with perfect correlations rather than statistical correlation, and one can completely dispense with inequalities [320, 321]. It enables the performance of experiments where the *quantum physical predictions are mutually contradictory with expectations based on local realism* [322]. Such a GHZ state forms an important class of entanglement and its generalization is given as

$$|\text{GHZ}\rangle_n^d = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |i\rangle^{\otimes n}, \quad (\text{A.1})$$

where n denotes the number of particles and d is the dimension for each particle.

Increasing the number of particles and dimensions in the GHZ states not only enlarges the Hilbert space but also leads to many new exciting classically paradoxical phenomena [150, 323, 324]. Besides their fundamental interest, high-dimensional multiparticle GHZ states have also served as an important resource for many quantum information applications [325].

With the size and dimension increasing, it is very challenging to know how to experimentally create GHZ states. Experimental progress has been made in this direction to push the size and dimension, especially in the linear optics regime [32, 105, 106, 326]. Until now, only the three-particle three-dimensional GHZ state has been demonstrated with linear

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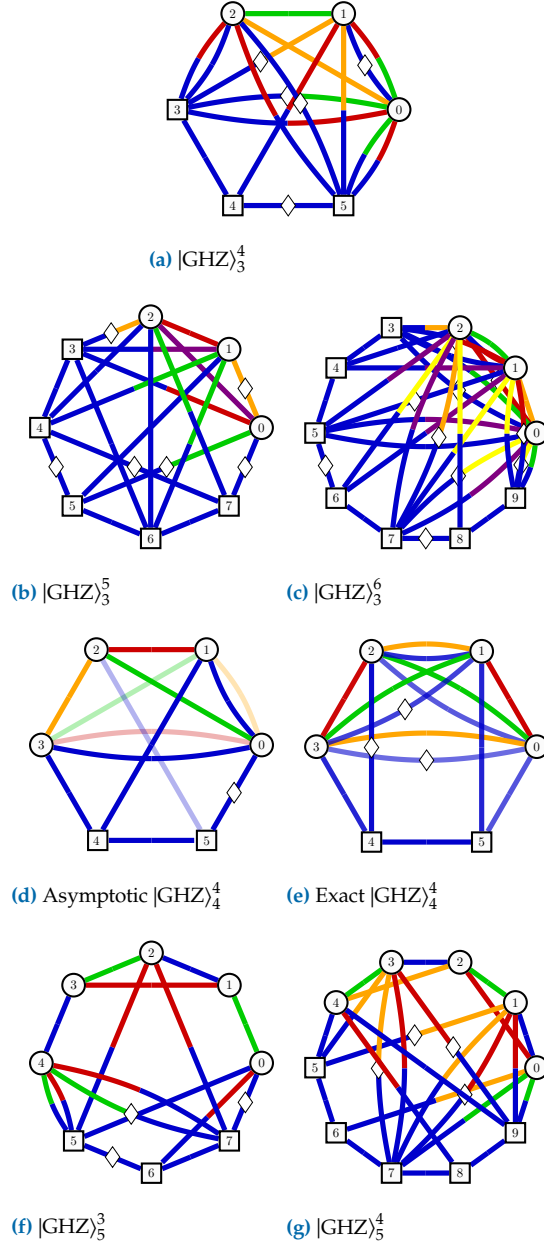


Figure A.1: High-dimensional multi-particle GHZ states with ancillas. The weights here in these graphs are real numbers and the diamonds inserted in the edges represent the negative sign. The vertices in square refer to the detection of an ancillary photon. For the state $|\text{GHZ}\rangle_4^4$ we have two possible solutions.

optics [32] and superconducting qutrits [327]. Going beyond the $|\text{GHZ}\rangle_3^3$ state with arbitrary particle numbers and higher dimensions will give rise to exciting new possible quantum applications and new insights on the foundations of quantum mechanics. This remains to be explored, in any experimental platform.

PyTheus enables the discovery of many new multi-particle high-dimensional GHZ states for which no experimental implementations are known yet. That includes the 3-particle GHZ state for 4, 5 and 6 dimensions [1] [2] [3], the 2 solutions (asymptotic and exact) for the 4-particle 4-dimension GHZ state [4] [5], and the 5-particle GHZ state for 3 and 4 dimensions [6] [7], as shown in Fig. A.1.

Bell Gems

Bell states offered the first proof of the non-locality of quantum mechanics. Later on, they became the cornerstone of many quantum communication schemes and are nowadays widely used in quantum computing. These maximally entangled bipartite states, described in the Hilbert space $\mathcal{H}_2 \otimes \mathcal{H}_2$, can be generalized for more particles and dimensions in multiple ways while keeping their main properties [328].

In 2004, Gregg Jaeger suggested a generalization for sets of 2^N qubits, the *Bell Gems* [329]. To construct these states we need to pick a pair of orthogonal states $|\alpha\rangle$ and $|\beta\rangle$ from the Bell basis: $|\Phi^\pm\rangle = |00\rangle \pm |11\rangle$ and $|\Psi^\pm\rangle = |01\rangle \pm |10\rangle$. Then, we iterate the following mappings in all possible ways:

$$\{|\alpha\rangle, |\beta\rangle\} \rightarrow (|\alpha\rangle|\alpha\rangle \pm |\beta\rangle|\beta\rangle) \quad (\text{A.2})$$

$$\{|\alpha\rangle, |\beta\rangle\} \rightarrow (|\alpha\rangle|\beta\rangle \pm |\beta\rangle|\alpha\rangle) \quad (\text{A.3})$$

Therefore, for 2^N qubits, we get a basis of 2^{2^N} states. For the case of a pair of 3 dimensional Bell states 84 states exist. Fig. A.2 shows how to generate one of these high-dimensional Bell gems which, up to normalization, reads [8]

$$|\text{GEM}\rangle = |\psi_a\rangle|\psi_a\rangle + |\psi_b\rangle|\psi_b\rangle + |\psi_c\rangle|\psi_c\rangle, \quad (\text{A.4})$$

where $|\psi_a\rangle = |00\rangle + |11\rangle + |22\rangle$, $|\psi_b\rangle = |01\rangle + |12\rangle + |20\rangle$, and $|\psi_c\rangle = |02\rangle + |10\rangle + |21\rangle$.

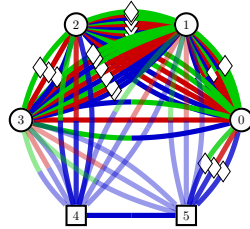


Figure A.2: Graph corresponding to the creation of a 3 qutrit Bell gem.

Nine Ways to Entangle Four Qubits

In 2001, F. Verstraete et. al. [330] introduced a classification of any form of entanglement between 4 qubits into nine categories. Their classification establishes an equivalence relation between states generated by reversible stochastic local quantum operations assisted by classical communication (SLOCC) operations. States inside the same category can perform the same quantum information tasks, albeit with different probabilities.

Out of the nine categories, six of them include well-known quantum states: a separable state of 4 qubits, the W states of 3 and 4 qubits, the product of 2 Bell states, and the GHZ states of 3 and 4 qubits. For the 3 qubit states, the fourth particle is separable (like an ancilla). Fig. A.3 shows the graphs to produce these states.

The remaining three categories, out of the initial nine ones, refer to less-known ways of entanglement. Verstraete's work refers to them as L_{a4} ,

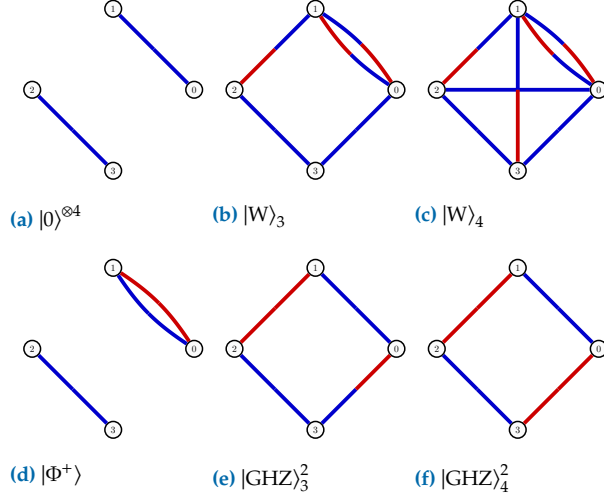


Figure A.3: Six well-known examples from different entanglement categories described by Verstraete et. al. [330]

$L_{0_{5\oplus\bar{3}}}$, and $L_{0_{7\oplus\bar{1}}}$. In Fig. A.4 we see a graph for each category, producing the states $\textcircled{9}$ $\textcircled{10}$ $\textcircled{11}$

$$|L_{a_4}\rangle = |0001\rangle + |0110\rangle + |1000\rangle, \quad (\text{A.5})$$

$$|L_{0_{5\oplus\bar{3}}}\rangle = |0000\rangle + |0101\rangle + |1000\rangle + |1110\rangle, \quad (\text{A.6})$$

$$|L_{7\oplus\bar{1}}\rangle = |0000\rangle + |1011\rangle + |1101\rangle + |1110\rangle. \quad (\text{A.7})$$

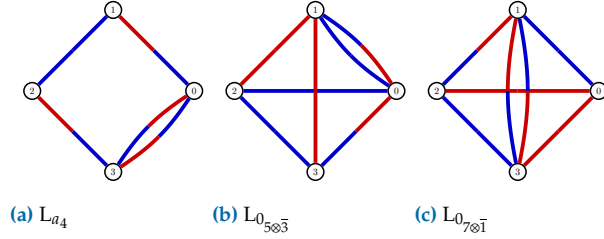


Figure A.4: Graphs for L_{a_4} , $L_{0_{5\oplus\bar{3}}}$, and $L_{0_{7\oplus\bar{1}}}$.

Quantum Error Correction

The path to the quantum computer requires qubits with low error rates independent of their realization platform. However, dealing with noise on any platform is inevitable. Thus, we need error-correcting codes to protect the information and achieve fault-tolerant quantum computation. This is where *Logical Qubits* play a crucial role.

Logical qubits are sets of N physical qubits that define two orthogonal states $|0_L\rangle$ and $|1_L\rangle$. These states are defined in such a way that errors in the physical qubits can be detected by applying a set of global measurements. These measurements will produce a different combination of outcomes for each potential error, allowing its correction. The graphs from this section produce 3 well-known logical qubits to detect and correct arbitrary errors on single qubits.

The first one, the Shor code [331], uses 9 physical qubits to define $\textcircled{12}$

$$|0_L^{(9)}\rangle = (|000\rangle + |111\rangle)^{\otimes 3}, \quad (\text{A.8})$$

$$|1_L^{(9)}\rangle = (|000\rangle - |111\rangle)^{\otimes 3}. \quad (\text{A.9})$$

The second example, the Steane code [332], uses 7 physical qubits [13]

$$\begin{aligned} |0_L^{(7)}\rangle &= |0000000\rangle + |1010101\rangle + |0110011\rangle \\ &\quad + |1100110\rangle + |0001111\rangle + |1011010\rangle \\ &\quad + |0111100\rangle + |1101001\rangle, \end{aligned} \quad (\text{A.10})$$

$$|1_L^{(7)}\rangle = X^{\otimes 7} |0_L^{(7)}\rangle = X_L |0_L^{(7)}\rangle. \quad (\text{A.11})$$

We go from the logical qubit to the other by applying a logical gate X_L , that is, by applying a Pauli X gate on each physical qubit.

The third and last one, with only five physical qubits [333], is the Laflamme code [14]

$$\begin{aligned} |0_L^{(5)}\rangle &= |00000\rangle + |11000\rangle + |01100\rangle + |00110\rangle \\ &\quad + |00011\rangle + |10001\rangle - |10100\rangle - |01010\rangle \\ &\quad - |00101\rangle - |10010\rangle - |01001\rangle - |11110\rangle \\ &\quad - |01111\rangle - |10111\rangle - |11011\rangle - |11101\rangle, \end{aligned} \quad (\text{A.12})$$

$$|1_L^{(5)}\rangle = X^{\otimes 5} |0_L^{(5)}\rangle = X_L |0_L^{(5)}\rangle. \quad (\text{A.13})$$

These two logical qubits constitute the smallest error-correcting code resilient to arbitrary single qubit errors [334].

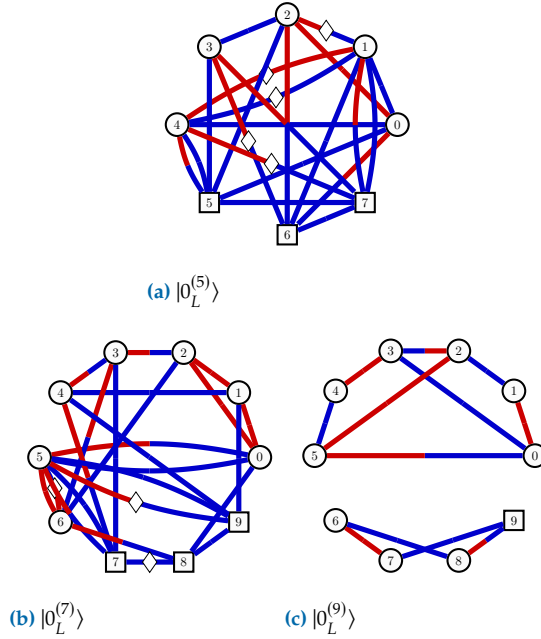


Figure A.5: Graphs for generating the logical 0 state using 5, 7, and 9 physical qubits (+ ancillas, which are in a fixed state).

Fig. A.5 shows the graphs for producing each of the logical 0 states for 5, 7, and 9 physical qubits. For the Shor code, the second logical qubit state can be obtained by multiplying the weights of every bicolored edge by -1 (or alternatively, taking the 3rd, 6th, and 9th qubits and applying a minus at each red incoming edge). For the other two codes, one can obtain the graphs associated with the logical 1 state by switching the colors blue and red. Such transformation is equivalent to applying the gate X on each qubit, the logical X_L .

Products of W States

Previous work with the graph representation showed how to create W states for an arbitrary even number of qubits without using any ancilla [10]. On the other hand, to generate a W state with an odd number of qubits, we always needed at least one ancillary detector. Accordingly, if we want to duplicate such odd-qubit states, we can simply duplicate the graph together with the ancillas. However, this straightforward method is not the most efficient, at least for the states $|W_3\rangle^{\otimes 2}$ and $|W_5\rangle^{\otimes 2}$. As shown in Fig. A.6, these two states can be obtained with no ancillas [15] [16].

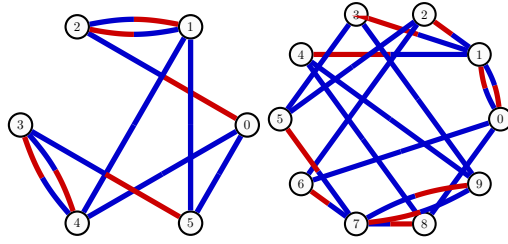


Figure A.6: Graphs for generating the states $|W_3\rangle^{\otimes 2}$ and $|W_5\rangle^{\otimes 2}$ without ancillas.

Dicke States

The equal superposition of every N -qubit state containing a single $|1\rangle$ gives us the W states. These states can be generalized, superposing all permutations of N qubits with exactly k $|1\rangle$ s. They are the *Dicke states*

$$|D_N^k\rangle = \sqrt{\frac{k!(N-k)!}{N!}} \sum_i P_i(|0\rangle^{\otimes N-k} |1\rangle^{\otimes k}), \quad (\text{A.14})$$

where $\sum_i P_i(\cdot)$ stands for all possible permutations of k $|1\rangle$ among N qubits. These states, which have applications in multiparty quantum communication and quantum metrology [335–338], can be produced for an even number of qubits with a general graph introduced in previous works [10]. Here we produce a pair of Dicke states with an odd number of qubits and two excitations [17] [18] (see Fig. A.7).

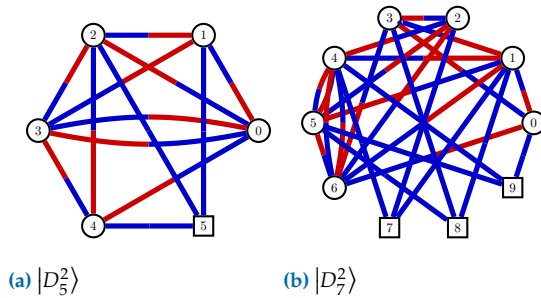


Figure A.7: Dicke states with two $|1\rangle$ for five and seven qubits.

Moreover, there exists a generalization of Dicke states for higher dimensions [339]. Given a set of integers $\{k_i\}$, such that $\sum_i k_i = N$, we describe its corresponding d-dimensional Dicke state as

$$|D(N, \{k_i\})\rangle = \sqrt{\frac{\prod_i k_i!}{N!}} \sum_i P_i(|0\rangle^{\otimes k_0} |1\rangle^{\otimes k_1} \dots |d-1\rangle^{\otimes k_{d-1}}) \quad (\text{A.15})$$

where $\sum_i P_i(\cdot)$ stands for all possible permutations of the N qudits. In Fig. A.8 we plot the graphs that produce the 3-dimensional Dicke state with three particles [19]

$$|D(3, (1, 1, 1))\rangle = |012\rangle + |021\rangle + |102\rangle \\ + |120\rangle + |201\rangle + |210\rangle, \quad (\text{A.16})$$

and with 4 particles (with uneven coefficients $\{k_i\}$) [20]

$$|D(4, (2, 1, 1))\rangle = |0012\rangle + |1200\rangle + |1020\rangle \\ + |0102\rangle + |1002\rangle + |0120\rangle \\ + |0021\rangle + |2100\rangle + |2010\rangle \\ + |0201\rangle + |2001\rangle + |0210\rangle. \quad (\text{A.17})$$

After plotting the results found by PyTheus to generate the last two states, we found a pattern on the graphs that led us to states like $|D(5, (2, 2, 1))\rangle$ and $|D(4, (1, 1, 1, 1))\rangle$: We start taking a fully connected graph of N particles, in which the vertices are connected by 2 bicolored edges, blue-red and red-blue, indicating two modes (the same as in Ref. [10]). Then one can add ancillas that are connected to the N first particles by bicolored edges, one for each of the N particles. These new edges introduce more modes, leading to the following series of states for an even number of particles, $2k$:

$$\begin{aligned} 0 \text{ ancillas, } 2 \text{ modes} &\rightarrow |D(2k, (k, k))\rangle \\ 2 \text{ ancillas, } 3 \text{ modes} &\rightarrow |D(2k, (k, k-1, 1))\rangle \\ 2 \text{ ancillas, } 4 \text{ modes} &\rightarrow |D(2k, (k-1, k-1, 1, 1))\rangle \\ &\dots \\ 2(k-1) \text{ ancillas, } 2k \text{ modes} &\rightarrow |D(2k, (1, \dots, 1))\rangle \end{aligned}$$

For an odd number of particles, $2k+1$, starting with 1 ancilla, we obtain

$$\begin{aligned} 1 \text{ ancillas, } 3 \text{ modes} &\rightarrow |D(2k+1, (k, k, 1))\rangle \\ 3 \text{ ancillas, } 4 \text{ modes} &\rightarrow |D(2k+1, (k, k-1, 1, 1))\rangle \\ 3 \text{ ancillas, } 5 \text{ modes} &\rightarrow |D(2k+1, (k-1, k-1, 1, 1))\rangle \\ &\dots \\ 2k-1 \text{ ancillas, } 2k+1 \text{ modes} &\rightarrow |D(2k+1, (1, \dots, 1))\rangle. \end{aligned}$$

Yeo-Chua State

In 1993, Bennett et. al. showed how to use Bell states to teleport arbitrary and unknown single-qubit states [340]. Going one step further, Yeo and Chua developed a protocol to teleport unknown states of two qubits [341], and instead of using Bell states, they employed the Yeo-Chua state

$$\begin{aligned} [21] |YC\rangle &= |0000\rangle - |0011\rangle - |0101\rangle \\ &\quad + |0110\rangle + |1001\rangle + |1001\rangle \\ &\quad + |1100\rangle + |1111\rangle. \end{aligned} \quad (\text{A.18})$$

This highly entangled state can be produced with the graph of Fig. A.9.

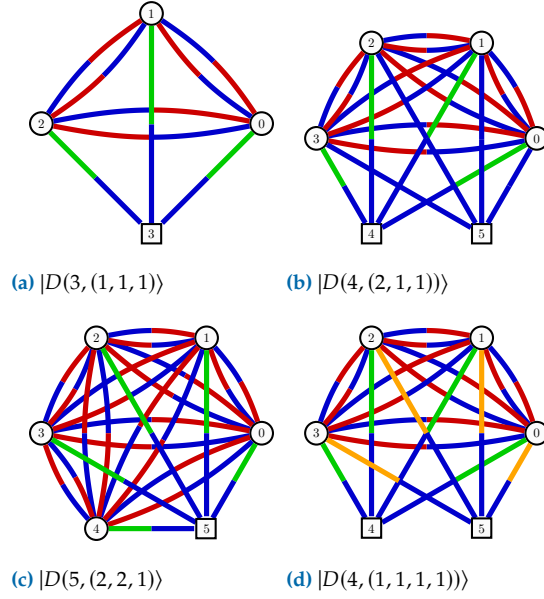


Figure A.8: The three-dimensional Dicke states for three and four particles (top) were found by PyTheus. The 2 graphs in the bottom row were manually produced following the observed pattern.

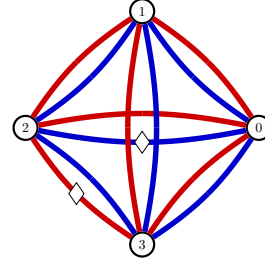


Figure A.9: Graph for generating the Yeochua state.

Complex Weights as a Resource

Exploring the applicability range of PyTheus, we searched for four-qubit states of the form

$$|\psi\rangle = c_1 |0000\rangle + c_2 |0001\rangle + \dots + c_{16} |1111\rangle, \quad (\text{A.19})$$

where $c_i \in \{0, 1\}$. A total of $2^{16} - 1 = 65535$ states. Graphs with ± 1 weights suffice to produce most of the states, many of them do not even require interference. Yet, some states, such as

$$\begin{aligned} |\psi\rangle = & |0011\rangle + |0100\rangle + |0111\rangle + |1000\rangle \\ & + |1100\rangle + |1101\rangle + |1110\rangle, \end{aligned} \quad (\text{A.20})$$

require not only destructive interference between creation terms, but very specific ratios between the weights. In particular, the weights to produce the state equation A.20 make use of the *Golden Ratio*, since they must fulfill the expression

$$x^2 + x = 1. \quad (\text{A.21})$$

The graph that employs the golden ratio is shown in Fig A.10, and similar graphs can produce a whole family of 64 states which are equivalent under permutations and bit-flips. However, there is an alternative complex-weighted graph, with less edges, to produce these states. As shown in Fig A.10, the absolute value is the same for all weights and the interference patterns due to the complex phases are shown in Fig A.11. Even though

they are computationally more expensive to obtain, complex-weighted solutions can provide new, interesting designs.

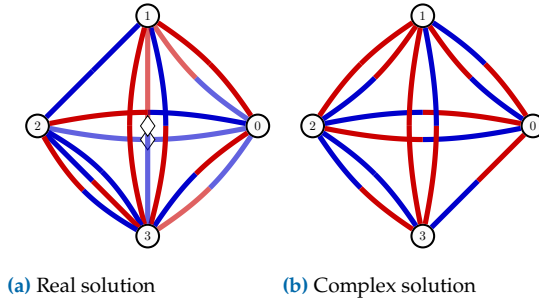


Figure A.10: Graphs for the four-qubit state defined in Eq. equation A.20. The left solution employs the golden ratio and the right one complex weights.

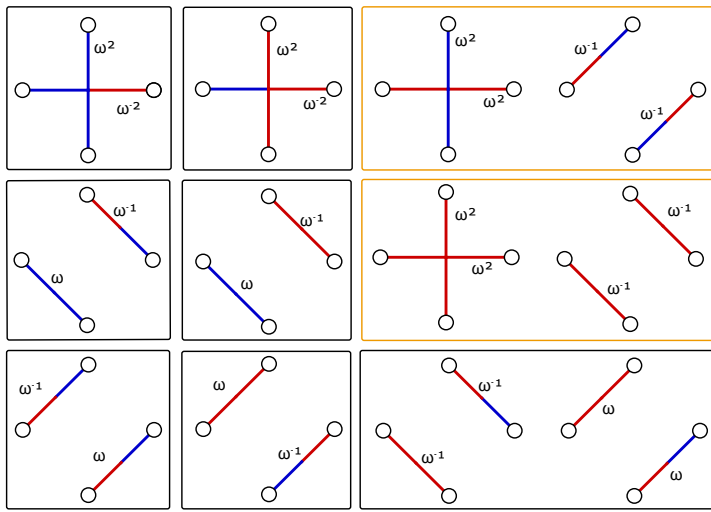


Figure A.11: Perfect matchings of graph shown in Fig. A.10 for the state defined in Eq. equation A.20. $\omega = e^{i\pi/6}$. Each green box contributes one term of the state. The perfect matchings in the orange boxes interfere destructively. It is especially notable that the weights of the two $|0111\rangle$ perfect matchings (bottom right) are complex conjugates and add to one as $\omega^2 + \omega^{-2} = 1$.

Single Photon Sources for State Creation

Significant progress has been made toward using deterministic single-photon sources in quantum optics experiments during the last years [58, 63, 113]. In this section, we present graphs corresponding to quantum experiments for the creation of several highly entangled states using single-photon sources. Deterministic sources do not suffer from the same trade-off between high fidelity and high count rate as probabilistic sources. When using deterministic sources, the number of photons required can be smaller than with probabilistic sources [178, 342].

Post-Selected. We have found graphs for creating different GHZ states with single-photon sources as a resource. One can create a six qubit GHZ from six input photons [22], a four qutrit GHZ state from six input photons [23], as well as the same four qutrit GHZ state from a combination of two single photon sources and two simultaneous SPDC events [24] (see Fig. A.12).

Heralded. There have been proposals for the creation of multi-particle high-dimensional entangled states from single photon sources using discrete Fourier transform [109]. In Fig. A.13 we show a graph [25], which achieves a heralded Bell state using two ancillary photons without the need for photon-number-resolving detectors with almost perfect fidelity

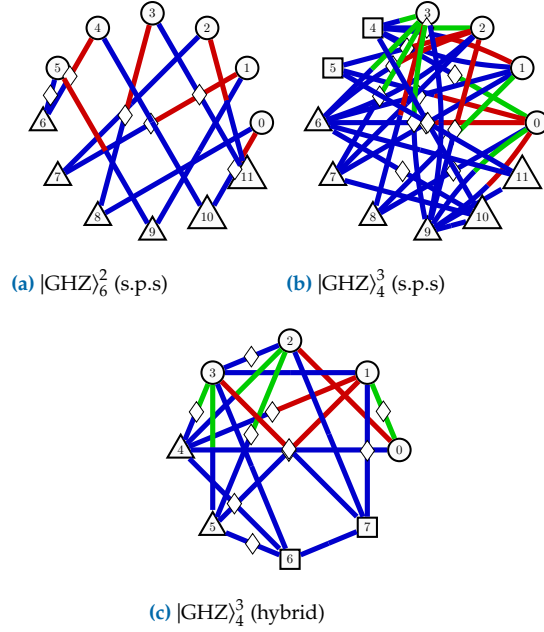
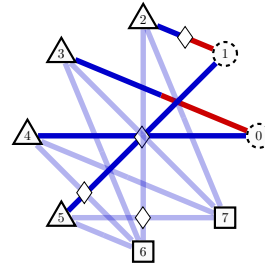


Figure A.12: Graphs corresponding to the creation of three-, four- and five-particle GHZ states using single-photon sources.

($F = 99\%$) for $w = 0.1$. There are cross-term that occur due to the lack of photon- number resolution. We want to acknowledge that the probability of a heralding event is very low for this example ($P = 0.08\%$) and it is thus unlikely that this setup is experimentally feasible. As w approaches 0, fidelity approaches 100% but count-rate approaches 0. The trade-off between high fidelity and high count-rate is interesting to consider.

Figure A.13: Graph corresponding to the creation of a heralded two-dimensional Bell state in the outgoing photons (0-1) from four single photon sources (2-5) and two heralding detectors (6-7). The heralding detectors do not require photon-number-resolution. The light edges have weights $w < 1$ corresponding to loss in the photon path.



A.2 Maximizing Entanglement

When a quantum state cannot be expressed as the product of its parts, we call it entangled. However, if one asks *how* entangled a state is, the answer is not unique, there are multiple measures of entanglement. Here we present some of these metrics, together with states that are known to maximize them. Finally, we choose a particular metric and optimize a graph, not to reach a particular state – as done so far by maximizing the fidelity – but to instead maximize the entanglement metric itself. In this way, we will recover some well-known entangled states and also find new ones.

Schmidt Rank Vector

Multiparticle entangled states in high dimensions allow for a much deeper structure underlying the potential ways in which the particles can be entangled. These structures can be characterized by the Schmidt Rank Vector (SRV) and give rise to new phenomena that only exist if both the number of particles and the number of dimensions go beyond two [343–345]. Here we consider tripartite states. The rank vector is a list of the ranks of the reduced density matrices. The state $\text{SRV}(A,B,C)$ refers to a state, where

$$A = \text{rank}(\rho_A), B = \text{rank}(\rho_B), C = \text{rank}(\rho_C),$$

and ρ_X is the density matrix of the system with the particle X traced out. In other words, the SRV is a vector of the dimensionalities of entanglement of every bipartition, which shows the dimensionality of entanglement between one particle and the rest of the quantum state. This master-slave-slave configuration is very useful for quantum applications such as layered quantum communication [346]. Taking a $\text{SRV}(3,3,2)$ state as an example, we then know that the first two particles are both three-dimensionally entangled with the third particle, whereas the third one is only two-dimensionally entangled with the rest. The dimensionality for each particle cannot be increased with linear operations and classical communication (LOCC).

Searching the $\text{SRV}(A,B,C)$ states has been investigated with the computer algorithm Melvin [5] and graph theory [9, 347]. Interestingly, it has been shown from graph theory that without using any ancilla particles, there are several $\text{SRV}(A,B,C)$ states that cannot be created. [9, 347]. Here we list these states (up to normalization):

- 26 $\text{SRV}(5,5,4)$:
 $|000\rangle + |111\rangle + |222\rangle + |333\rangle + |443\rangle$
- 27 $\text{SRV}(6,3,2)$:
 $|000\rangle + |101\rangle + |210\rangle + |311\rangle$
 $+ |420\rangle + |521\rangle$
- 28 $\text{SRV}(6,5,5)$:
 $|000\rangle + |111\rangle + |222\rangle + |334\rangle$
 $+ |443\rangle + |544\rangle$
- 29 $\text{SRV}(7,3,3)$:
 $|000\rangle + |101\rangle + |210\rangle + |311\rangle$
 $+ |422\rangle + |520\rangle + |621\rangle$

It has been unknown whether we can experimentally create these states with additional particles, and this challenging question has been open for more than three years. Also, until now, $\text{SRV}(A,B,C)$ states in high dimensional systems have only been recently demonstrated for $\text{SRV}(3,3,2)$ [70], $\text{SRV}(3,3,3)$ [348], and $\text{SRV}(4,4,2)$ [349]. By using PyTheus, one can learn how to generate much more complex SRV states of high dimensions and more particles, such as the impossible states described in [9, 347]. We show the results in Fig. A.14.

Hyperdeterminant

Expanding the definition of determinant to tensors of arbitrary dimensions, we obtain *Hyperdeterminants* [350], which can also be used as an entanglement metric. In particular, the hyperdeterminant of a tensor $2 \times 2 \times 2 \times 2$ characterizes the entanglement of a 4 qubit state, which is maximal for the state (up to normalization) [30]

$$|\text{HD}\rangle = |\Psi^+00\rangle + |00\Psi^+\rangle + |1111\rangle, \quad (\text{A.22})$$

where $\sqrt{2}|\Psi^+\rangle = |01\rangle + |10\rangle$, and whose associated graph is shown in Fig. A.15.

Negative Partial Trace

Given a state ρ_{AB} , its *partial transpose* with respect to the subsystem A is

$$(\rho^{TA})_{ab,mn} = \rho_{mb,an}. \quad (\text{A.23})$$

As shown by Peres, it is a necessary condition for the separability of A and B that none of the eigenvalues of ρ^{TA} are negative [351]. Moreover, it is also a sufficient condition for bipartite states of dimensions 2×2 and 2×3 [352].

Aiming for highly entangled states, Brown et. al. looked for states of 2, 3, 4, and 5 qubits, whose partial traces for each bipartition had the lowest possible eigenvalues [353]. They called this metric *Negative Partial Trace*, and it leads to the following states of 4 and 5 qubits [31] [32]

$$\begin{aligned} |\text{BSSB4}\rangle &= |0000\rangle + |+011\rangle \\ &\quad + |1101\rangle + |-110\rangle \end{aligned} \quad (\text{A.24})$$

$$\begin{aligned} |\text{BSSB5}\rangle &= |000\Phi^+\rangle + |100\Phi^-\rangle \\ &\quad + |010\Psi^+\rangle + |111\Psi^-\rangle. \end{aligned} \quad (\text{A.25})$$

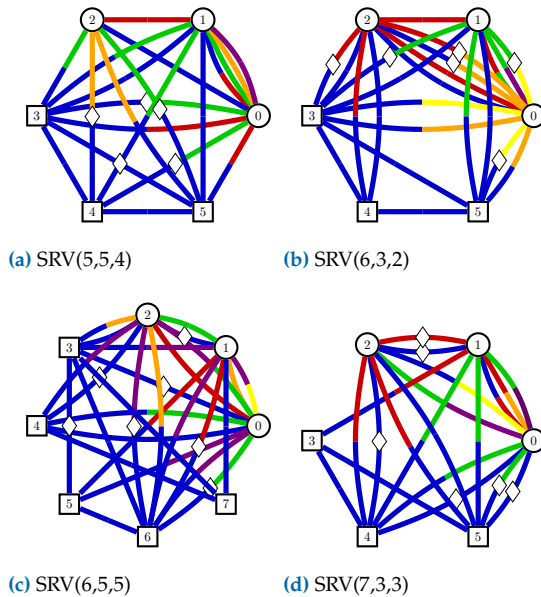


Figure A.14: Graphs for producing SRV(A,B,C) states using ancillas.

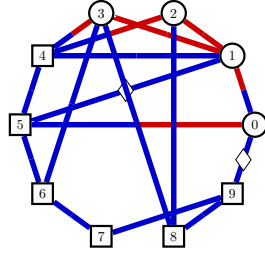


Figure A.15: Graph to generate the state $|\text{HD}\rangle$, which maximizes the hyperdeterminant.

Where $\sqrt{2}|\pm\rangle = |0\rangle \pm |1\rangle$, $\sqrt{2}|\Phi^\pm\rangle = |00\rangle \pm |11\rangle$, and $\sqrt{2}|\Psi^\pm\rangle = |01\rangle \pm |10\rangle$. These states can be produced with the graphs shown in Fig. A.16.

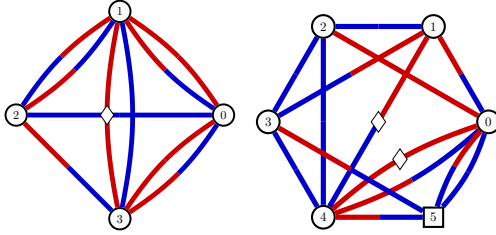


Figure A.16: Graphs to generate the BSSB states of 4 and 5 qubits, respectively. The latter requires an ancilla.

Rényi Entropy

Among all the entanglement metrics in this document, as well as the ones that are left out, the *Rényi Entropy* is one of the most important [354]. Given a state with ρ_{AB} , the Rényi entropy of order α between the subsystems A and B is

$$S_\alpha(\rho_A) = \frac{1}{1-\alpha} \log \text{Tr}(\rho_A^\alpha) = S_\alpha(\rho_B), \quad (\text{A.26})$$

where $\rho_A = \text{Tr}_B(\rho_{AB})$ and $\rho_B = \text{Tr}_A(\rho_{AB})$ are reduced density matrices. Similarly, the *Tsallis entropy* also describes the entanglement between A and B

$$\mathcal{T}_\alpha(\rho_A) = \frac{1}{1-\alpha} (\text{Tr}(\rho_A^\alpha) - 1) = \mathcal{T}_\alpha(\rho_B). \quad (\text{A.27})$$

Notice that in the limit $\alpha \rightarrow 1$, both expressions approach the Von Neumann entropy.

A relevant difference between both metrics is that, for $\alpha > 1$, only the Tsallis entropy is convex while the Rényi one is Schur convex [355]. Taking this difference into account, Gour and Wallach showed that the 4-qubit state $|L\rangle$ maximizes the average of Tsallis entropies with $\alpha = 2$ for 2-qubit partitions [356], a metric of entanglement also referred to as the *Meyer-Wallach measure* [357, 358]. This state, created by the graph shown in Fig. A.17, reads [33]

$$\begin{aligned} |L\rangle = & (1 + \omega)(|0000\rangle + |1111\rangle) \\ & + (1 - \omega)(|0011\rangle + |1100\rangle) \\ & + \omega^2(|0101\rangle + |0110\rangle + |1001\rangle + |1010\rangle), \end{aligned} \quad (\text{A.28})$$

where $\omega = \exp(2i\pi/3)$.

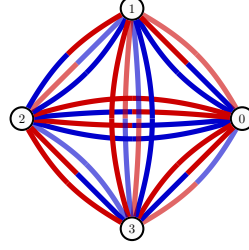


Figure A.17: Graph to generate the 4-qubit L-state. All the weights in the graph are complex.

Rényi–Ingarden–Urbanik Entropy

Minimizing the Rényi entropy over all local unitary transformations U_{LOC} , Enriquez et. al. described the minimal *Rényi–Ingarden–Urbanik* entropy [359]

$$S_{\alpha}^{\text{RIU}}(\phi) = \min_{U_{\text{LOC}}} S_{\alpha}(p(U_{\text{LOC}}|\phi)), \quad (\text{A.29})$$

where S_{α} is the Rényi entropy of order $\alpha \geq 0$, and p is a normalized probability vector resulting from expanding the state to a product basis.

Together with the definition, the authors numerically find two states of three qubits that minimize the entropy for $\alpha = 1$ and $\alpha = 2$ [34] [35]

$$|\Phi\rangle_{\alpha=1} = 0.27|000\rangle + 0.377|100\rangle + 0.326|010\rangle + 0.363|001\rangle + 0.74e^{-0.79\pi i}|111\rangle, \quad (\text{A.30})$$

$$|\Phi\rangle_{\alpha=2} = 0.438|000\rangle + 0.29|100\rangle + 0.371|010\rangle + 0.316|001\rangle + 0.698e^{-0.826\pi i}|111\rangle. \quad (\text{A.31})$$

These states correspond to the graphs of Fig. A.18.

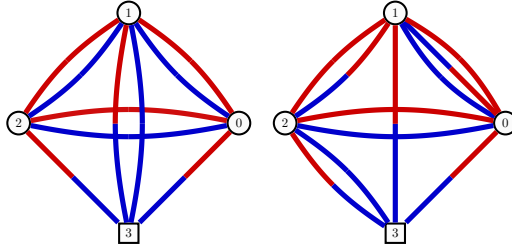


Figure A.18: Graphs to generate the states that optimize the RIU entropy for $\alpha = 1$ and $\alpha = 2$.

Maximizing Entanglement for Each Partition

The previous states were maximally entangled according to different metrics. Indeed, many of them were found numerically when maximizing such metrics. Similarly, we can use PyTheus to maximize arbitrary physical properties of the states produced by a graph, which can be immediately translated into experiments. This is a great advantage over previous work, in which an analytically described quantum state may not be experimentally feasible.

We optimize the graph according to the Tsallis entropy for $\alpha = 2$ (see Eq. equation A.27) using as loss function

$$\min_{\omega} \mathcal{L}(\rho(\omega)) = \sum_A \text{Tr} \rho_A^2, \quad (\text{A.32})$$

Dim	Number of particles									
	2	3	4	5	6	7	8	9	10	
2	✓	✓	✗	✓	✓	✗	✗(3)	✓(3)	✗(3)	
3	✓	✓	✓	✓	✓	✓	✗(3)	✓	✓	
4	✓	✓	✓	✓	✓	✓	?	✓	✓	
5	✓	✓	✓	✓	✓	✓	✓	✓	✓	

Table A.1: Known AME states up to 10 particles and 5 dimensions. ✗: no AME exist for the 2-dimensional systems of 4, 7, 8, and 10 particles, nor for the 3-dimensional system of 8 particles. ? : it is unknown whether there is an AME for the 4-dimensional system of 8 particles. ✓: there exist an AME state for the rest of systems. For the 2-dimensional systems of 8, 9, and 10 particles, as well as for the 3-dimensional system of 8 particles, there exist a state with maximally entangled partitions of 3 particles [361, 362]. The existence of some of these state does not imply that they have been (or can be) experimentally realized.

where $\rho = |\psi(\omega)\rangle\langle\psi(\omega)|$ is the state defined by a graph with weights $\{\omega\}$, ρ_A is the reduced density matrix with respect to a subsystem A . A subsystem and its complement A^c are a bipartition of the full system. The loss function is minimized for the sum of all bipartitions.

Given a bipartition, the entanglement between them is maximal if

$$\rho_A = \text{Tr}_{A^c}(\rho) = \frac{1}{D} \mathbb{I}, \quad (\text{A.33})$$

with D being the dimension of the Hilbert space in which ρ_A is defined, and $\mathbb{I}$ the identity matrix. Accordingly, multipartite states with maximal entanglement for every bipartition are called *Absolutely Maximally Entangled* (AME); such kinds of states have applications in quantum protocols like threshold secret sharing or open-destination teleportation [360]. On the other hand, given a state of N qudits, $\rho \in \mathcal{H}_d^{\otimes N}$, we call it k -uniform when the entanglement is maximal for all bipartitions of k qudits [361]. GHZ states of N -qudits (see Eq. equation A.1) are examples of 1-uniform states.

With PyTheus we can pick types of bipartitions the entanglement is to be maximized. Choosing all bipartitions targets the production of an AME state, bipartitions of size $|A| = k$ target k -uniform states. A mathematical result of entanglement theory is that AMEs only exist for certain combinations of N and d (see table A.1). In such situations we choose to optimize for k -uniform states.

AME States. Since no AME exist for a system of four qubits (see table A.1), we start optimizing the entanglement for all possible partitions for five qubits, obtaining two different AME states [36] [37]

$$\begin{aligned} |\text{AME}(5, 2)\rangle_a &= |00000\rangle + |01101\rangle + |01110\rangle \\ &\quad + |10110\rangle + |11000\rangle + |11011\rangle \\ &\quad - |00011\rangle - |10101\rangle, \end{aligned} \quad (\text{A.34})$$

$$\begin{aligned} |\text{AME}(5, 2)\rangle_b &= |01011\rangle + |01100\rangle + |10110\rangle \\ &\quad + |11010\rangle - |00000\rangle - |00111\rangle \\ &\quad - |10001\rangle - |11101\rangle. \end{aligned} \quad (\text{A.35})$$

For six qubits we found an asymptotic state [38]

$$\begin{aligned}
 |\text{AME}(6, 2)\rangle \approx & |000010\rangle + |000100\rangle + |001001\rangle \\
 & + |010101\rangle + |011000\rangle + |011110\rangle \\
 & + |101010\rangle + |110000\rangle + |111011\rangle \\
 & + |111101\rangle - |001111\rangle - |010011\rangle \\
 & - |100001\rangle - |100111\rangle - |101100\rangle \\
 & - |110110\rangle + \varepsilon(|000000\rangle + |000110\rangle \\
 & + |010001\rangle + |010111\rangle + |110010\rangle \\
 & + |110100\rangle + |111001\rangle + |111111\rangle). \quad (\text{A.36})
 \end{aligned}$$

This is not exactly the AME state of eight qubits, it has eight terms which vanish asymptotically as ε goes to zero. However, by doing so, we reduce the creation rate of the whole state. The graphs to produce this and the 5-qubits AME states are shown in Fig. A.19.

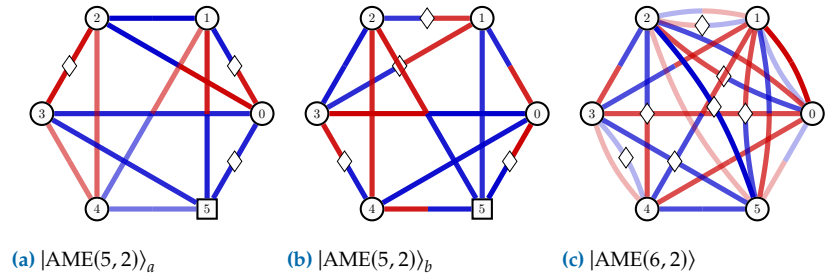


Figure A.19: Absolutely Maximally Entangled states with 5 and 6 qubits. The 6-qubit graph produces an asymptotic state.

k-Uniform States. While the AME solution for six qubits was asymptotic, we can generate the exact 2-uniform state of such a system [39]

$$\begin{aligned}
 |U2\rangle_6^2 = & |100011\rangle + |101110\rangle - |000000\rangle \\
 & - |001101\rangle - |010111\rangle - |011010\rangle \\
 & - |110100\rangle - |111001\rangle. \quad (\text{A.37})
 \end{aligned}$$

Twelve out of twenty three-qubit bipartitions of the above state are maximally entangled.

Similarly, we generated a 7-qubit state that maximizes the entanglement between each of the 21 possible $k = 2$ partitions except one [40]

$$\begin{aligned}
 |\sim U2\rangle_7^2 = & |0011110\rangle + |0101000\rangle + |0110111\rangle \\
 & + |1001101\rangle + |1010010\rangle + |1100100\rangle \\
 & - |0000001\rangle - |1111011\rangle. \quad (\text{A.38})
 \end{aligned}$$

The graph to produce this ‘almost’ uniform state and the previous one are shown in Fig. A.20.

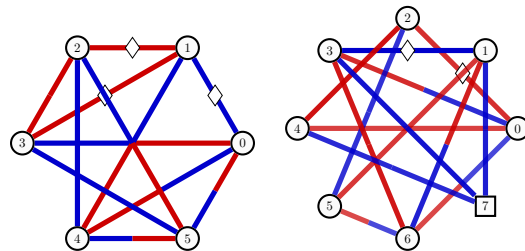


Figure A.20: $k=2$ uniform state for 6 qubits and $k=2$ (almost) uniform state for 7 qubits (+ one ancilla).

Other States. As we have seen, even when an AME or k -uniform state for a given system of qudits exists, it is not always realizable with linear optics. However, even if not perfect, we have found several highly entangled states which are worth mentioning. We refer to them as

$$|\text{Ent}(n,d,k)\rangle, \quad (\text{A.39})$$

where n stands for the number of particles, d for their dimensions, and k for the size of the partitions.

For a system of four qubits, we find two interesting states when maximizing the entanglement (see Eq. equation A.32) for every 2-qubit partition (see Fig. A.21). We first optimize using real weights (41)

$$|\text{Ent}(4,2,2)\rangle_{\mathbb{R}} = |1010\rangle + |1101\rangle - |0011\rangle - |0100\rangle, \quad (\text{A.40})$$

obtaining a state which maximizes all possible partitions except one. For the second solution, we extend the weights to the complex domain, finding a local optimal (42)

$$\begin{aligned} |\text{Ent}(4,2,2)\rangle_{\mathbb{C}} = & |0011\rangle + e^{-i0.54\pi} |0101\rangle \\ & + e^{-i0.93\pi} |0110\rangle + e^{i0.45\pi} |1001\rangle \\ & + e^{i0.71\pi} |1010\rangle + e^{i0.86\pi} |1100\rangle. \end{aligned} \quad (\text{A.41})$$

This state minimizes the sum of partial traces for $k = 2$ partitions, following Eq. equation A.32. However, while the sum is (locally) minimal, and all qubit partitions are equally entangled, they are not maximally entangled for $k = 2$. and all are equally for all $k = 1$ partitions and for none of the $k = 2$. However, all $k = 2$ partitions lead to the same value for the partial trace defined in Eq. equation A.32. The entanglement seems to be ‘equally distributed’. The complex state in Eq. equation A.41 gives

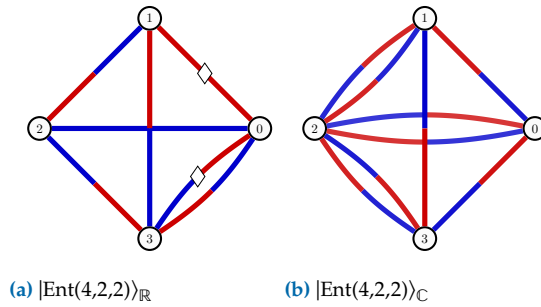


Figure A.21: The graphs produced maximally entangled states for single qubit partitions on a 4 qubit system, they also present high entanglement between 2 qubit partitions (even if not maximal). The weights on the left graph are real, and the weights on the right graph are complex (the phase of every edge can be found in the repository).

the same entanglement structure as the Higuchi-Sudbery state presented in [363]

$$\begin{aligned} |\text{HS}\rangle = & |0011\rangle + |1100\rangle + \omega(|1010\rangle + |0101\rangle) \\ & + \omega^2(|1001\rangle + |0110\rangle), \end{aligned} \quad (\text{A.42})$$

where $\omega = e^{i2\pi/3}$.

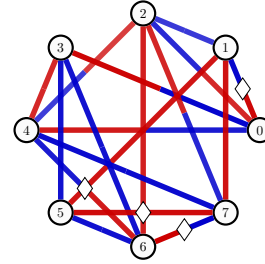
Finally, for a system of 8 qubits, PyTheus found a state for which 48 out

of 56 total three-qubit bipartitions are maximally entangled. [43](#)

$$\begin{aligned}
 |\text{Ent}(8,2,3)\rangle = & |00010000\rangle + |00010111\rangle \\
 & + |01101011\rangle + |01101100\rangle \\
 & + |01110001\rangle + |10100010\rangle \\
 & + |10100101\rangle + |10111111\rangle \\
 & + |11000100\rangle + |11011001\rangle \\
 & - |00001010\rangle - |00001101\rangle \\
 & - |01110110\rangle - |10111000\rangle \\
 & - |11000011\rangle - |11011110\rangle.
 \end{aligned} \tag{A.43}$$

The graph to produce this state is shown in Fig. A.22.

It would be interesting to extend the maximization of other entanglement measures, such as experimental feasible, strong forms of all-vs-nothing violations [\[150\]](#), Hardy's version of local realism experiments which goes beyond the violations in Ref. [\[364, 365\]](#) (for example, by involving more particles), or other measures of quantum correlations. Such extensions would contribute to the study of artificial intelligence for the foundations of quantum mechanics [\[366\]](#).



(a) $|\text{Ent}(8,2,3)\rangle$

Figure A.22: The state generated by this graph is maximally entangled for 48 of the 56 possible 3-qubit partitions.

A.3 Generation of Mixed States

Werner State

The Werner state with the density matrix

$$\rho_\alpha = \alpha |\Phi^+\rangle \langle \Phi^+| + (1 - \alpha) \frac{\mathbb{I}}{4}, \tag{A.44}$$

has different properties for different values of α . For $\alpha < 0.683$, it is known to be Bell-local. For $\alpha > 0.697$, it is known to be Bell nonlocal [\[367\]](#). This leaves the gap between 0.683 and 0.697 open without a mathematical proof of its properties. By creating this state, it could be possible to test the Bell-locality of a state in the middle of the gap, $\alpha = 0.69$. The graph for the creation of this mixed state is shown in Fig. A.23 [44](#).

Counter-Example to the Peres Conjecture

Relations between different notions of quantum correlations are studied in quantum information theory. These insights are important to understand

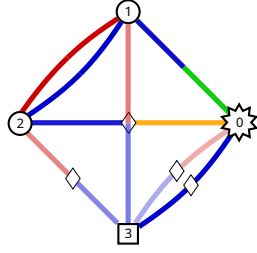


Figure A.23: Graph for the Werner state with $\alpha = 0.69$. Information at vertex 0 is to be discarded. Vertex 3 is an ancillary detector

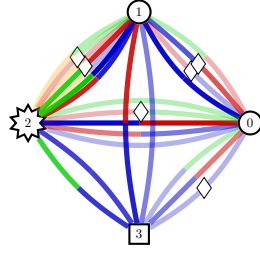


Figure A.24: Graph for the Moroder-Gittsovich-Huber-Gühne-Vertesi-Brunner (MGHG-VB) state, which is a counter-example to the Peres conjecture. Here, vertex 3 is an ancilla photon and vertex 2 is the environment vertex. Due to the structure of the MGHG-VB state, the edge weights are not simple fractions.

the underlying resources of quantum correlations and their applicability in quantum protocols. In 1999, Asher Peres conjectured that the notion of Bell nonlocality and distillability is equivalent [152]. Bell nonlocality, as witnessed by a violation of Bell's inequality, says that no local hidden-variable model can explain the correlations observed in an experiment. A state is called distillable if a maximally entangled Bell pair can be extracted from multiple copies. The conjecture was open for 15 years until a counter-example was discovered in 2014, first for a strong version of the conjecture [153] and shortly after in its full form [154]. These violations are based on the uncovering of a two-particle bound state (a state that cannot be distilled) which can be used for quantum steering and violation of Bell's inequality. The Moroder-Gittsovich-Huber-Gühne-Vertesi-Brunner (MGHG-VB) state can be written as a mixed state that is an incoherent superposition of four entangled two-qutrit states, [45]

$$\rho = \sum_{i=1}^4 \lambda_i |\psi_i\rangle \langle \psi_i| \quad (\text{A.45})$$

with $\lambda = (\frac{3257}{6884}, \frac{450}{1721}, \frac{450}{1721}, \frac{27}{6884})$ and

$$\begin{aligned} |\psi_1\rangle &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ |\psi_2\rangle &= \frac{\sqrt{131}}{12\sqrt{2}} (|01\rangle + |10\rangle) + \frac{1}{60} |02\rangle - \frac{3}{10} |21\rangle \\ |\psi_3\rangle &= \frac{\sqrt{131}}{12\sqrt{2}} (|00\rangle - |11\rangle) + \frac{1}{60} |12\rangle - \frac{3}{10} |20\rangle \\ |\psi_4\rangle &= \frac{1}{\sqrt{3}} (-|01\rangle + |10\rangle + |22\rangle). \end{aligned}$$

The graph that corresponds to this state is shown in Fig. A.24. It might be interesting to generate this state and show experimentally that the state can violate Bell's inequality and thereby demonstrate an experimental falsification of Peres's conjecture.

A.4 Generation of Entanglement in the Photon-Number Basis

NOON States

The ability to perform precise measurements is always affected by several limitations, some of them avoidable by careful design of experiments and others of a fundamental nature. Classically, one can estimate a phase parameter more precisely by increasing the number of particles N in the measurement process; the precision is typically limited to $1/\sqrt{N}$, also known as the standard quantum limit (SQL). With the aid of quantum resources, one can beat the SQL, and the measurement precision can be enhanced to scale as $1/N$, approaching the Heisenberg limit that is governed by the physical law of quantum mechanics. One prominent representative of quantum states with the ability to break the SQL is the well-known NOON states [53, 368], as described in Eq. equation 3.8.

These NOON states can not only be used to test fundamental physics via violating Bell-type inequalities [369] but also play an important role in quantum-enhanced applications, quantum-enhanced microscopes and imaging systems [370–373], super-resolving phase measurements [374–377], and quantum lithography [378], to name a few. Interestingly, as the two-mode NOON states are formed in a superposition of two distinct states $|N, 0\rangle$ and $|0, N\rangle$, one could consider that the two terms correspond to the “dead cat” and “alive cat” in the “Schrödinger-cat” form [122]. With high-NOON states, we might step closer to a better understanding of macroscopic entanglement envisioned by Schrödinger [379]. In the optical regime, achieving an efficient photonic source of NOON states with large N is very challenging. Several schemes and experiments have been toward the direction of generating photonic high-NOON states in different ways, such as by using strong optical nonlinearities [380], by linear optics and feed-forward [381], and by mixing coherent light with SPDC photon-pair sources on a standard beam splitter [122, 382–384].

Additionally, interest has also been growing in the simultaneous estimation of multiple parameters using multi-mode quantum states [118, 119]. Using a multimode NOON state, one can reach the Heisenberg limit with an $O(d)$ advantage over what is obtained with d copies of a two-mode NOON state [385, 386]. The generalization of the two-mode NOON states in Eq. equation 3.8 is given as [118, 119]

$$| \text{NOON} \rangle_d^N := \frac{1}{\sqrt{d}} (|N, 0, \dots, 0, 0\rangle \pm |0, N, \dots, 0, 0\rangle \pm \dots \pm |0, 0, \dots, N, 0\rangle \pm |0, 0, \dots, 0, N\rangle)_{1, \dots, d}. \quad (\text{A.46})$$

Here we show how to produce perfect NOON states without using any coherent states or feed-forward. We list some examples that might give a different conceptual understanding and new insight. For more path-entangled states, one can directly employ PyTheus. For the ancillary detectors, we always assume they are reached by single photons, like in previous sections.

Two Particles. As we have introduced in section 3.2.1, the simplest NOON state is the $| \text{NOON} \rangle_2^2$ state that can be deterministically created via HOM interference when two indistinguishable photons are incident on a standard beam splitter simultaneously [114]. From the graph in Fig. 3.6, we can easily generalize the state to an arbitrary number of modes by

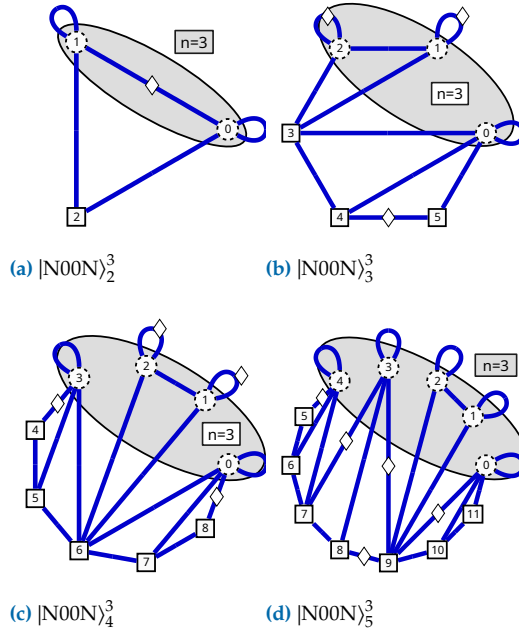


Figure A.25: Graphs for multi-mode three-particle N00N states $|N00N>_d^3$ with increasing number of ancillas.

adding more vertices, each one with a loop. The coherent superposition of a loop being in one of the vertices indicates the resulting multi-mode two-particle N00N states.

Three Particles. For producing 2-mode 3-photon N00N states $|N00N>_2^3$, we additionally employ an ancilla particle in the experiment. We show the graph found by PyTheus in Fig. A.25, which uses only one color, thus corresponding to the standard path-entangled N00N states. The edge can also be in different colors such that we are able to construct graphs for polarization-entangled N00N states [387–389]. We further explore the states in multi-mode cases, as described in Fig. A.25 (b), (c) and (d) for 3-, 4-, 5-mode three-photon N00N states. Interestingly, the graphs found by PyTheus exhibit a nice structure, which may be amenable to generalization, to systematically produce other multi-mode NOON states without any optimization process.

Four Particles. For the previous N00N states, we optimized the graphs using real weights, finding the states $|40> - |04>$ and $|400> - |040> - |004>$ [46] [47]. However, to generate the same N00N states with zero phase difference between the terms we need to, either add a $\pi/2$ phase on the solitary loops from Fig. A.26 or, using only real weights, add 2 more ancillas. Additionally, after generating the 4-particle N00N states for 2 and 3 modes, it became obvious how to generate the graphs for an arbitrary number of modes.

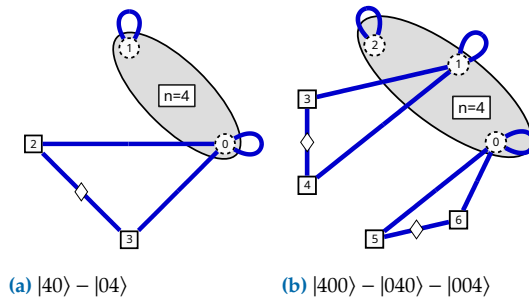


Figure A.26: Graphs with only real weights to generate 4-particle N00N states for 2 and 3 modes. Multiplying the i the loops of the disconnected vertices all the amplitudes of the state's terms become positive. To increase the number of modes, we add a triangular subgraph connected to a loop. The edge connecting the two new ancillas must be negative.

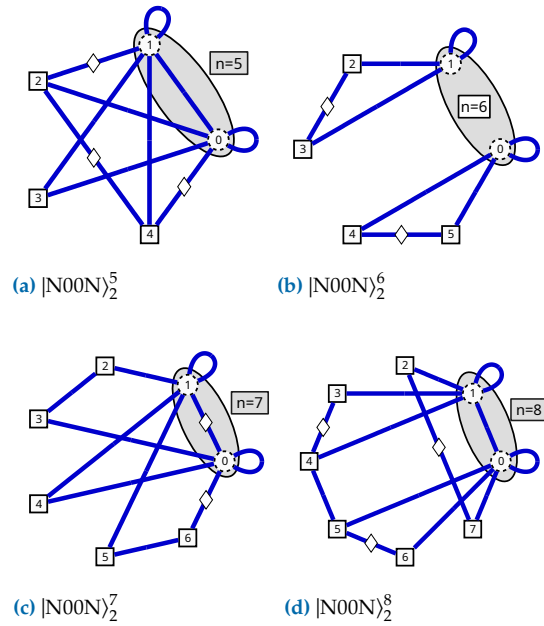


Figure A.27: Graphs for two-mode N00N states of 5, 6, 7, and 8 photons.

It is worth noting that given a ket $|40\rangle$, each combination of non-repeating edges contributes twice as much as one with duplicate edges (assuming equal weights). This is a consequence of the multinomial theorem applied to the creation operators described in Eq. equation 3.2.

Other Two-Mode States. For the previous N00N states, the associated graphs have shown some patterns. The more obvious was the increasing need for ancillas, whether we were adding more photons or modes. Moreover, based on a few graphs, we realized how to generate N00N states with 2 and 4 photons for an arbitrary number of modes. In Fig. A.27 we show how to produce the two-mode N00N state for 5, 6, 7, and 8 photons [48] [49] [50] [51]. As with the 3-photon N00N states, it seems to be an underlying pattern in the graphs solutions when increasing the number of photons.

Platonic Solid States

We have been considering states where all N photons are in the same path, with the other paths empty. However, PyTheus can produce path-entangled states which do not fulfill such restrictions. Among them, a very special case leads to the highly symmetric Platonic Solid State [390–392], which can resolve rotations around any axis equally well. In the Platonic picture, shown in Fig. A.28, an N -photon state is mapped onto N points on the Poincaré sphere [393], offering a systematic way for visualization. This also relates to a long-lasting problem of distributing N points on the Poincaré sphere in a highly symmetric fashion, where one can have many different solutions based on the function one tries to optimize [394, 395]. Additionally, the concept of Platonic solids has also been used for fundamental investigations in quantum physics [396, 397]. Apart from their symmetric elegance, there is plenty of room for applications of these states, for instance, in magnetometry, polarimetry, and metrology. Researchers have started investigating how to generate these Platonic solid state. In this part, we show the examples found by PyTheus.

Tetrahedron. Our first Platonic solid state is the tetrahedron [398], which is the unique optimal four-photon state for characterizing polarization rotations. The state written in the Fock basis reads as

$$|\Psi^{(1)}\rangle = \frac{1}{\sqrt{3}}(|4, 0\rangle + \sqrt{2}|1, 3\rangle). \quad (\text{A.47})$$

It can be produced by the graph in Fig. A.28 (top panels), and has already been implemented [399].

The similarity between the graphs that produce the tetrahedron and the three-photon N00N state discovered by PyTheus (see Fig. A.25 a) shows a nice pattern: to generate states in the Fock basis, we can add an arbitrary number of photons to any path. In this way, starting with a general state $|M, N\rangle \pm |N, M\rangle$, we can obtain

$$|M, N\rangle \pm |N, M\rangle \Rightarrow |M+x, N+y\rangle \pm |N+x, M+y\rangle. \quad (\text{A.48})$$

We only need to connect the first and second paths to x and y ancillary photons, respectively. This construction can be used for an arbitrary number of modes.

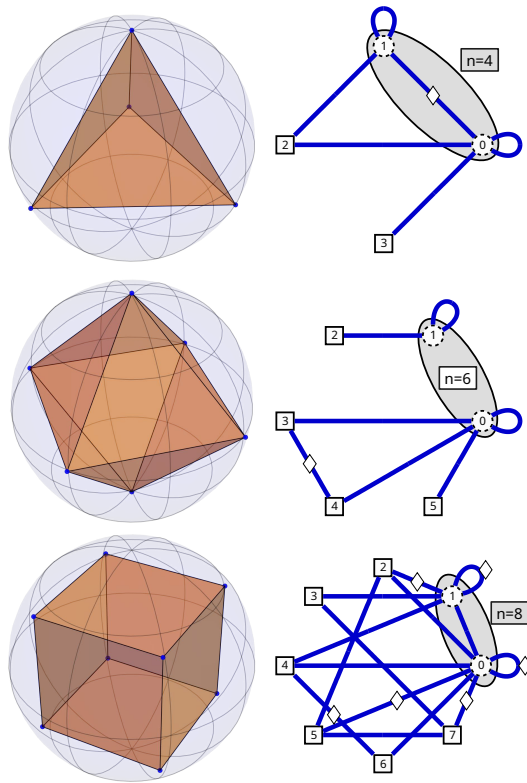


Figure A.28: Graphs for producing tetrahedron, octahedron, and cube quantum states.

Octahedron. The octahedron state reads

$$|\Psi^{(2)}\rangle = \frac{1}{\sqrt{2}}(|5, 1\rangle - |1, 5\rangle), \quad (\text{A.49})$$

and was found by PyTheus using only 4 ancillary detectors (see Fig. A.28). Alternatively, we can apply the pattern described in Eq. equation A.48 to the graph that produces $|40\rangle - |04\rangle$, which is shown in Fig. A.26 a).

Cube. The last platonic solid we are able to produce is the cube state,

which reads [52](#)

$$|\Psi^{(3)}\rangle = \frac{\sqrt{3}}{12}(\sqrt{10}|8,0\rangle + 2\sqrt{7}|4,4\rangle + \sqrt{10}|0,8\rangle). \quad (\text{A.50})$$

In Fig. [A.28](#), we show the graph that can be used to generate the cube state. As we can see from the expression and Fig. [A.28](#) (bottom), the graph cannot be obtained by modifying one of the previous graphs for a two-mode N00N state.

A.5 Towards Quantum Simulation

In this section, we present quantum state generation for states from condensed matter physics which could be interesting for quantum simulation[\[400\]](#). The quantum entangled states of few- or many-body systems can generally be expressed by the tensor network formulations [\[401, 402\]](#). One of the most successful members of this family with a vast application in condensed matter physics is the 1D matrix product state (MPS). Here one represents the quantum state of a system with periodic boundary conditions and N particles as

$$|\psi\rangle = \sum_s \text{tr} \left[A_1^{(s_1)} A_2^{(s_2)} \dots A_N^{(s_N)} \right] |s_1 \dots s_N\rangle, \quad (\text{A.51})$$

where $s_i = \{0, \dots, d-1\}$ denotes the local state of the i th particle with local physical dimension d . Here $A_i^{(s_i)}$ is a complex matrix with dimension χ , also known as the bond (virtual) dimension. The matrix $A_i^{(s_i)}$ can be viewed as a projector from a χ -dimensional virtual (correlation) vector space into the physical d -dimensional space [\[403\]](#). While in non-interacting systems, described by product states, the bond dimension χ is one, this quantity grows exponentially with N in most strongly correlated systems. This results in employing numerical techniques to (approximately) obtain the ground states of most many-body systems. Nevertheless, one can find multiple few- or many-body ground states with $\chi > 1$ but independent of the particle number. Here, we present a collection of these ground states with a concise description of their host Hamiltonian and their physical implication upon realization. While our focus is on zero-temperature ground states, we emphasize that the applicability of PyTheus is not limited to these states. This is because both mixed states [3.2.1](#), which describe open quantum states, and states constitute combinations of (various) Fock states [3.2.1](#), can be used to study the (grand) canonical ensemble, are treatable by PyTheus.

Spin-1/2 Systems

Due to the simplicity of spin-1/2 systems and the surge of interest in qubit quantum computation, these many-body systems received ever-growing attention. The local physical space of these systems spanned by $\{|\uparrow\rangle, |\downarrow\rangle\}$, or equivalently $\{|0\rangle, |1\rangle\}$, results in the local physical dimension $d = 2$. Despite such a small physical dimension, the obtained ground states can be rich and exotic. In the following, we list some of these states.

Spin-1/2 Wire. A computational quantum spin-1/2 wire governed by nearest-neighbor interactions, associated with non-zero two-point correlation functions and arbitrary local entanglement, is described by [\[403\]](#),

404]

$$|\psi\rangle = \sum_{s_i=0}^1 \text{tr}[A(s_{n-1}) \dots A(s_1)] |s_1 \dots s_n\rangle, \quad (\text{A.52})$$

where

$$A(\uparrow) = \frac{1}{\sqrt{2}} G, \quad G = \exp(i\pi\sigma^x/\tau), \quad (\text{A.53})$$

$$A(\downarrow) = \frac{1}{\sqrt{2}} GT(\phi), \quad T = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}, \quad (\text{A.54})$$

where ϕ and τ denote the entanglement factor and period, respectively.

Fig. A.29 displays the associated graph for the ground state of the spin-1/2 wire for four particles with $\phi = \pi/2$. The state, up to normalization, reads [53]

$$\begin{aligned} |\psi\rangle = & \sqrt{2} |0000\rangle + |0001\rangle + |0010\rangle + |0100\rangle \\ & - |0111\rangle + |1000\rangle - |1011\rangle \\ & - |1101\rangle - |1110\rangle - \sqrt{2} |1111\rangle. \end{aligned} \quad (\text{A.55})$$

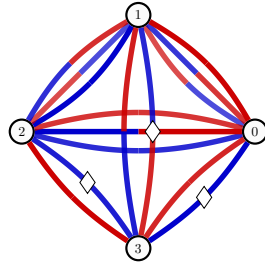


Figure A.29: The associated graph for the ground state of the spin-1/2 wire with four particles.

Spin-1/2 States with No Adjacent Spin-ups. Aside from the previous spin-1/2 state, one can compute entangled states where no two neighboring spin-ups appear in the ground state. One can expect to detect these states in spin systems with nearest-neighbor interactions. In the Rydberg-atom experiments, this situation occurs due to the Rydberg blockade [405]. The matrix product representation of these states reads

$$|\psi\rangle = \sum_{s_i=0}^1 \text{tr}[A(s_{n-1}) \dots A(s_1)] |s_1 \dots s_n\rangle, \quad (\text{A.56})$$

where

$$A(\uparrow) = \frac{1}{\sqrt{2}} (\mathbb{I} + \sigma^x), \quad (\text{A.57})$$

$$A(\downarrow) = 2\sigma^+, \quad (\text{A.58})$$

with $\sigma^+ = \sigma^x + i\sigma^y$.

Fig. A.30 displays the associated graphs for this system with various numbers of particles whose ground states read

- [54] Three particles with one ancillary particle:

$$|\psi_3\rangle = |000\rangle + |001\rangle + |010\rangle + |100\rangle. \quad (\text{A.59})$$

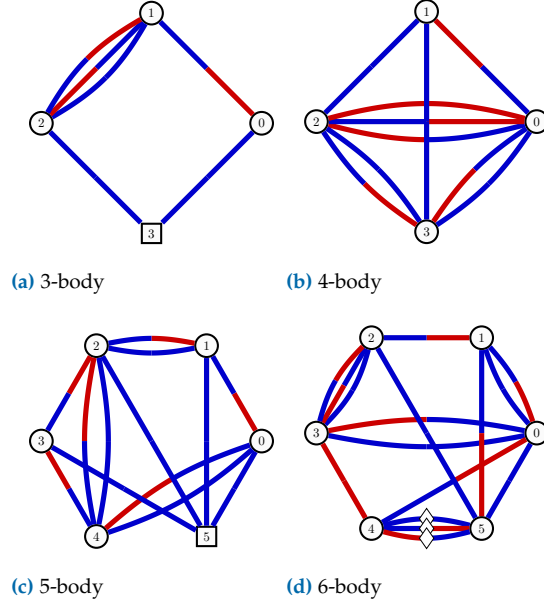


Figure A.30: Associated graphs for spin-1/2 states with no adjacent spin-ups for three, four, five, and six particles. One ancillary particle is included in states with an odd number of particles (left column).

- **55** Four particles:

$$|\psi_4\rangle = |0000\rangle + |0001\rangle + |0010\rangle + |0100\rangle + |0101\rangle + |1000\rangle + |1010\rangle. \quad (\text{A.60})$$

- **56** Five particles with one ancillary particle:

$$|\psi_5\rangle = |00000\rangle + |00001\rangle + |00010\rangle + |00100\rangle + |00101\rangle + |01000\rangle + |01001\rangle + |01010\rangle + |10000\rangle + |10010\rangle + |10100\rangle. \quad (\text{A.61})$$

- **57** Six particles:

$$|\psi_6\rangle = |000000\rangle + |000001\rangle + |000010\rangle + |000100\rangle + |000101\rangle + |001000\rangle + |001001\rangle + |001010\rangle + |010000\rangle + |010001\rangle + |010010\rangle + |010100\rangle + |010101\rangle + |100000\rangle + |100010\rangle + |100100\rangle + |101000\rangle + |101010\rangle. \quad (\text{A.62})$$

Majumdar-Gosh Model. The one-dimensional quantum Heisenberg spin model is known as the Majumdar-Gosh model when the value of the next-nearest-neighbor interaction is half the value of the nearest-neighbor antiferromagnetic exchange interaction. The Hamiltonian of this model casts [406]

$$H = \sum_i 2\vec{\sigma}_i \cdot \vec{\sigma}_{i+1} + \vec{\sigma}_i \cdot \vec{\sigma}_{i+2}, \quad (\text{A.63})$$

where $\vec{\sigma}_i = (\sigma_i^x, \sigma_i^y, \sigma_i^z)$.

The ground states of this model Hamiltonian are dimerized states given by products of singlet configurations of spins on neighboring sites. The

linear combination of these states reads [406]

$$|\psi\rangle = \sum_{s_i=0}^1 \text{tr}[A(s_{n-1}) \dots A(s_1)] |s_1 \dots s_n\rangle, \quad (\text{A.64})$$

where

$$A^\uparrow = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 \end{pmatrix}, \quad A^\downarrow = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{A.65})$$

This ground state of the Majumdar-Gosh model is one of the simplest spin-1/2 valence-bond solids in one-dimensional systems.

Fig. A.31 displays the associated graphs for the four- and six-particle systems whose ground states read

- [58] Four particles:

$$|\psi^{(4)}\rangle = |0011\rangle - 2|0101\rangle + |0110\rangle \\ + |1100\rangle - 2|1010\rangle + |1001\rangle. \quad (\text{A.66})$$

- [59] Six particles with two ancillary particles:

$$|\psi^{(6)}\rangle = |001011\rangle - |001101\rangle \\ - |010011\rangle + |010110\rangle \\ + |011001\rangle - |011010\rangle \\ + |100101\rangle - |100110\rangle \\ - |101001\rangle + |101100\rangle \\ + |110010\rangle - |110100\rangle. \quad (\text{A.67})$$

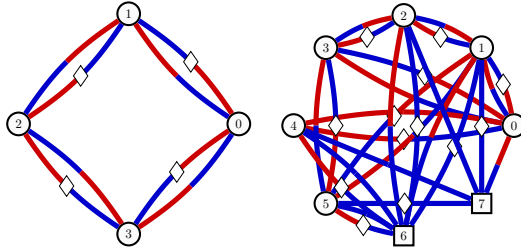


Figure A.31: Ground states of the Majumdar-Gosh model with four (left panel) and six (right) particles. Two ancillary particles are included in the right panel.

Dyck Words. The Fredkin spin-1/2 model with a three-body interaction term reads [407–410]

$$\mathcal{H} = \sum_i U_{i-1} P_{i,i+1} + P_{i-1,i} D_{i+1}, \quad (\text{A.68})$$

where $U_i = |\uparrow_i\rangle\langle\uparrow_i|$, $D_i = |\downarrow_i\rangle\langle\downarrow_i|$ and the spin-singlet projector reads $P_{i,i+1} = |S_{i,i+1}\rangle\langle S_{i,i+1}|$ with $|S_{i,j}\rangle = (|\uparrow_i\rangle\langle\downarrow_j| - |\downarrow_i\rangle\langle\uparrow_j|)$. The ground state of this model is an equally weighted superposition of spin configurations forming Dyck words. Here one may use the notation $|\uparrow\rangle = |(\rangle$ and $|\downarrow\rangle = |)\rangle$ to translate the spin states into the Dyck words. In this notation, the ground state forms balanced strings whose segments contain equal numbers of open and closed parentheses. The ground states of the Fredkin model with six and eight particles cast

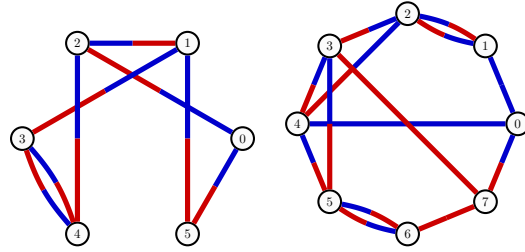
► **[60]** Six particles:

$$|\psi^{(6)}\rangle = \frac{1}{\sqrt{5}} [|000\rangle + |0(0)\rangle + |(0)0\rangle + |(00)\rangle + |((0))\rangle]. \quad (\text{A.69})$$

► **[61]** Eight particles:

$$|\psi^{(6)}\rangle = \frac{1}{\sqrt{14}} [|(00)0\rangle + |(000)\rangle + |(0((0)))\rangle + |(((0)))0\rangle + |(((0)0))\rangle + |(((00)))\rangle + |0000\rangle + |00(0)\rangle + |0(0)0\rangle + |0(00)\rangle + |0((0))\rangle + |(0)00\rangle + |(0)(0))\rangle]. \quad (\text{A.70})$$

The associated graphs for these states are shown in Fig. A.32.



(a) Dyck Word with six letters (b) Dyck Word with eight letters

Figure A.32: The graphs corresponding to the creation of Dyck word states.

Spin-1 Systems

The quest to go beyond the two-level qubit systems in quantum computation put forward other proposals based on three-level qutrit quantum states [411, 412]. In spin systems, this inquiry is translated into exploring the $S = 1$ space spanned by $\{|-1\rangle, |0\rangle, |1\rangle\}$ with physical dimension $d = 3$. While spin-1 states can be realized experimentally in a controlled fashion [413], often, one may encounter these states as emergent phenomena in various spin-1/2 condensed matter systems, e.g., in describing the low-energy physics of chiral threefold fermions [414]. As a result, aside from technological implications, exploring the higher-spin systems may shed light on a better understanding of some emergent phenomena in other fields of physics.

Spin-1 Wire. The antiferromagnetic ground state of a spin-1 chain with nearest-neighbor interactions can be represented in the matrix product form as [415]

$$|\psi\rangle = \text{tr}(A_1 A_2 \dots A_N), \quad (\text{A.71})$$

where

$$A_i = \begin{pmatrix} |0\rangle & -\sqrt{a}|+1\rangle \\ \sqrt{a}|-1\rangle & -\sigma|0\rangle \end{pmatrix}, \quad (\text{A.72})$$

with nonvanishing a and σ .

Fig. A.33 displays the associated graph with one ancilla, for the ground states of the spin-1 chain with three particles, which, up to normalization, reads (62)

$$\begin{aligned} |\psi\rangle = & 0.3(|012\rangle + |120\rangle + |201\rangle) - 0.875|111\rangle \\ & - 0.15(|021\rangle + |102\rangle + |210\rangle). \end{aligned} \quad (\text{A.73})$$

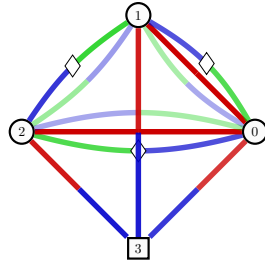


Figure A.33: The associated graph for the ground states of the spin-1 chain with three particles.

Affleck-Kennedy-Lieb-Tasaki Model. A particular spin-1 bilinear bi-quadratic Hamiltonian introduced by Affleck, Kennedy, Lieb, and Tasaki (AKLT) is given by [416, 417]

$$H = \sum_i \vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2. \quad (\text{A.74})$$

The ground state of this solvable model, known as the AKLT state, is short-range entangled and classified in symmetry-protected-topological states [418]. The MPS representation of this state reads [401, 406]

$$|\psi\rangle = \sum_{s_i=0}^1 \text{tr}[A(s_{n-1}) \dots A(s_1)] |s_1 \dots s_n\rangle, \quad (\text{A.75})$$

where $s_i \in \{0, 1\}$ and

$$\{A_{-1}, A_0, A_{+1}\} = \left\{ \sqrt{\frac{2}{3}}\sigma^+, \frac{1}{\sqrt{3}}\sigma^z, -\sqrt{\frac{2}{3}}\sigma^- \right\}. \quad (\text{A.76})$$

Here $\sigma^\pm = \sigma^x \pm i\sigma^y$. The AKLT state is the spin-1 valance bond solids [419].

Fig. A.34 displays the associated graphs for the three- and four-particle

AKLT system whose ground states read, up to normalization [63] [64]

$$\begin{aligned} |\psi^{(3)}\rangle &= |0, -1, +1\rangle - |-1, 0, +1\rangle \\ &\quad + |-1, +1, 0\rangle - |0, +1, -1\rangle \\ &\quad + |+1, 0, -1\rangle - |+1, -1, 0\rangle, \end{aligned} \quad (\text{A.77})$$

$$\begin{aligned} |\psi^{(4)}\rangle &= 2(|-1, 1, -1, 1\rangle + |1, -1, 1, -1\rangle) \\ &\quad + |1, 0, -1, 0\rangle - |1, -1, 0, 0\rangle \\ &\quad + |0, 1, 0, -1\rangle - |0, 1, -1, 0\rangle \\ &\quad - |0, 0, 1, -1\rangle + |0, 0, 0, 0\rangle \\ &\quad - |0, 0, -1, 1\rangle - |0, -1, 1, 0\rangle \\ &\quad + |0, -1, 0, 1\rangle - |-1, 1, 0, 0\rangle \\ &\quad - |1, 0, 0, -1\rangle + |-1, 0, 1, 0\rangle \\ &\quad - |-1, 0, 0, 1\rangle. \end{aligned} \quad (\text{A.78})$$

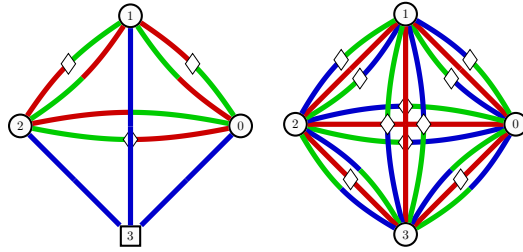


Figure A.34: The associated graphs for the ground state of the spin-1 AKLT state with three and four particles. One ancillary particle is included in the first state.

Motzkin State. The spin-1 generalization of the Dyck words is known as the Motzkin State [408, 420, 421]. The spin configurations, in this case, can be translated into strings using $|-1\rangle = | \rangle$, $|0\rangle = |-\rangle$, and $|1\rangle = |() \rangle$. Here, similar to the Dyck words, the number of open and closed parentheses is equal. Examples of such states cast

► [65] Three particles:

$$|\psi\rangle = \frac{1}{2}[|---\rangle + |()--\rangle + |(-)\rangle + |-(())\rangle]. \quad (\text{A.79})$$

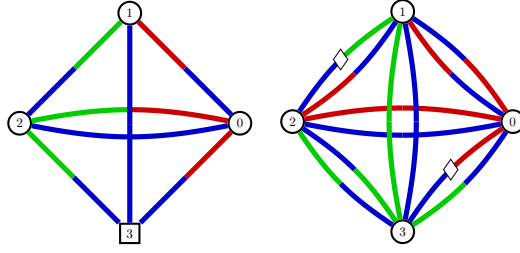
► [66] Four particles:

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{9}}[|----\rangle + |()--\rangle + |-(())\rangle \\ &\quad + |---() \rangle + |(--)\rangle + |()() \rangle \\ &\quad + |-(())\rangle + |(-)-\rangle + |((-))\rangle]. \end{aligned} \quad (\text{A.80})$$

The associated graphs for these states are presented in Fig. A.35.

Spin-3/2 Systems

Sharing the same motivation as spin-1 systems, the four-level spin-3/2 systems have putative implications in qudits with $d = 4$ quantum computation [422]. The local Hilbert space of these spin-3/2 systems are spanned by $\{|1\rangle, |-1\rangle, |3\rangle, |-3\rangle\}$ resulting in local physical dimension $d = 4$.



(a) Motzkin with three letters (b) Motzkin with four letters

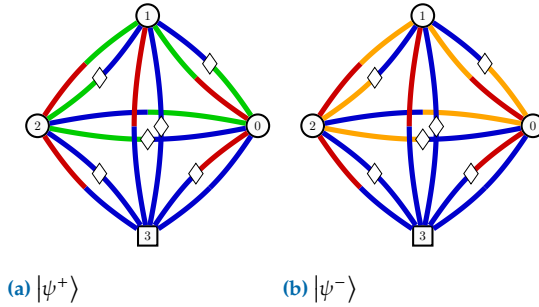
Figure A.35: The graphs corresponding to the creation of Motzkin states.

Spin-3/2 Wire. Two exact ground states for a spin-3/2 chain with ferromagnetic character are shown to have the following matrix product representation [423, 424]

$$|\psi^\pm\rangle = \text{tr}(A_1^\pm A_2^\pm \dots A_N^\pm), \quad (\text{A.81})$$

where

$$A^+ = \begin{pmatrix} |1\rangle & -\sqrt{3}|3\rangle \\ |-1\rangle & -|1\rangle \end{pmatrix}, \quad A^- = \begin{pmatrix} -|-1\rangle & |1\rangle \\ -\sqrt{3}|-3\rangle & |-1\rangle \end{pmatrix}. \quad (\text{A.82})$$



(a) $|\psi^+\rangle$

(b) $|\psi^-\rangle$

Figure A.36: The associated graphs for the ground states of the spin-3/2 chain for three particles.

The ground states of the spin-3/2 chain, up to normalization, reads [67]

$$\begin{aligned} |\psi^\pm\rangle = & |-1, 1, \pm 3\rangle - |1, -1, \pm 3\rangle + |1, \pm 3, -1\rangle \\ & - |-1, \pm 3, 1\rangle + |\pm 3, -1, 1\rangle - |\pm 3, 1, -1\rangle. \end{aligned} \quad (\text{A.83})$$

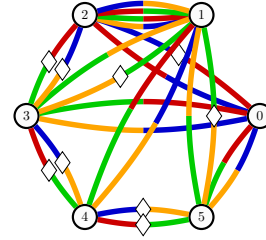
As one ground state, say $|\psi^+\rangle$, can be achieved from the other ground state, $|\psi^-\rangle$, by merely replacing $|3\rangle$ with $|-3\rangle$, we only present the associated graph for $|\psi^+\rangle$ with three particles and one ancilla in Fig. A.36.

Spin-3/2 Letter State. As a straightforward generalization of the Dyck words and Motzkin letter states, discussed previously, one can introduce the letter states for the spin-3/2 states. Here one should translate the spin state into the letters as $|-3\rangle = |[\rangle$, $|-1\rangle = |(\rangle$, $|1\rangle = |)\rangle$ and $|3\rangle = |]\rangle$.

Using this language, the six particle state is [68]

$$\begin{aligned}
 |\psi\rangle = \frac{1}{\sqrt{40}} [& |000\rangle + |010\rangle + |100\rangle + |001\rangle \\
 & + |110\rangle + |101\rangle + |011\rangle + |111\rangle \\
 & + |(0)0\rangle + |(0)1\rangle + |(1)0\rangle + |(1)1\rangle \\
 & + |[0]1\rangle + |[1]0\rangle + |[1]1\rangle + |[1]1\rangle \\
 & + |0((\rangle + |1((\rangle + |0((\rangle + |0[(\rangle \\
 & + |[1[(\rangle + |[1[(\rangle + |0[(\rangle + |[1[(\rangle \\
 & + |(00)\rangle + |(00)\rangle + |(10)\rangle + |(01)\rangle \\
 & + |(11)\rangle + |[11)\rangle + |[01)\rangle + |[11)\rangle \\
 & + |((0))\rangle + |((1))\rangle + |((0))\rangle + |((0))\rangle \\
 & + |(((\rangle + |(((\rangle + |([0))\rangle + |([1))\rangle].
 \end{aligned} \tag{A.84}$$

The associated graph for this state is shown in Fig. A.37.



(a) Spin-3/2 letter state with six letters

Figure A.37: The graphs corresponding to the creation of the spin-3/2 letter state.

Spin-2 Systems

Five-level computational units can further be of interest in quantum computation beyond qubits. The local physical space of these spin-2 units is spanned by $\{|-2\rangle, |-1\rangle, |0\rangle, |1\rangle, |2\rangle\}$ with local physical dimension $d = 5$. Similar to other higher-spin systems in condensed matter systems, spin-2 states can also be viewed as emergent states in some (lower-spin with $S < 2$) many-body systems [425].

Spin-2 Wire. The anisotropic spin-2 chain with nearest-neighbour interactions hosts various phases depending on three parameters (a, x, σ) with $a \in \mathbb{R}, \sigma = \pm 1$ [426]. These phases consist of three distinct antiferromagnetic Haldane phases and a weak antiferromagnetic phase.

The ground states of the weak-antiferromagnetic phase have the following matrix product representation:

$$|\psi_0^{(1)}\rangle = \text{tr} \left[\prod_{i=1}^{L/2} m_{2i-1} g_{2i} \right], \tag{A.85}$$

$$|\psi_0^{(2)}\rangle = \text{tr} \left[\prod_{i=1}^{L/2} g_{2i-1} m_{2i} \right], \tag{A.86}$$

where

$$m = \begin{pmatrix} |1\rangle & x\sqrt{a}|2\rangle \\ \sqrt{a}|0\rangle & |1\rangle \end{pmatrix}, \quad g = \begin{pmatrix} |-1\rangle & \sqrt{a}|0\rangle \\ x\sqrt{a}|-2\rangle & |-1\rangle \end{pmatrix}. \tag{A.87}$$

Here we set $a = 0.5$.

When the translational symmetry is not broken, i.e., $x = 1$, two more states can also be identified for this phase, given by

$$\left| \psi_0^{(3)} \right\rangle = \text{tr} \left[\prod_{i=1}^L m_i^{x=1} \right], \quad \left| \psi_0^{(4)} \right\rangle = \text{tr} \left[\prod_{i=1}^L g_i^{x=1} \right]. \quad (\text{A.88})$$

Fig. A.38 displays the associated graphs for the three-particle ground states of the weak-antiferromagnetic phase when the translational symmetry is broken (top panel) and respected (bottom). The states with broken translation symmetry, up to normalization, read [69](#)

$$\begin{aligned} \left| \psi_0^{(1)} \right\rangle = & 0.5(|2, -1, 0\rangle + |0, -1, 2\rangle) \\ & + 0.25(|2, -2, 1\rangle + |1, -2, 2\rangle) \\ & + |1, 0, 0\rangle + |0, 0, 1\rangle + 4|1, -1, 1\rangle, \end{aligned} \quad (\text{A.89})$$

$$\begin{aligned} \left| \psi_0^{(2)} \right\rangle = & |-1, 0, 0\rangle + |0, 0, -1\rangle + 4|-1, 1, -1\rangle \\ & + 0.25(|-2, 2, -1\rangle + |-1, 2, -2\rangle) \\ & + 0.5(|-2, 1, 0\rangle + |0, 1, -2\rangle). \end{aligned} \quad (\text{A.90})$$

When the symmetry is respected, the states, up to normalization, are [70](#)

$$\begin{aligned} \left| \psi_0^{(3)} \right\rangle = & |0, 1, 2\rangle + |0, 2, 1\rangle + |1, 0, 2\rangle + |2, 1, 0\rangle \\ & + |1, 2, 0\rangle + |2, 0, 1\rangle + 4|1, 1, 1\rangle, \end{aligned} \quad (\text{A.91})$$

$$\begin{aligned} \left| \psi_0^{(4)} \right\rangle = & |0, -1, -2\rangle + |0, -2, -1\rangle + |-1, 0, -2\rangle \\ & + |-2, -1, 0\rangle + |-1, -2, 0\rangle \\ & + |-2, 0, -1\rangle + 4|-1, -1, -1\rangle. \end{aligned} \quad (\text{A.92})$$

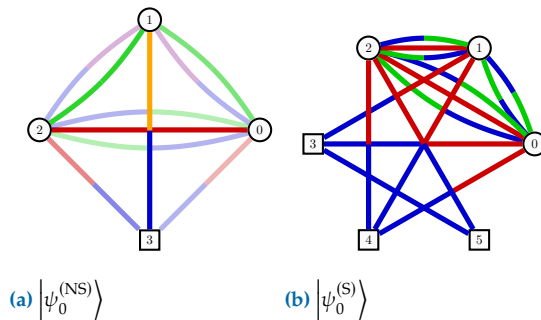


Figure A.38: The associated graphs for the three-particle ground states of the weak-antiferromagnetic phase when the translational symmetry is broken (left) and respected (right).

The Haldane-antiferromagnetic-A is another possible phase in this spin-2 system described by the ground state

$$|\psi\rangle = \sum_{s_i=-2}^2 \text{tr} [A_1^{s_1} A_2^{s_2} \dots A_N^{s_N}] |s_1, s_2, \dots, s_N\rangle, \quad (\text{A.93})$$

where

$$\sum_{s_m=-2}^2 A_m^{s_m} = \begin{pmatrix} |0\rangle & \sqrt{x} |1\rangle & a |2\rangle \\ \sqrt{x} |-1\rangle & \gamma |0\rangle & \sqrt{x} |1\rangle \\ a |-2\rangle & \sqrt{x} |-1\rangle & |0\rangle \end{pmatrix}. \quad (\text{A.94})$$

Fig. A.39 displays the graph associated with the ground state, with three particles and five ancillas, for the Haldane-antiferromagnetic-A phase. Up to normalization, the state reads [71](#)

$$\begin{aligned} |\psi_A\rangle = & |-1, 0, 1\rangle + |-1, 1, 0\rangle + |0, -1, 1\rangle \\ & + |0, 1, -1\rangle + |1, -1, 0\rangle + |1, 0, -1\rangle \\ & + 0.5(|-2, 0, 2\rangle + |0, -2, 2\rangle + |-2, 2, 0\rangle) \\ & + 0.5(|2, 0, -2\rangle + |0, 2, -2\rangle + |2, -2, 0\rangle) \\ & + 0.25(|1, -2, 1\rangle + |-1, -1, 2\rangle + |-1, 2, -1\rangle) \\ & + 0.25(|1, 1, -2\rangle + |-2, 1, 1\rangle + |2, -1, -1\rangle) \\ & + 6|0, 0, 0\rangle. \end{aligned} \quad (\text{A.95})$$

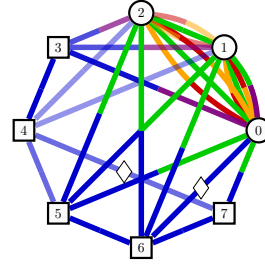


Figure A.39: The associated graph for the three-particle ground state of the Haldane-antiferromagnetic-A phase.

The ground states of the second antiferromagnetic Haldane phase, referred to as Haldane-antiferromagnetic-B, reads

$$|\psi_B\rangle = \sum_{s_i=-1}^1 \text{tr} [A_1^{s_1} A_2^{s_2} \dots A_N^{s_N}] |s_1, s_2, \dots, s_N\rangle, \quad (\text{A.96})$$

where

$$\sum_{s_m=-1}^1 A_m^{s_m} = \begin{pmatrix} |0\rangle & \sqrt{a} |1\rangle \\ \sqrt{a} |-1\rangle & \sigma |0\rangle \end{pmatrix}. \quad (\text{A.97})$$

Similarly, the ground state of the third Haldane-antiferromagnetic phase, known as Haldane-antiferromagnetic-C, is

$$|\psi_C\rangle = \sum_{s_i \in \{0, \pm 2\}} \text{tr} [A_1^{s_1} A_2^{s_2} \dots A_N^{s_N}] |s_1, s_2, \dots, s_N\rangle, \quad (\text{A.98})$$

where

$$\sum_{s_i \in \{0, \pm 2\}} A_m^{s_m} = \begin{pmatrix} |0\rangle & \sqrt{a} |2\rangle \\ \sqrt{a} |-2\rangle & \sigma |0\rangle \end{pmatrix}. \quad (\text{A.99})$$

Notice that the last Haldane phases, B and C, have essentially the same Hamiltonian as the weak-antiferromagnetic phase when the translational symmetry is preserved (see Eq. equation A.92). One can obtain one from the other by performing $|\pm 1\rangle \leftrightarrow |\pm 2\rangle$.

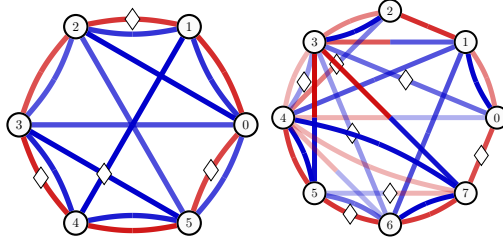


Figure A.40: The associated graphs for the Onsager's scar in a system with six (left panel) and eight (right) qubits. Here we set $\beta = 1/\sqrt{2}$.

Quantum Many-Body Scars

The phenomena in which weakly entangled nonthermal quantum eigenstates are embedded in the eigensystem of non-integrable (thermal) systems is dubbed 'quantum many-body scars' [427]. The well-known examples of these scar states are shown to be present in the AKLT spin chain models [428, 429]. The experimental realizations of such states are also reported in Rydberg-atom quantum simulators [405, 430]. In the following, we suggest that one may also detect these states using quantum optics experiments.

Onsager's Scars in Disordered Spin Chains. A non-integrable quantum spin chain that exhibits quantum many-body scars can be described by the coherent state with parameter β as [431]

$$|\psi_\beta\rangle = \text{tr}[A_{p_1} B_{p_1} \dots A_{p_n} B_{p_n}] |p_1, \dots, p_n\rangle, \quad (\text{A.100})$$

where A, B are $n \times n$ matrices with matrix elements

$$(A_p)_{ij} = \beta^p \delta_{ip} \delta_{j0} + \frac{(-1)^{j+1} \beta^p}{\sin(\pi(n-j)/n)} \delta_{n-p, j-i}, \quad (\text{A.101})$$

$$(B_p)_{ij} = \beta^p \delta_{ip} \delta_{j0} + \frac{(-1)^{n-j} \beta^p}{\sin(\pi(n-j)/n)} \delta_{n-p, j-1}. \quad (\text{A.102})$$

Here $0 \leq i, j \leq n-1$ with $n \geq 2$. This representation ensures that no certain spin configurations occur over three consecutive sites in $|\psi_\beta\rangle$.

Fig. A.40 displays the graph associated with the Onsager's scar in a system with six and eight spins, with $\beta = 1/\sqrt{2}$. Up to normalization, the states read [72, 73]

$$\begin{aligned} |\psi_\beta^{(6)}\rangle = & +2(|000011\rangle - |000110\rangle + |001100\rangle \\ & + |110000\rangle - |011000\rangle - |100001\rangle) \\ & + |011110\rangle + |001111\rangle + |100111\rangle \\ & - |101101\rangle - |011011\rangle + |110011\rangle \\ & - |110110\rangle + |111001\rangle + |111100\rangle \\ & + 4|000000\rangle, \end{aligned} \quad (\text{A.103})$$

$$\begin{aligned}
|\psi_\beta^{(8)}\rangle = & 8|00000000\rangle + 4(|00000011\rangle - |00000110\rangle \\
& + |00001100\rangle - |00011000\rangle + |00110000\rangle \\
& - |01100000\rangle + |11000000\rangle - |10000001\rangle) \\
& + 2(|00001111\rangle - |00011011\rangle + |00011110\rangle \\
& + |00110011\rangle - |00110110\rangle + |00111100\rangle \\
& - |01100011\rangle + |01100110\rangle - |01101100\rangle \\
& + |01111000\rangle + |10000111\rangle - |10001101\rangle \\
& + |10011001\rangle - |10110001\rangle + |11000011\rangle \\
& - |11000110\rangle + |11001100\rangle - |11011000\rangle \\
& + |11100001\rangle + |11110000\rangle) + |00111111\rangle \\
& + |10110111\rangle - |10111101\rangle - |01101111\rangle \\
& - |10011111\rangle + |11001111\rangle - |11011011\rangle \\
& + |11011110\rangle + |01111011\rangle - |01111110\rangle \\
& - |11100111\rangle + |11101101\rangle + |11110011\rangle \\
& - |11110110\rangle - |11111001\rangle + |11111100\rangle \\
& + |11111111\rangle.
\end{aligned} \tag{A.104}$$

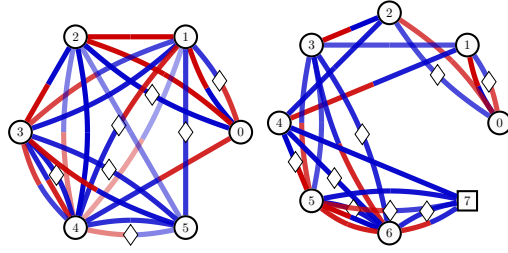


Figure A.41: The graphs associated with the ground state of the PXP model with six (left panel) and eight (right) particles.

Scars in the PXP Model. Another platform where quantum scars emerge is in the *PXP* model. This model on a chain with L sites and with the periodic boundary condition is [432]

$$\mathcal{H} = \sum_i P_{i-1} X_i P_{i+1} + P_L X_1 P_2 + P_{L-1} X_L P_1, \tag{A.105}$$

where $P = |0\rangle\langle 0|$ and $X = |0\rangle\langle 1| + |1\rangle\langle 0|$. One of the ground states of this model in the matrix product representation yields

$$|\text{PXP}\rangle = \sum_{\{\sigma\}} \text{tr}[B^{\sigma_1} C^{\sigma_2} \dots B^{\sigma_{L-1}} C^{\sigma_L}] |\sigma_1 \sigma_2 \dots \sigma_L\rangle, \tag{A.106}$$

where

$$B^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad B^1 = \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \tag{A.107}$$

$$C^0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad C^1 = \sqrt{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ -1 & 0 \end{pmatrix}. \tag{A.108}$$

Fig. A.41 displays the graph associated with the ground state of the PXP model with six and eight particles. Up to normalization, the states are

74 75

$$\begin{aligned}
|PXP6\rangle = & |000000\rangle - |000010\rangle \\
& - |101000\rangle - |100000\rangle \\
& + |101010\rangle - 2|010100\rangle \\
& + \sqrt{2}(|100100\rangle + |010000\rangle), \tag{A.109}
\end{aligned}$$

$$\begin{aligned}
|PXP8\rangle = & |00000000\rangle + |00001000\rangle \\
& + |10000000\rangle + |10101010\rangle \\
& - |00001010\rangle - |10000010\rangle \\
& - |10100000\rangle - |10101000\rangle \\
& - 2(|01010000\rangle + |10010100\rangle) \\
& + \sqrt{2}(|01000010\rangle - |00000100\rangle) \\
& + |10010000\rangle - |01000000\rangle \\
& + |10100100\rangle + 2|01010100\rangle. \tag{A.110}
\end{aligned}$$

A.6 Quantum Communication

Quantum communication refers to communication protocols that involve the transfer of quantum states. Machine learning has been previously employed for the artificial discovery of quantum communication schemes [84].

The first experimental realization of entanglement swapping was presented as a way of entangling two photons that never interacted [124]. In this section we show graphs corresponding to experiments that create entanglement between two parties (each party with single or multiple photons) without interaction between the parties. Entanglement swapping has a strong connection to other quantum communication tasks, e.g. teleportation. This opens the door to use PyTheus for discovering related quantum information protocols.

Entangling Two Photons that Never Interacted without Bell Pairs.

Respecting the constraints that are placed on two particles that do not interact (presented in section 3.2.2) we show a graph that entangles two photons from independent sources in Fig. A.42. The corresponding setup constructed by path identity does not require the initial creation of entangled Bell pairs [76].

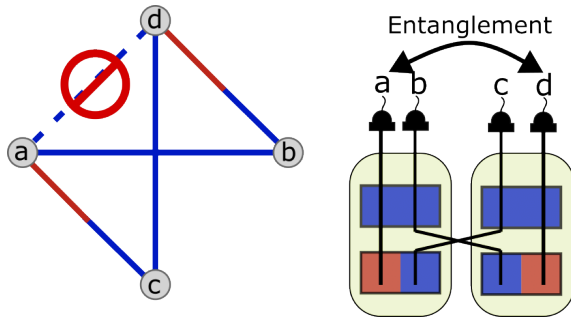


Figure A.42: Entanglement between two qubits without a common source. Because the vertices a and d have no connection in the graph, each of the corresponding photons is created in one of two disjoint subsystems (green boxes).

Multiparticle Entanglement Swapping. Performing simultaneous entanglement swapping on multiple pairs of particles is one of the key players in achieving resource-efficient quantum communication. Entanglement

Figure A.43: Entanglement swapping of 3 Bell pairs measuring only 4 photons. Left: abstract graph for multi-particle entanglement swapping. Right: outline of a multi-particle Entanglement swapping experiment. Alice and Bob each receive three particles from the two independent sources 1 and 2 (each a collection of SPDC crystals). When Charlie measures a coincidence in all four detectors Alice and Bob share three Bell pairs (0&3, 1&4, 2&5) without them having interacted. This is possible due to a superposition of events where Charlie either receives three photons from Alice and one photon from Bob or one photon from Alice and three photons from Bob.

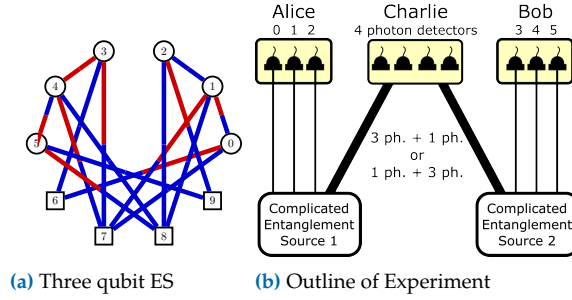
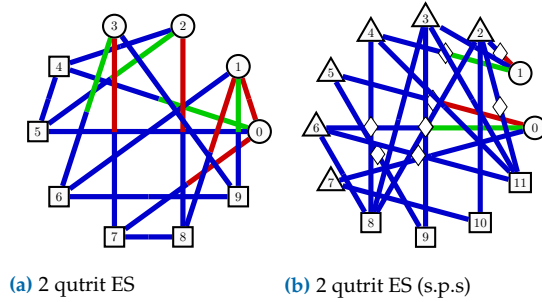


Figure A.44: Graphs for entanglement swapping experiments.



swapping between two qubits requires two additional particles for a Bell state measurement. Performing this swapping experiment in parallel for n qubit pairs would require $2n$ additional photons. We prompted PyTheus to find experiments that beat this naive baseline. With this, we found a three-qubit entanglement swapping experiment (Fig. A.43), which produces the state [77]

$$|\Phi^+\rangle_{03} \otimes |\Phi^+\rangle_{14} \otimes |\Phi^+\rangle_{25}, \quad (\text{A.111})$$

where the photons 0, 1, 2 are separated from 3, 4, 5. This experiment requires four additional particles instead of the six particles necessary for the parallel case described above.

Similarly, entanglement swapping between a pair of qutrits requires four ancillary particles. We found that "performing entanglement swapping for two pairs of qutrits could also beat the naive baseline by only requiring six ancillas instead of eight, in total, to produce [78]

$$|\Phi^+\rangle_{02}^{3d} \otimes |\Phi^+\rangle_{13}^{3d}, \quad (\text{A.112})$$

where the photons 0, 1 are separated from 2, 3, and $|\Phi^+\rangle^{3d}$ is the first three-dimensional Bell state.

Entanglement Swapping with Single-Photon Sources. It is also possible to use single-photon sources for entanglement swapping [126, 433, 434]. In Fig. A.44, we show a graph [79] for a higher-dimensional case, performing three-dimensional entanglement swapping with single photon sources. In total, six single photon sources are necessary.

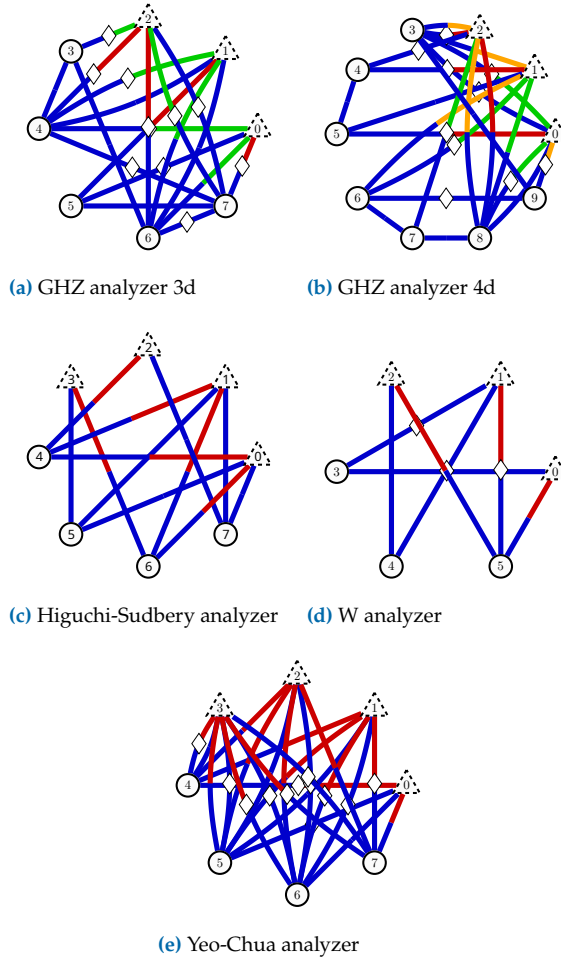


Figure A.45: Graphs for analyzers. When all detectors click, the incoming photons were in the corresponding state.

A.7 Quantum Measurements

An important tool in quantum information is multi-particle measurements such as the Bell state measurement [129] or the GHZ analyzer [435]. Here we present a wider range of measurements that PyTheus has discovered. Many others, such as POVM (positive operator-valued measure) or their symmetric, informationally complete special case (SIC-POVMs) could be additional targets for future research, given their exceptional importance for quantum information tasks [436, 437] as well as their connection to exciting number theoretical questions [438].

Analyzers

Here, we use the word *analyzer* to refer to experimental setups that confirm a collection of photons to be in a particular state. Their formulation in terms of graphs is described in 3.2.3. The two-dimensional GHZ analyzer realized in Ref. [435] can be extended towards higher dimensions, giving the 3d GHZ analyzer [80] and the 4d GHZ analyzer [81]. Further, we show analyzers for the W state [82], the Higuchi-Sudbery state (shown in Eq. equation A.42) [83] and the Yeo-Chua state (shown in Eq. equation A.18) [84] in Fig. A.45.

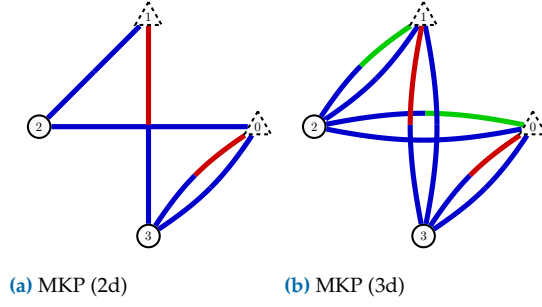


Figure A.46: Graphs realizing the VAA state measurement for the MKP in the 2-dimensional case and 3-dimensional case.

Mean King's Problem

In 1987, L. Vaidman, Y. Akaronov, and D. Z. Albert [151] devised an interesting quantum communication task that can be solved only with quantum resources. Later referred to as the Mean King's Problem (MKP), it involves two parties. Alice, who sends a quantum state which she created to another party – the Mean King. The Mean King then performs a projective measurement on the state in a basis of his choice out of a collection of mutually unbiased bases (MUBs). Alice is then allowed to perform one more measurement, after which the King declares his measurement basis and Alice must correctly guess his result. Should Alice guess incorrectly even once, she will suffer a cruel fate, for the Mean King is exceptionally intolerant of poor guesses. This task has applications in quantum key distribution, wherein even the slightest discrepancy between the shared secret key of Alice and Bob implicates (in principle) the presence of an eavesdropper.

Over the years, various generalizations to the initial solution, proposed in the original paper [151], have been introduced. Here, we employ a generalization proposed by Hayashi, Horibe, and Hashimoto [439] for quantum states of dimension D which is a prime power. This solution proposes that Alice first prepares a maximally entangled, two-photon state $|\psi_0\rangle$ and sends one of her photons to the Mean King. After the King makes a projective measurement in one of the $(D + 1)$ MUBs, Alice retrieves her photon and performs a measurement in the basis of states credited to Vaidman, Akaronov, and Albert (VAA states). There are D^2 VAA states in all. For $D = 2$, the first VAA state can be written as

$$|\phi_1\rangle = \frac{1}{2}(\sqrt{2}|00\rangle + e^{-i\frac{\pi}{4}}|01\rangle + e^{i\frac{\pi}{4}}|10\rangle) \quad (\text{A.113})$$

For $D = 3$, the third VAA state can be written as

$$|\phi_3\rangle = \frac{1}{\sqrt{3}}(|00\rangle + \alpha(|02\rangle + |10\rangle + |01\rangle + |20\rangle) + \beta(|12\rangle) + \gamma(|21\rangle)) \quad (\text{A.114})$$

where $\omega = e^{-i2\pi/3}$, $\alpha = (\omega^2 + 2\omega)/3$, $\beta = (\omega^2 + 2)/3$, and $\gamma = (\omega + 2)/3$,

Experimentally realizing Alice's measurement in this basis is non-trivial. Setups that recreate the two-dimensional VAA measurement using two qubits encoded in a single photon [440, 441] and that realize extensions to the Mean King's problem [442] have been proposed; but the original solution to the Mean King's Problem involves two photons, to which no experimental realization has been proposed for any prime-power D . Here we provide examples discovered by PyTheus for select prime-dimensional cases.

Two-dimensional Case. The left side of Fig. A.46 shows the graph of Alice’s measurement in the VAA basis [85]. This graph was found with PyTheus by optimizing for the VAA state projection into $|\phi_1\rangle$. Surprisingly however, during the translation of the graph to an optical setup, we found that several other VAA states can be projected with the same setup, by adding detectors at unused beam splitter ports. Several of the remaining VAA states can then be distinguished by the set of possible simultaneous two-detector click events that they each trigger. Since any result of the Mean King’s measurement can be expressed in terms of a superposition of VAA states, Alice is able to guess the result by performing the measurement of her input state in the VAA basis and guessing the state that can be expressed in terms of Alice’s measurement result. This procedure works regardless of the Mean King’s choice of MUB in his measurement.

Three-Dimensional Case. The right-hand side of Fig. A.46 shows the graph for the projection of the state into the one of the three-dimensional VAA states $|\phi_3\rangle$ [86]. As with the two-dimensional case, we add again detectors at the empty port of beam splitters and find that we can distinguish between more three of the nine states without any additional modifications. After post-selection, Alice has a guaranteed chance to correctly guess the Mean King’s measurement result so long as the King does not choose the second MUB for his measurement. Were the King to pick that MUB, two of the three results of the measurement are expressed in terms of VAA states that Alice cannot distinguish, giving her a 50% chance of escaping the Mean King’s cruelty.

A.8 Quantum Gates

Universal quantum gates, which rely on the interaction between two or more photons, can be realized with non-linearities induced by measurements [132]. Quantum gates based on this approach, such as a CNOT between two qubits, have been realized experimentally for almost twenty years [131]. However, since then, the attempt to experimentally discover a wide range of quantum gates has continued due to the advancement of experimental resources which can deal with higher dimensional systems.

Heralded. We call a quantum gate heralded when only ancillary particles are detected, meaning that the outgoing particles are not measured. The photons exiting such a gate can continue into further components of a longer circuit. At the end of the circuit it is necessary to confirm that two photons have exited the gate, but not which path they followed. We find that a higher dimensional version of the CNOT gate with a control qubit and a target qutrit (CNOT(2, 3) [87]) can also be realized with two ancilla photons. Further, we find a heralded Toffoli gate with four ancillas [88]; see Fig. A.47.

Reduced Input Space. For some tasks in which quantum gates are applied, there is some prior knowledge about the input state. This knowledge can be exploited to reduce the experimental resources required to perform the transformation [139, 443]. A wide range of constraints on the input state can be encoded by our framework. Here we show examples of heralded quantum gates where a target qubit is prepared in the zero state before entering the gate.

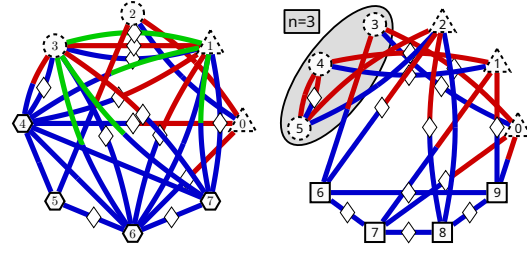


Figure A.47: Graphs for heralded quantum gates.

(a) Heralded CNOT(2,3)

(b) Heralded Toffoli

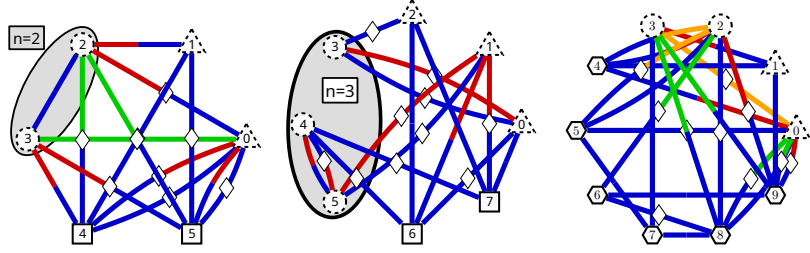


Figure A.48: Graphs for heralded quantum gates with reduced input space.

(a) heralded CNOT(3,3)

(b) heralded Fredkin

(c) heralded CNOT(4,4)

PyTheus found gates CNOT(3, 3) [89], CNOT(4, 4) [90] and the Fredkin gate [91] acting on a target input photon in the computational zero state; see Fig. A.48. A post-selected Fredkin gate on the full input space has been experimentally realized [140].

Post-Selected. When post-selecting a gate, all outgoing photons are detected and thus destroyed. Such a gate does not require as many resources as a heralded gate, but it imposes restrictions, such as not mixing the two output paths, on the remainder of the circuit, to ensure that the presence of a photon in each output of the gate can be ascertained. PyTheus found different post-selected high-dimensional CNOT gates, such as CNOT(2, 3) [92], CNOT(2, 4) [93], CNOT(3, 3) [94] and the Toffoli gate on a qutrit [95]; see Fig. A.49.

Single Photon Sources for Quantum Gates

Experiments exploiting single-photon sources as an additional resource for the construction of quantum gates have been previously demonstrated [146]. Fig. A.50 displays graphs for a heralded CNOT(2, 2) with two additional input photons [96], a heralded CNOT(2, 3) with three additional input photons [97], and a post-selected CNOT(2, 3) with two additional input photons [98].

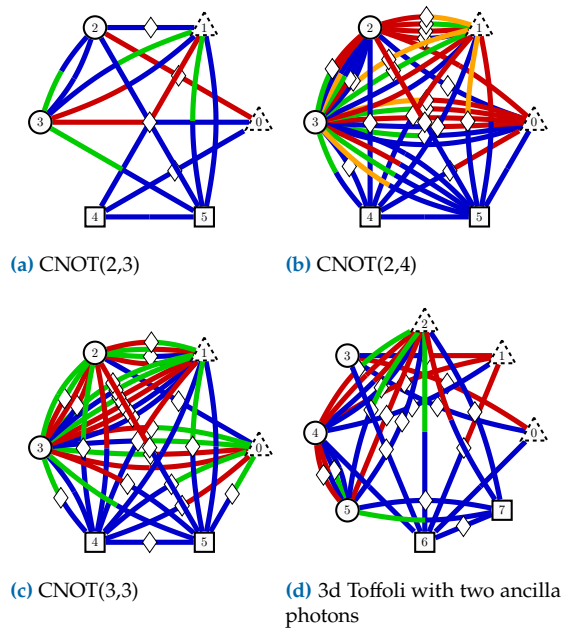


Figure A.49: Graphs for quantum gates with post-selection.

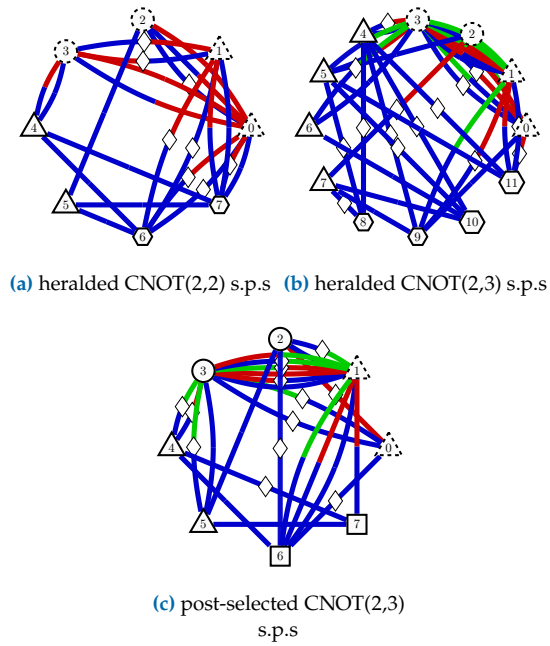


Figure A.50: Graphs for quantum gates with single-photon sources as an additional resource.

A.9 Combinatorial Measures

In this section, we demonstrate how PyTheus can discover exceptional structures without computing the quantum state. This ability might be useful for finding experimental configurations with particular symmetries or properties that are interesting, independent of the resulting quantum state. Such metrics can also be combined with other objectives mentioned in the previous sections.

Assembly Index

As an example for structural property discovery, we consider the *assembly index*, which has been invented in chemistry for the search of extraterrestrial life [444, 445]. Specifically, the assembly index counts the complexity of building up a combinatorial structure, for instance, molecules or — in our case — graphs. It counts the number of independent processes that are necessary to create a structure. The hypothesis is that a structure with a large assembly index cannot be formed by natural processes and would require complex (living) systems for its generation. Strong indications in favor of this hypothesis have been found in the study of millions of molecules on earth [445].

Our goal here is to discover graphs with very high assembly indices. The assembly index is a discrete structural metric that cannot be optimized directly using gradients. However, we can use the weights of edges to transform the discrete metric into a continuously differentiable metric. A general procedure is to compute the average combinatorial value of the weighted graph from sampled discrete graphs. Here, the weights $|\omega_i|^2$ are used as the probability for sampling discrete graphs. While this procedure is differential, it is computationally expensive as many sampled graphs need to be evaluated to obtain an average assembly index of the whole graph.

Alternatively, here we restrict ourselves to graphs with exactly eight edges (with 4 [99] and 6 [100] vertices). Here, the eight highest-weighted edges are used to generate the graph, and in addition, weights are used to change the order of edges. The graphs in Fig. A.51 present graphs with very high assembly indices with 4 vertices and 8 edges as well as 6 vertices and 10 edges.

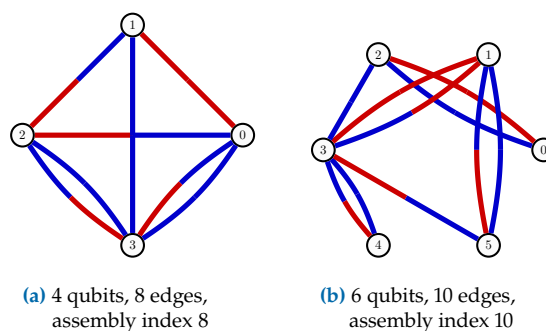
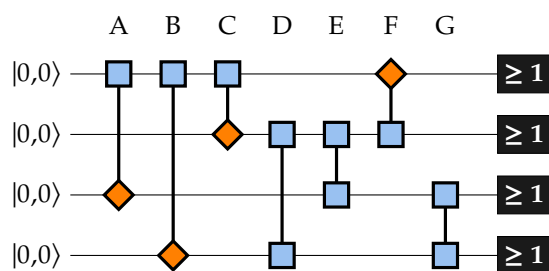


Figure A.51: Graphs for assembly index.

B.

B.1 4-qubit W state	133
B.2 Heralded Bell state . . .	134
B.3 2-mode NOON states . .	135

For the W state, the baseline we surpassed was introduced as the following path identity experiment [10].



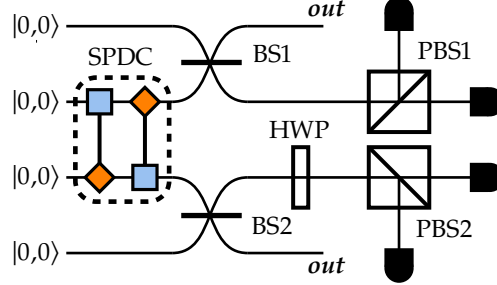
W state's baseline: 54 sortings.

Fidelity	G (Counts)	G*F (Counts)	G**F (Counts)	G**F**F (Counts)
0.70	2.25	2.20	2.15	2.10
0.72	1.85	1.80	1.75	1.70
0.74	1.50	1.45	1.40	1.35
0.76	1.22	1.20	1.18	1.15
0.78	1.05	1.02	1.00	0.98
0.80	0.90	0.88	0.85	0.82
0.82	0.75	0.72	0.70	0.68
0.84	0.60	0.58	0.55	0.52
0.86	0.45	0.43	0.40	0.38
0.88	0.35	0.33	0.30	0.28
0.90	0.25	0.23	0.20	0.18
0.92	0.18	0.16	0.14	0.12
0.94	0.12	0.10	0.08	0.06
0.96	0.08	0.06	0.04	0.02
0.98	0.05	0.03	0.01	0.00
1.00	0.02	0.01	0.00	0.00

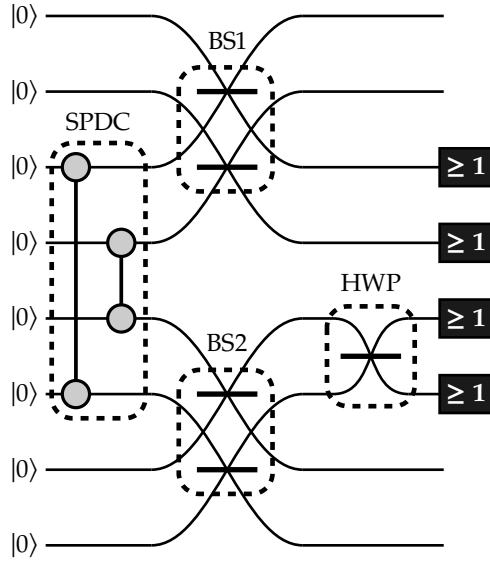
Figure B.1: 54 canonical sortings for the baseline design. When generating the W state, the sources’ ordering in the baseline design leads to different tradeoffs between fidelity and success probability. The positions of sources F and G are a good proxy for performance.

B.2 Heralded Bell state

The heralded Bell state we surpassed was not presented as an undirected graph, but as the following setup [178].



Equivalently, each optical path can be split into two polarization modes, making the polarizing beam splitters redundant. Additionally, each of the beam splitters (BS1 and BS2) turns into 2 beam splitters, one per mode, and the half-wave plate becomes a beam splitter between the two modes within the same optical path.



Notice that the two undetected modes in the top (bottom) correspond to a single optical path, like in the heralded experiment in Fig. 3 of the main text. The original photon source was a type-II SPDC [183], modeled as

$$\begin{aligned} S_2(r) &= \exp \left(r(h_1 v_2 - h_1^\dagger v_2^\dagger) - r(h_2 v_1 - h_2^\dagger v_1^\dagger) \right) \\ &= \exp \left(r(h_1 v_2 - h_1^\dagger v_2^\dagger) \right) \exp \left(-r(h_2 v_1 - h_2^\dagger v_1^\dagger) \right), \end{aligned} \quad (\text{B.1})$$

following the notation of Eq. (1) of the main text. As depicted in the design, this is equivalent of having two sources acting on the same paths with a phase difference of π . Thus, the two sources share a common pump $r \in \mathbb{R}$. Additionally, the beamsplitters (BS1 and BS2) were characterized by a common transmission factor t . To obtain the baseline data shown in Fig. 3 of the main text, we optimized both parameters for different fidelity values f_0 (see Eq. (5) of the main text). In contrast, the crystals'

pumps of the other 2 baselines had a predefined ratio, and we rescaled them (like in Fig. B.3).

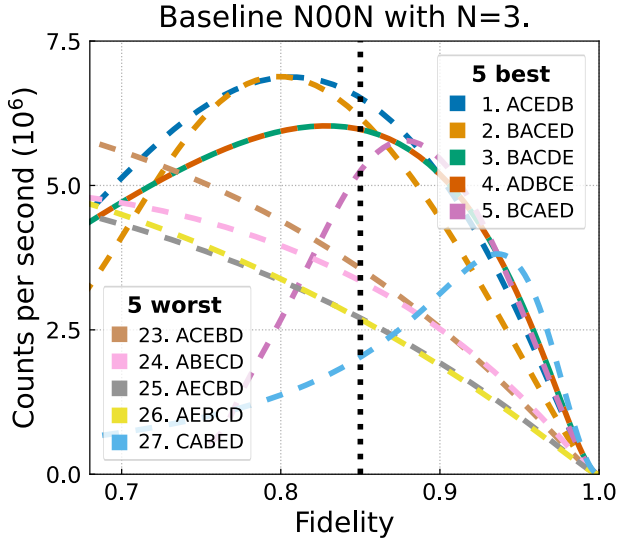
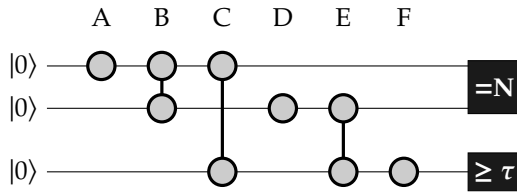


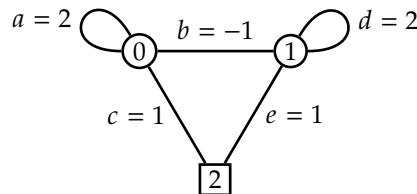
Figure B.2: best & worst sortings for the 3-photon N00N. We ranked the operators' sortings based on their successful detection counts at fidelity 85%. Yet, no sequence is always superior for all fidelity values.

B.3 2-mode N00N states

The N00N state experiments use two main modes and one ancillary mode. Therefore, without repetition, we have at most 6 sources with 180 unique sequences, 89 if we consider the equivalence between the first 2 modes.



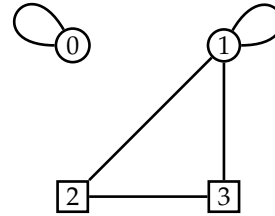
The baselines were even smaller [6]. For $N=3$, the baseline used only 5 sources, introduced as the graph:



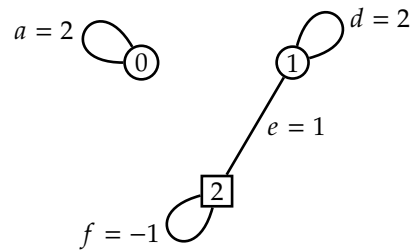
The square node shows the ancillary mode, and the labels the ratio between crystal pumps. Following the partial commutation rules, the operators can be sorted in 54 distinct ways. However, as the two main modes can be switched, the canonical sortings are only 27. While they converge to fidelity 1, the tradeoff curves diverge for lower fidelities. In Fig. B.2, we show the “best” and “worst” orderings based on the success

probability at fidelity 85%. Yet, the worst classified reached the highest probability at high fidelity (>95%).

For $N=4$, the original baseline used 5 sources and 2 ancillary modes (see the following diagram)



Here, we realized that there is no need for a second ancillary mode. As photons are always created in pairs and we know that the main modes receive exactly 2 photons, a single path with a threshold photodetector suffices. To have a fair comparison, we simplified the baseline into a more efficient graph, with only 4 sources and 1 ancilla.



The remaining sources can be sorted in four ways. As shown in Fig. B.3, one sorting surpasses the others for all fidelities higher than 73%.

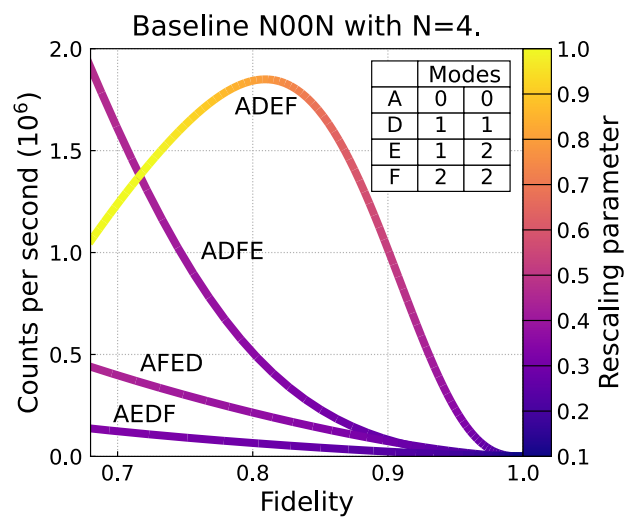


Figure B.3: all possible sortings for the 4-photon N00N. Keeping the original ratio between sources, we show how the tradeoff changes with the crystal pump. Among 4 possible sortings, one stands as the best design for fidelity > 73%.

Additional material for Chapter 6

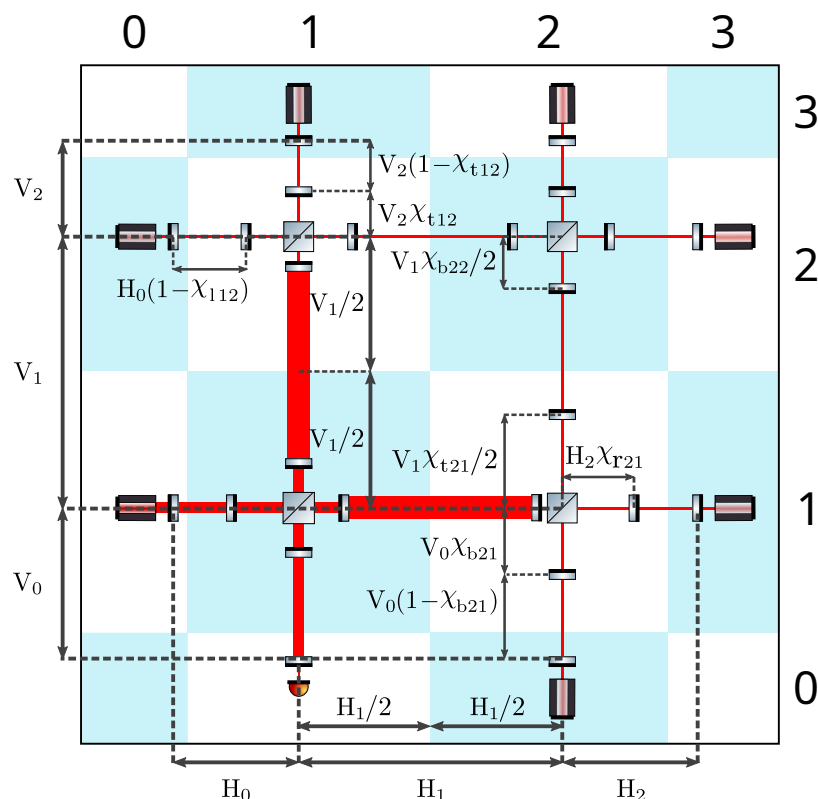
C.

C.1 UIFO characterization

C.1.1 Consistent distances

In the UIFO framework, the optical components are located within an irregular rectangular grid (see Fig. C.1). To locate each element, we set the distances between columns and rows of beam splitters and then locate the mirrors in between based on fractions of the largest distances. We fix the distances between the outer mirrors and their closest light source (or detector), since they do not affect the GWD performance. The values of the vertical and horizontal distances, V_i and H_j , are also fractions of a larger tunable length, which can take up to 4km.

With the fraction based location the mirrors could overlap. To prevent that nonphysical situation, we impose a minimum value ϵ to all the distances described. This works because the number of elements between beam splitters is always the same, and therefore the number of distances. If we wanted to remove elements we should be cautious with these fixed gaps.



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Figure C.1: positioning of the optical elements. To keep the grid structure, we locate the mirrors of the optical path based on fractions of larger distances, vertical $\{V_i\}$ and horizontal $\{H_j\}$, which are also fractions of a maximum vertical and horizontal size, up to 4km. For the sake of readability, we keep the mirrors within the cells, so the mirrors between beam splitters can be localized as a fraction of half the gap between beam splitters. The distance between light sources and their nearest mirrors is fixed, since that does not affect the design performance. Same applies for the detector and its nearest mirror.

C.1.2 Discrete design choices

Having set the UIFO topology and distances there are a few discrete design choices to make, like where to locate the detector, or which type

of light sources to use. We also have to choose if we place beam splitters or Faraday isolators in the grid intersections, and how to orient them. We could also decide, for each optical component, whether we want it to be there at all.

In our experiments, we always place the detector in the same grid cell and only use lasers as light sources, not squeezers. We do not use Faraday isolators either and the orientation of the beam splitters is always the same. We are keeping it simple for now.

C.1.3 Optical components

Except for the detector, all optical components have, at least, one continuous parameter to tune. Table C.1 summarizes the range of parameter values for the UIFO designs, before they are normalized (from 0 to 1) for training. Finesse can also simulate other elements and parameters, all listed and documented [206].

It is worth noticing that when you ‘turn off’ a light source, both for squeezers and lasers, by setting their parameters to 0, their effect on the system ceases. That is not the case for mirrors and beam splitters. We can never ignore them because they have a minimum loss, and because they are still susceptible to power damages. To properly ‘turn them off’ we must remove them completely.

Table C.1: Range of values (before normalization) for different optical components. In Finesse, the mass of mirrors and beam splitters is infinite by default, we leave it like that in all our experiments. Mirrors and beam splitters have a minimum loss and transmission of 5ppm and 15ppm, respectively, which limits their reflectivity.

Components	Parameters	Min	Max
Mirrors and beam splitters	Reflectivity	0	0.99998
	Phase (°)	0	360
Lasers	Power (W)	0	200
Squeezers	Factor (db)	0	20
	Angle (°)	0	360

C.2 Datasets

The 1 million training samples and 100K testing samples are stored in H5 data files with 1000 pairs of inputs-outputs each, normalized from 0 to 1. The inputs are the linearly normalized parameters of distances (lengths up to 4km and fractions of these) and optical components (described in Table C.1). The loss and masses of mirrors and beamsplitters are not included as they are constant.

The outputs are the sensitivity values of 101 frequency values (a geometric sequence from 1Hz to 10kHz), and the light power acting on the detector, mirrors, and beam splitters. As with the loss and masses, we do not include the frequency values in the dataset, as they are always the same. Since both sensitivities and powers range across several orders of magnitude, we must not rescale them directly from 0 to 1. Instead, and only during training/testing, we rescale the logarithm of their values

$$\mathcal{N}(x) \equiv \frac{\log_{10}(x + \epsilon) + b}{S}, \quad (\text{C.1})$$

where the values of ϵ , b , and S are (0, 30, 30) for the sensitivities, and (10^{-6} , 6, 16) for the powers. These choices are motivated by the expected

range of sensitivities, usually from 10^{-25} to 10^{-10} (by definition, never 0 or higher than 1), and the expected range of powers, from 0 to 10^7 (beyond that, everything burns).

C.3 Surrogate model architecture

To mimic the behavior of the Finesse simulator we trained a transformer-based surrogate model grids [294, 295]. The parameters of neighbor elements were merged into tokens. We divided the grid by the beamsplitters (see Fig. C.2) and add their parameters to the four neighboring subgraphs. These overlapping patches were projected into a higher-dimensional space via a linear layer followed by a sinusoidal activation. This transformation, also known as Fourier feature mapping [261], proved essential for capturing the high-frequency features of the GWD (see Fig. 6.1), overcoming the spectral bias of standard neural networks [296, 297]. The tokens are then processed by 8 transformer layers. After processing the tokens, they were projected into a lower dimension and concatenated. With a final multilayer perceptron, we obtain an array of predicted sensitivities and optical powers.

Since the transformer encoder is invariant under permutation of patches, we add 2 trainable positional encoding to the patches: before and after they are processed by the Fourier mapping.

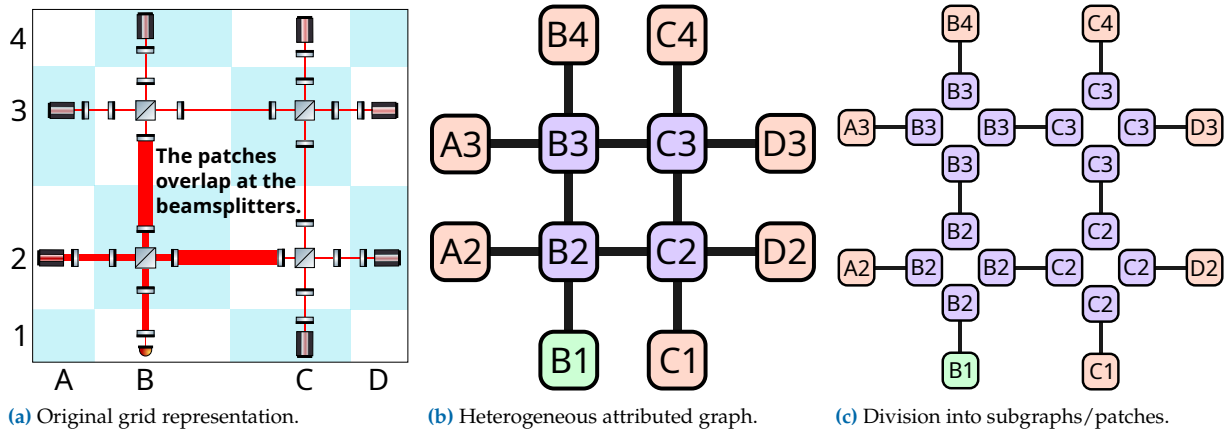


Figure C.2: patch extraction from the gravitational wave detector. We encoded GWD into rectangular grids (a), which can be seen as graphs with different types of nodes, and multiple attributes at nodes and edges (b). Dividing the rectangular graph into overlapping subgraphs, we group the parameters that characterize them into patches (c). The three types of nodes, light sources (orange), beam splitters (purple), and detector (green), while the mirror properties are encoded into the edges that connect the nodes. Since the patches have different number of attributes we pad the patches up to a common length. For simplicity, we group the vertical and horizontal distances, $\{V_i\}$ and $\{H_j\}$, into 2 additional patches.

C.4 GPU resources used for surrogate-based inverse design

The training times showed in Table 6.2 are averaged from the data shown in Table C.2. The inverse design of 20K samples over 16384 optimization steps took 16, 40 and 85 minutes on average, for the 3 grid sizes. We used *AMD Instinct MI300A APU* units.

Table C.2: training times of the surrogate models for the 3 UIFO sizes.

Generation of surrogate	Seed	UIFO size (# parameters)		
		1x1 (29)	2x2 (85)	3x3 (169)
0th	0	20' 12"	27' 00"	47' 00"
	1	20' 05"	31' 23"	40' 57"
1st	0	45' 09"	56' 04"	1h 30' 32"
	1	44' 29"	56' 44"	1h 13' 20"
2nd	0	1h 07' 12"	1h 36' 27"	2h 25' 46"
	1	1h 15' 58"	1h 47' 48"	1h 39' 36"
3rd	0	1h 42' 31"	2h 04' 59"	3h 06' 26"
	1	1h 43' 32"	2h 25' 05"	3h 12' 27"
4th	0	2h 20' 13"	2h 34' 05"	4h 09' 31"
	1	2h 12' 22"	2h 48' 26"	3h 25' 32"
Total time	0	6h 15' 17"	7h 38' 35"	11h 59' 15"
	1	6h 16' 26"	8h 29' 26"	10h 11' 52"

C.5 Optimization with Finesse

Fig. C.3 shows the detailed results of the design optimization run with Finesse for 5 days in 256 parallel jobs for each size of the UIFO.

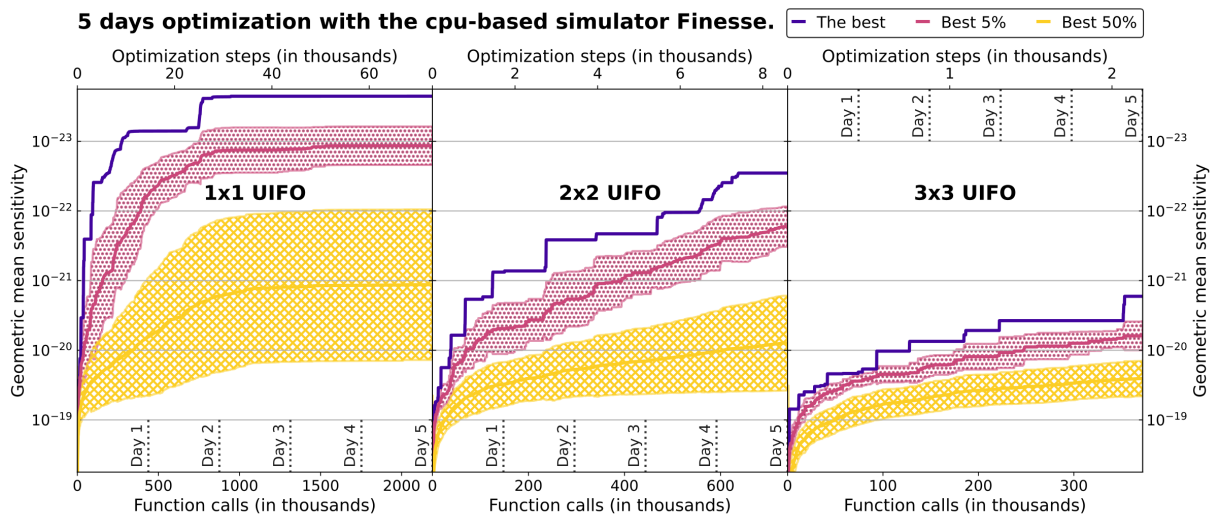


Figure C.3: optimization of 256 designs with Finesse during 5 days. Using the Adam optimizer with annealed noise (same than for the surrogates) we optimize 256 independent input parameter sets, randomly initialized, to minimize the loss from Eq. equation 6.2. Lacking automatic differentiation, we must use finite differences to compute the gradients, taking 30, 86, and 170 evaluations per gradient for the different UIFO sizes: 1x1, 2x2, and 3x3. With approximate running times of 0.2, 0.6, and 1.2 seconds respectively, each optimization step might take more than 3 minutes for the largest system. The plotted data considers the best sensitivity ever achieved by the samples at a given optimization step, ignoring the decreases from the gradient computations and/or the optimization process.

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